Cochrane Library

**Cochrane** Database of Systematic Reviews

# Long-term effects of weight-reducing drugs in people with hypertension (Review)

Spary-Kainz U, Posch N, Radl-Karimi C, Jeitler K, Krenn C, Siebenhofer A, Berghold A, Horvath K, Winterholer S, Semlitsch T

Spary-Kainz U, Posch N, Radl-Karimi C, Jeitler K, Krenn C, Siebenhofer A, Berghold A, Horvath K, Winterholer S, Semlitsch T.
Long-term effects of weight-reducing drugs in people with hypertension.
*Cochrane Database of Systematic Reviews* 2026, Issue 6. Art. No.: CD007654.
DOI: 10.1002/14651858.CD007654.pub6.

**www.cochranelibrary.com**




WILEY

# TABLE OF CONTENTS

[Intervention Review]

# Long-term effects of weight-reducing drugs in people with hypertension

Ulrike Spary-Kainz[1], Nicole Posch[1a], Christina Radl-Karimi[1], Klaus Jeitler[1,2], Cornelia Krenn[1], Andrea Siebenhofer[1,3], Andrea Berghold[2], Karl Horvath[1,4], Sebastian Winterholer[1], Thomas Semlitsch[1]

[1]Institute of General Practice and Evidence-Based Health Services Research, Medical University of Graz, Graz, Austria. [2]Institute for Medical Informatics, Statistics and Documentation, Medical University of Graz, Graz, Austria. [3]Institute for General Practice, Goethe University, Frankfurt am Main, Germany. [4]Klinikum Bad Gleichenberg, Bad Gleichenberg, Austria

[a]ORCID iD: https://orcid.org/0000-0001-7939-1435

**Contact:** Nicole Posch, nicole.posch@medunigraz.at.

**Editorial group:** Cochrane Central Editorial Service.
**Publication status and date:** New search for studies and content updated (no change to conclusions), published in Issue 6, 2026.

**Citation:** Spary-Kainz U, Posch N, Radl-Karimi C, Jeitler K, Krenn C, Siebenhofer A, Berghold A, Horvath K, Winterholer S, Semlitsch T. Long-term effects of weight-reducing drugs in people with hypertension. *Cochrane Database of Systematic Reviews* 2026, Issue 6. Art. No.: CD007654. DOI: 10.1002/14651858.CD007654.pub6.



## ABSTRACT

### Rationale

Long-term weight management drugs have the potential to reduce body weight and blood pressure in obese or overweight individuals. Being overweight or obese is often associated with hypertension, which is linked to an increased risk of cardiovascular mortality and morbidity. The effects of weight-reducing drugs on patient-relevant outcomes, especially in people with hypertension, are currently unclear.

This review is the fourth update of one first published in July 2009.

### Objectives

#### Primary objective

To assess the effects of drugs approved for long-term weight management on all-cause mortality, cardiovascular morbidity and adverse events in adults with essential hypertension.

#### Secondary objective

To assess the effects of drugs approved for long-term weight management on changes in systolic and diastolic blood pressure and body weight in adults with essential hypertension.

### Search methods

For this updated review, the Cochrane Hypertension Information Specialist searched the following databases to 22 April 2024: the Cochrane Hypertension Specialised Register, Cochrane CENTRAL, MEDLINE, Embase and trial registries, with no language restrictions. We also checked references, searched citations, and contacted manufacturers and authors of relevant papers.

### Eligibility criteria

We included randomised controlled trials (RCTs) of at least 24 weeks' duration that compared approved long-term weight management drugs to placebo in non-pregnant adults with essential hypertension.

**Outcomes**

Critical outcomes were all-cause mortality, cardiovascular morbidity and adverse events. Important outcomes were changes in systolic and diastolic blood pressure and changes in body weight.

**Risk of bias**

We assessed the risk of bias using the Cochrane RoB 1 tool.

**Synthesis methods**

We used standard Cochrane methods. For meta-analyses, we used risk ratios (RRs) for dichotomous outcomes and mean differences (MDs) for continuous outcomes. We undertook both fixed-effect and random-effects meta-analysis, presenting the random-effects results. We assessed the certainty of evidence using GRADE.

**Included studies**

We identified no additional trials of drugs included in the previous version of the review. We included two new RCTs evaluating the recently approved drugs semaglutide and tirzepatide. In total, eight RCTs reported results on different weight-reducing drugs compared to placebo in people with hypertension (approximately 13,000 hypertensive participants in total). The interventions evaluated were orlistat (4 trials, 3132 participants), phentermine/topiramate (1 trial, 1305 participants), naltrexone/bupropion (1 trial, 8283 participants), semaglutide (1 trial, 274 participants) and tirzepatide (1 trial, 153 participants with stage 2 hypertension and 635 participants taking antihypertensive medication). Study duration ranged from six to 48 months. Seven of the eight trials were funded by the pharmaceutical industry.

We were unable to include any liraglutide results, as no RCTs have provided separate data for participants with hypertension. Rimonabant, sibutramine and lorcaserin are no longer considered relevant, since their marketing approval has been withdrawn.

**Synthesis of results**

**Orlistat versus placebo**

Orlistat may have little to no effect on all-cause mortality compared to placebo, but the evidence is very uncertain (3 versus 0 deaths; 3 trials, 1488 participants; very low certainty evidence). Orlistat may have little to no effect on cardiovascular morbidity compared to placebo, but the evidence is very uncertain (1.9% versus 0.7% in one trial and 17% versus 19% in another trial; 2 trials, 1721 participants; very low certainty evidence).

Orlistat probably increases serious adverse events (SAEs) compared to placebo (RR 1.45, 95% CI 1.10 to 1.91; 3 trials, 1476 participants; moderate-certainty evidence). There may be little to no difference in all adverse events (AEs) between orlistat and placebo, but the evidence is very uncertain (RR 1.13, 95% CI 0.84 to 1.54; 2 trials, 1386 participants; very low-certainty evidence).

**Phentermine/topiramate versus placebo**

Phentermine/topiramate may have little to no effect on all-cause mortality compared to placebo, but the evidence is very uncertain (no deaths in either group; 1 trial, 1305 participants; very low-certainty evidence). Phentermine/topiramate may have little to no effect on cardiovascular morbidity compared to placebo, but the evidence is very uncertain (treatment-emergent cardiac events: high-dose versus placebo: RR 2.39, 95% CI 0.82 to 6.69; 1 trial, 1044 participants; low-dose versus placebo: RR 1.51, 95% CI 0.43 to 5.27; 1 trial, 785 participants; both very low-certainty evidence).

Phentermine/topiramate may have little to no effect on SAEs compared to placebo, but the evidence is very uncertain (RR 0.85, 95% CI 0.49 to 1.48; 1 trial, 1305 participants; very low-certainty evidence). Phentermine/topiramate probably increases all AEs compared to placebo (RR 1.13, 95% CI 1.08 to 1.20; 1 trial, 1305 participants; moderate-certainty evidence).

**Naltrexone/bupropion versus placebo**

Naltrexone/bupropion may have little to no effect on all-cause mortality (RR 0.99, 95% CI 0.70 to 1.40; 1 trial, 8283 participants; very low-certainty evidence) and cardiovascular morbidity (RR 1.11, 95% CI 0.87 to 1.41; 1 trial, 8283 participants; very low-certainty evidence) compared to placebo, but the evidence is very uncertain. There is probably little to no difference between naltrexone/bupropion and placebo in SAEs (RR 1.05, 95% CI 0.96 to 1.14; 1 trial, 8283 participants; moderate-certainty evidence), but naltrexone/bupropion probably increases all AEs compared to placebo (RR 1.69, 95% CI 1.58 to 1.80; 1 trial, 8283 participants; moderate-certainty evidence).

**Semaglutide versus placebo**

No results were reported for all-cause mortality, cardiovascular morbidity, SAEs and AEs.

**Tirzepatide versus placebo**

No results were reported for all-cause mortality, cardiovascular morbidity, SAEs and AEs.

**Authors' conclusions**

There have been multiple developments in the domain of medications for long-term weight reduction in recent years. Several pharmaceutical products have been removed from the market due to safety concerns, whilst new drugs have been introduced. Our critical outcomes of death and cardiovascular complications were not the focus of the trials included in this review but were sometimes presented as part of an 'adverse events' outcome, with the resultant data providing only very low certainty evidence. Overall, the evidence for people with hypertension remains insufficient to draw conclusions regarding the benefits of pharmacological weight loss in terms of reducing the risk of mortality or cardiovascular morbidity. Further trials are needed. Moreover, separate results for participants with hypertension should be made available from any completed trials involving both normotensive and hypertensive people.

**Funding**

This Cochrane review had no dedicated funding.

**Registration**

Protocol and previous version DOIs: 10.1002/14651858.CD007654; 10.1002/14651858.CD007654.pub2; 10.1002/14651858.CD007654.pub3; 10.1002/14651858.CD007654.pub4; 10.1002/14651858.CD007654.pub5

## PLAIN LANGUAGE SUMMARY

**Do medicines that help people lose weight reduce the harmful effects of high blood pressure?**

**Key messages**

- The effects of three weight-reducing medicines (orlistat, phentermine/topiramate and naltrexone/bupropion) on death rates and cases of heart disease are unclear for people with hypertension (high blood pressure) due to a lack of reliable evidence. There is no information about these measures for the medicines semaglutide and tirzepatide.
- Taking phentermine/topiramate or naltrexone/bupropion probably causes more unwanted events in people with hypertension than taking a placebo (inactive 'dummy' medicine). Taking orlistat probably causes an increase in the number of serious unwanted effects compared to taking a placebo. We did not find any information about side effects for people with hypertension taking semaglutide or tirzepatide.
- Long-term studies are needed to assess the effects of medicines used for weight management on death rates and heart and blood vessel problems in people with hypertension. It is also important that results for people with hypertension be provided by authors of any completed study that included people with and without hypertension.

**What is high blood pressure (hypertension)?**

Blood pressure is a measure of the force that your heart uses to pump blood around your body. It is usually given as two numbers: the pressure when your heart pushes blood out (systolic pressure), and the pressure when your heart rests between beats (diastolic pressure). Blood pressure is considered to be high when systolic pressure is over 140 and/or diastolic pressure is over 90. The risk of developing high blood pressure (hypertension) increases with age.

**Why is hypertension a problem?**

Hypertension can increase the risk of developing serious health problems, such as heart attack or stroke. Lowering blood pressure in people with hypertension reduces the number of people who develop diseases of the heart and blood vessels (cardiovascular diseases), which leads to fewer deaths and cardiovascular problems.

**Is there a link between being overweight and hypertension?**

Being overweight is associated with increased blood pressure. Hypertension treatment guidelines recommend maintaining a healthy weight and losing weight when needed. Some people may take medicine specifically to help reduce their weight.

**What did we want to find out?**

We wanted to find out if weight-loss medicines could reduce the negative effects of hypertension on people's health.

**What did we do?**

We updated a systematic review. We looked for studies on how weight-loss medicines affect people with hypertension. We wanted to know how many people experienced unwanted events (e.g. side effects), how many developed heart disease or stroke, and if any died.

We looked for studies called 'randomised controlled trials', which usually give the most reliable evidence about the effects of a treatment. We rated how certain we were about the evidence, taking into account how the studies were done, how many people took part and whether results were consistent. All studies compared a weight-loss medicine with a placebo, which is a 'dummy' medicine, one that looks like the same medicine but is inactive.

**What did we find?**

We added two new studies of medicines recently approved for weight reduction, so, in total, this updated review includes eight studies. They involved around 13,000 people with hypertension (average age 46 to 62 years). Three studies were conducted in the USA, three in European countries and two were multinational. The studies lasted between six and 48 months. Pharmaceutical companies funded seven of the eight studies.

For people with hypertension, three drugs that are approved to help people lose weight over time (orlistat, phentermine/topiramate and naltrexone/bupropion) may make little to no difference to how many people die or how many people have heart problems, but we are very uncertain about the results. People taking orlistat, phentermine/topiramate and naltrexone/bupropion probably have more serious unwanted events than those taking placebo. However, there may be little to no difference in unwanted events in general between orlistat and placebo, and little to no difference in serious unwanted events between phentermine/topiramate and placebo, but we are very uncertain about these results. The most common unwanted events were digestive problems (for orlistat, phentermine/topiramate), and dry mouth and skin tingling or numbness (for naltrexone/bupropion).

The studies that investigated semaglutide or tirzepatide did not present information on the number of deaths, heart problems or safety issues in people with hypertension.

**What are the limitations of the evidence?**

We are not confident in the evidence regarding overall deaths and heart problems for orlistat, phentermine/topiramate or naltrexone/bupropion, or regarding serious unwanted events for phentermine/topiramate, because of our concerns about how the studies were conducted and the low number of the events. We are not confident in the evidence about all unwanted events for orlistat because of our concerns about how the studies were conducted and because the results were inconsistent across different studies. We have more confidence in the evidence regarding serious unwanted events for orlistat and naltrexone/bupropion, and regarding all unwanted events for phentermine/topiramate or naltrexone/bupropion, but we do not have high certainty about these because of our concerns about how some of the studies were conducted.

**How up to date is this evidence?**

We included evidence published up to 22 April 2024.

## SUMMARY OF FINDINGS

### Summary of findings 1. Orlistat versus placebo for weight reduction

**Orlistat compared with placebo for weight reduction**

**Patient or population:** men and non-pregnant women ≥ 18 years old with essential hypertension

**Setting:** outpatient clinics, private endocrinologists, medical centres, USA, Europe

**Intervention:** orlistat

**Comparison:** placebo

| Outcomes | Anticipated absolute effects (95% CI) | | Relative effect (95% CI) | No of participants (studies) | Certainty of the evidence (GRADE) | Comments |
|---|---|---|---|---|---|---|
| | Risk with placebo | Risk with orlistat | | | | |
| **All-cause mortality**<br>Follow-up: 24 - 208 weeks | 2 deaths (OD subgroup) and 1 death (OS subgroup) with orlistat, and no deaths with placebo in 1 trial; no deaths with orlistat or placebo in 2 other trials | | - | 1488 (3 studies) | ⊕⊝⊝⊝<br>**very low**[a] | Very low event rate |
| **Cardiovascular morbidity**<br>Follow-up: 24 weeks | see comment | | - | 1721 (2 studies) | ⊕⊝⊝⊝<br>**very low**[b] | Very low event rate. Outcomes only mentioned in the context of adverse events and probably incomplete. Reporting of cardiovascular morbitity not comparable between the 2 trials (singular cardivascular events vs percentage of vascular complications), therefore no meta-analysis possible. Small number of events. The effects of orlistat compared with placebo for this outcome are uncertain. |
| **Serious adverse events**<br>Follow-up: 24 - 208 weeks | 101 per 1000 | 146 per 1000 (111 to 192) | RR 1.45 (1.10 to 1.91) | 1476 (3 studies) | ⊕⊕⊕⊝<br>**moderate**[c] | - |
| **All adverse events**<br>Follow-up: 24-208 weeks | 865 per 1000 | 977 per 1000 (727 to 1000) | RR 1.13 (0.93 to 1.37) | 1386 (2 studies) | ⊕⊝⊝⊝<br>**very low**[d] | 95% heterogeneity |

**CI:** confidence interval; **MD:** mean difference; **OD**: orlistat and diastolic blood pressure ≥ 90 mm Hg; **OS**: orlistat and systolic blood pressure ≥ 140 mm Hg; **RR**: risk ratio

GRADE Working Group grades of evidence

**High certainty:** we are very confident that the true effect lies close to that of the estimate of the effect.
**Moderate certainty:** we are moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.
**Low certainty:** our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect.
**Very low certainty:** we have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of effect.

[a]Downgraded by two levels because of serious imprecision (very low event rates) and by one level because of increased risk of bias
[b]Downgraded by two levels because of serious imprecision (very low event rates, inconclusive reporting within the trials) and by two levels because of high risk of bias
[c]Downgraded by one level because of increased risk of bias
[d]Downgraded by one level because of increased risk of bias and by two levels because of severe inconsistency (98% heterogeneity)

### Summary of findings 2. Phentermine/topiramate versus placebo for weight reduction

**Phentermine/topiramate compared with placebo for weight reduction**

**Patient or population:** men and non-pregnant women ≥ 18 years old with essential hypertension

**Setting:** outpatient clinics, USA

**Intervention:** phentermine/topiramate

**Comparison:** placebo

| Outcomes | Anticipated absolute effects (95% CI) | | Relative effect (95% CI) | No of participants (studies) | Certainty of the evidence (GRADE) | Comments |
|---|---|---|---|---|---|---|
| | Risk with placebo | Risk with phentermine/topiramate | | | | |
| **All-cause mortality**<br>Follow-up: 56 weeks | see comment | | - | 1305 (1 study) | ⊕⊝⊝⊝<br>**very low**[a] | No death occurred in the hypertensive subgroup of the included RCT. |
| **Cardiovascular morbidity**<br>Follow-up: 56 weeks | 25 per 1000 | 37 per 1000 (11 to 48) | RR 1.51 (0.43 to 5.27) | 785 (1 study) | ⊕⊝⊝⊝<br>**very low**[b] | 2.3% of the hypertensive participants in the low-dose phen/top group, 3.7% in the high-dose phen/top group, and 1.7% in the placebo group experienced treatment-emergent cardiac adverse events. No information on vascular events. |
| | | 59 per 1000 (20 to 166) | RR 2.39 (0.82 to 6.69) | 1044 (1 study) | ⊕⊝⊝⊝<br>**very low**[b] | |

| | | | | | | |
|---|---|---|---|---|---|---|
| **Serious adverse events**<br><br>Follow-up: 56 weeks | 42 per 1000 | 36 per 1000 (21 to 62) | RR 0.85 (0.49 to 1.48) | 1305 (1 study) | ⊕⊝⊝⊝<br>**very low**[a] | - |
| **All adverse events**<br><br>Follow-up: 56 weeks | 773 per 1000 | 873 per 1000 (835 to 927) | RR 1.13 (1.08 to 1.20) | 1305 (1 study) | ⊕⊕⊕⊝<br>**moderate**[c] | - |

**CI:** confidence interval; **MD:** mean difference; **RCT:** randomised controlled trial; **RR**: risk ratio

GRADE Working Group grades of evidence
**High certainty:** we are very confident that the true effect lies close to that of the estimate of the effect.
**Moderate certainty:** we are moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.
**Low certainty:** our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect.
**Very low certainty:** we have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of effect.

[a]Downgraded by two levels because of serious imprecision (very low event rate) and by one level because of increased risk of bias
[b]Downgraded by two levels because of serious imprecision (very low event rate) and by two levels because of high risk of bias
[c]Downgraded by one level because of increased risk of bias

### Summary of findings 3. Naltrexone/bupropion versus placebo for weight reduction

**Naltrexone/bupropion compared with placebo for weight reduction**

**Patient or population:** men and non-pregnant women ≥ 18 years old with essential hypertension

**Setting:** medical sites, USA

**Intervention:** naltrexone/bupropion

**Comparison:** placebo

| Outcomes | Anticipated absolute effects (95% CI) | | Relative effect (95% CI) | No of participants (studies) | Certainty of the evidence (GRADE) | Comments |
|---|---|---|---|---|---|---|
| | Risk with placebo | Risk with naltrexone/bupropion | | | | |
| **All-cause mortality**<br><br>Follow-up: 121 weeks | 15 per 1000 | 15 per 1000 (11 to 21) | RR 0.99 (0.70 to 1.40) | 8283 (1 study) | ⊕⊝⊝⊝<br>**very low**[a] | - |

| | | | | | | |
|---|---|---|---|---|---|---|
| **Cardiovascular morbidity**<br><br>Follow-up: 121 weeks | 29 per 1000 | 32 per 1000 (25 to 40) | RR 1.11 (0.87 to 1.41) | 8283 (1 study) | ⊕⊝⊝⊝<br>**very low**[b] | MI, stroke and hospitalisation for unstable angina |
| **Serious adverse events**<br><br>Follow-up: 121 weeks | 205 per 1000 | 215 per 1000 (196 to 233) | RR 1.05 (0.96 to 1.14) | 8283 (1 study) | ⊕⊕⊕⊝<br>**moderate**[c] | - |
| **All adverse events**<br><br>Follow-up: 121 weeks | 256 per 1000 | 432 per 1000 (404 to 460) | RR 1.69 (1.58 to 1.80) | 8283 (1 study) | ⊕⊕⊕⊝<br>**moderate**[c] | - |

**CI:** confidence interval; **MD:** mean difference; **MI:** myocardial infarction; **RR**: risk ratio

GRADE Working Group grades of evidence
**High certainty:** we are very confident that the true effect lies close to that of the estimate of the effect.
**Moderate certainty:** we are moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.
**Low certainty:** our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect.
**Very low certainty:** we have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of effect.

[a]Downgraded by two levels because of serious imprecision (very low event rate) and by one level because of increased risk of bias
[b]Downgraded by two levels because of imprecision (very low event rate) and by one level because of high risk of bias (other bias: only 50% of the planned number of cardiovascular events were reported)
[c]Downgraded by one level because of increased risk of bias

### Summary of findings 4. Semaglutide versus placebo for weight reduction

**Semaglutide compared with placebo for weight reduction**

**Patient or population:** men and non-pregnant women ≥ 18 years old with uncontrolled hypertension (i.e. systolic BP ≥ 140 mm Hg and/or diastolic BP ≥ 90 mm Hg)

**Setting:** medical sites, USA

**Intervention:** semaglutide 2.40 mg

**Comparison:** placebo

| Outcomes | Anticipated absolute effects (95% CI) | Relative risk effect<br><br>(95% CI) | No of participants (studies) | Certainty of the evidence (GRADE) | Comments |
|---|---|---|---|---|---|

| | Risk with placebo | Risk with semaglutide | | | | |
|---|---|---|---|---|---|---|
| **All-cause mortality** | - | - | - | - | - | No results reported |
| **Cardiovascular morbidity** | - | - | - | - | - | No results reported |
| **Serious adverse events** | - | - | - | - | - | No results reported |
| **All adverse events** | - | - | - | - | - | No results reported |

**BP:** blood pressure; **CI:** confidence interval

### Summary of findings 5. Tirzepatide versus placebo for weight reduction

**Tirzepatide compared with placebo for weight reduction**

**Patient or population:** men and non-pregnant women ≥ 18 years old with essential hypertension

**Setting:** medical sites, USA

**Intervention:** tirzepatide

**Comparison:** placebo

| **Outcomes** | **Anticipated absolute effects (95% CI)** | | **Relative effect (95% CI)** | **No of participants (studies)** | **Certainty of the evidence (GRADE)** | **Comments** |
|---|---|---|---|---|---|---|
| | **Risk with placebo** | **Risk with tirzepatide** | | | | |
| **All-cause mortality** | - | - | - | - | - | No results reported |
| **Cardiovascular morbidity** | - | - | - | - | - | No results reported |
| **Serious adverse events** | - | - | - | - | - | No results reported |
| **All adverse events** | - | - | - | - | - | No results reported |

**BP:** blood pressure; **CI:** confidence interval

# BACKGROUND

## Description of the condition

Hypertension is a chronic condition associated with an increased risk of cardiovascular mortality and morbidity. It is estimated that raised blood pressure affects 1.3 billion people worldwide and leads to an estimated 10.8 million deaths each year [1]. Lowering blood pressure levels in hypertensive people has been shown to be an effective means of reducing cardiovascular morbidity and mortality [2].

Epidemiological investigations have consistently found an association between high blood pressure and different lifestyles, some of which lead to excess body weight, for example, sedentary lifestyles [3, 4]. Major clinical guidelines recommend weight reduction as a first-step intervention in the therapy of hypertensive people [5, 6, 7, 8]. Body weight may be reduced by lifestyle modifications as well as pharmacological and invasive interventions.

## Description of the intervention and how it might work

Dietary-intervention studies in hypertensive people have shown a positive association between weight loss and blood pressure reduction [9]. It therefore seems reasonable to suppose that medical weight-reducing treatment may also lead to a fall in blood pressure, and subsequently, a reduction in cardiovascular risk.

For a select group of overweight or obese people for whom lifestyle interventions (e.g. physical activity or dietary interventions) are unsuccessful, drugs for long-term weight management may be an option to help reduce body weight. Orlistat, sibutramine, and rimonabant were formerly the most commonly used weight management drugs. But of these, only orlistat, which was approved in 1998, still has market approval for the long-term treatment of overweight and obesity. Sibutramine and rimonabant both lost their marketing approval about 10 years ago. Sibutramine had been approved by the US Food and Drug Administration (FDA) in the late 1990s, but based on the results of the Sibutramine Cardiovascular Outcomes Trial (SCOUT 2010 [10]), which showed an increased risk of serious cardiovascular events with sibutramine, the FDA and European Medical Agency (EMA) recommended suspension of the marketing authorisation [11, 12, 13]. In 2010, the manufacturer agreed to voluntarily withdraw sibutramine from the market [14, 15]. Rimonabant received regulatory approval in several European countries in 2006, but failed to receive FDA approval after concerns about an association between rimonabant and an increased incidence of psychiatric adverse events [16]. In 2008, the EMA recommended the suspension of rimonabant from the market because post-marketing analyses demonstrated detrimental effects compared with placebo [17, 18]. In 2009, the European Commission decided to withdraw market authorisation for rimonabant in all countries of the European Union [19].

Since 2012, four additional drugs (lorcaserin, liraglutide, phentermine/topiramate and naltrexone/bupropion) have been approved by the FDA for long-term weight management in obese (body mass index (BMI) ≥ 30 kg/m$^2$) and overweight (BMI ≥ 27 kg/m$^2$) people with at least one obesity-related comorbidity [20, 21, 22, 23]. These medications have been recommended in guidelines for the long-term pharmacological treatment of obesity [24]. Between 2015 and 2018, liraglutide, which had originally been approved for the treatment of diabetes mellitus type 2, and naltrexone/bupropion have also received market approval for long-term weight management in many countries all over the world, including the EU countries [25, 26], Canada [27, 28], Australia [29, 30] and South Korea [31, 32]. In 2013, the manufacturer of lorcaserin withdrew its application to the EMA after the Committee for Medicinal Products for Human Use raised safety concerns [33]. In 2020, lorcaserin was withdrawn from the USA market by the manufacturer after a request by the FDA [34]. This action was taken after an FDA analysis of data from a safety clinical trial showed an increased occurrence of cancer in participants treated with lorcaserin [35]. For phentermine/topiramate, marketing authorisation was refused by the EU and other countries because of major concerns about the long-term effects of the drug on the heart and blood vessels, and particularly because phentermine is known to increase heart rate [36]. There were also concerns about long-term psychiatric and cognitive effects related to the topiramate component [37].

Since 2021, two additional drugs, semaglutide and tirzepatide, have received market approval for long-term weight management around the world, such as the USA [38, 39], EU countries [40, 41], Canada [42, 43], Australia [44, 45] and South Korea [46, 47].

Drugs for long-term weight management aim to reduce body weight and to maintain weight reduction over a longer period. Orlistat is a gastric and pancreatic lipase inhibitor that blocks the absorption of about 30% of dietary fat [48]. The combination of phentermine, a neurostabiliser, and topiramate, an antiseizure medication, appears to have an additive effect on weight reduction [49]. In combination with naltrexone, bupropion, a dopamine and norepinephrine reuptake inhibitor, reduces appetite and increases energy expenditure [50]. The mechanisms by which these three medications cause weight loss are not yet fully understood. Liraglutide, a glucagon-like peptide 1 (GLP-1) receptor agonist, appears to regulate appetite by increasing feelings of satiety [51]. Semaglutide, another GLP-1 receptor agonist, suppresses appetite, improves control of eating and reduces food cravings and ad libitum energy intake [52]. Tirzepatide, which combines the dual agonism of Glucose-Dependent Insulinotropic Polypeptide (GIP) and GLP-1 receptors, reduces food intake and delays gastric emptying [53, 54].

## Why it is important to do this review

For overweight or obese people with established hypertension, blood pressure should first be managed with non-pharmacological interventions such as weight reduction [5, 6, 7, 8]. Since drugs for long-term weight management may support the efforts of people to reduce body weight, it is important that physicians be informed about their efficacy and about the potential harms of these drugs before prescribing them.

Systematic reviews and meta-analyses have shown that pharmacological weight-reducing interventions with orlistat, phentermine/topiramate, liraglutide, semaglutide and tirzepatide reduce both blood pressure and body weight in the general population [55, 56, 57, 58, 59]. Treatment with naltrexone/bupropion reduces body weight but does not lower blood pressure [50, 56, 57]. None of these reviews investigated the efficacy and safety of weight-management drugs specifically in the population of overweight or obese people with hypertension. Furthermore, only limited data are available on whether pharmacological weight reduction lowers the risk of mortality and other patient-relevant endpoints. Results from a few RCTs that investigated mixed populations of hypertensive and normotensive people showed no beneficial effect on cardiovascular events for liraglutide or phentermine/topiramate compared to placebo [57]. People with hypertension have an increased risk for a variety of cardiovascular diseases, including stroke, coronary artery disease, heart failure, atrial fibrillation [60, 61]. In addition, long-term weight loss in overweight people with hypertension leads to a greater reduction in blood pressure than in those without hypertension [62]. It is therefore particularly interesting to determine whether people with hypertension benefit from weight loss in terms of reducing their cardiovascular risk.

Rimonabant, sibutramine and lorcaserin have been withdrawn from the market after safety concerns. These three drugs are therefore no longer considered relevant for this review update. This leaves orlistat, liraglutide, naltrexone/bupropion, semaglutide and tirzepatide as currently approved for the long-term treatment of obesity in most countries worldwide, and phentermine/topiramate as approved for the long-term treatment of obesity in the USA only.

This is an update of a Cochrane review first published in 2009, and previously updated in 2013, 2016 and 2021 [63]. We have chosen to update it again now because of the approval of new weight-management drugs.

# OBJECTIVES

## Primary objective

To assess the effects of drugs approved for long-term weight management on all-cause mortality, cardiovascular morbidity, and adverse events in adults with essential hypertension.

## Secondary objective

To assess the effects of drugs approved for long-term weight management on change from baseline in systolic blood pressure, change from baseline in diastolic blood pressure, and change from baseline in body weight in adults with essential hypertension.

# METHODS

## Differences between protocol and review or between review versions

We conducted this review in accordance with the published protocol [64]. In conducting the review, we followed the Methodological Expectations for Cochrane Intervention Reviews (MECIR) [65]. For the reporting of this 2026 review update, we followed PRISMA 2020 guidance [66].

Since guidelines for pharmacological management of obesity cite four additional medications (liraglutide, lorcaserin, phentermine/topiramate, naltrexone/bupropion) for long-term weight reduction [24], we extended our search to include these drugs in the 2016 version of this review [67].

Since the market approvals of rimonabant and sibutramine were withdrawn in 2009 and 2010, respectively, we no longer consider these two drugs relevant for long-term weight management, and we therefore excluded them from the 2021 version of this review [63].

In 2020, the market approval for lorcaserin was withdrawn after safety concerns. Therefore, we no longer consider this drug relevant for long-term weight management and have excluded it from this 2026 update of the review.

Since current guidelines for the pharmacological management of obesity cite two additional medications (semaglutide and tirzepatide) for long-term weight reduction [68], we extended our search to include these drugs in this 2026 update of the review.

We also made the following changes in this 2026 update.

- We described the objectives of the review more clearly with regard to the included intervention. We used the term 'drugs approved for long-term weight management' instead of 'pharmacologically-induced reduction in body weight'.
- We added a detailed definition of the critical outcome 'cardiovascular morbidity'. We considered both major and minor cardiovascular events in relation to this outcome. We did not consider changes in surrogate endpoints (e.g. laboratory measures), which may be associated with cardiovascular morbidity, as relevant.
- We shortened the summary of findings tables to include only our critical outcomes. We provided additional summary of findings tables with all important outcomes for all comparisons in Supplementary material 10.

## Criteria for considering studies for this review

### Types of studies

Randomised controlled trials (RCTs), including those of parallel-group, cross-over and cluster-randomised design, that compare pharmacological interventions approved for long-term weight management versus placebo, with a follow-up of at least 24 weeks. Studies spanning several years are particularly relevant for assessing morbidity and mortality. However, shorter studies may also be useful for evaluating therapy quality, provided the blood pressure-lowering effect can be reliably assessed over several months and compared with the effect on patient-relevant therapy goals (e.g. reduction of drug side effects or physical complaints). Therefore, we included only studies with a minimum duration of 24 weeks. For cross-over studies, an observation period of 24 weeks was required for each study arm.

We only included RCTs where the randomised intervention was the sole difference between the treatment arms, and where all participants were otherwise treated equally. Any additional active care, such as antihypertensive medication or a lifestyle intervention, must have been applied to both the active treatment and control groups in the same way throughout the study.

### Types of participants

We included men and non-pregnant women aged 18 years or older with essential hypertension according to the WHO definition (i.e. a baseline blood pressure of at least 140 mm Hg systolic or a diastolic blood pressure of at least 90 mm Hg, or both, or people on antihypertensive treatment) [1].

### Types of interventions

We included drugs approved for long-term weight management according to international regulatory agencies at the time of the review update. For drugs also approved for other indications, we took into account only those dosages that are approved for weight loss.

- Orlistat (60 or 120 mg)
- Phentermine (7.5 or 15 mg)/topiramate (46 or 92 mg)
- Naltrexone (8 mg)/bupropion (90 mg)
- Liraglutide (3.0 mg)
- Semaglutide (2.4 mg)
- Tirzepatide (5, 10 or 15 mg)

## Outcome measures

### Critical outcomes

- All-cause mortality
- Cardiovascular morbidity
  - Non-fatal major cardiovascular events
  - Minor cardiovascular events
- Adverse events
  - Serious adverse events (SAEs)
  - All adverse events (AEs)
  - AEs leading to withdrawals
  - AEs related to a particular anti-obesity drug
  - Specific AEs

### Important outcomes

- Change in systolic blood pressure
- Change in diastolic blood pressure
- Change in body weight

## Search methods for identification of studies

### Electronic searches

The Cochrane Hypertension Information Specialist searched the following databases, without language, publication year or publication status restrictions.

- The Cochrane Hypertension Specialised Register via the Cochrane Register of Studies (CRS-Web) (searched 22 April 2024)
- The Cochrane Central Register of Controlled Trials (CENTRAL) via the Cochrane Register of Studies (CRS-Web; searched 22 April 2024)
- Ovid MEDLINE ALL (from 1946 onwards) (searched 22 April 2024)
- Embase Ovid (from 1974 onwards) (searched 22 April 2024)
- US National Institutes of Health Ongoing Trials Register (www.clinicaltrials.gov) (searched 17 March 2020)
- World Health Organisation International Clinical Trials Registry Platform (https://trialsearch.who.int/) (searched 17 March 2020)

For the update search conducted in 2024, we searched ClinicalTrials.gov and WHO ICTRP within Cochrane CENTRAL.

The Information Specialist modelled subject strategies for databases on the search strategy designed for MEDLINE. Where appropriate, they were combined with subject strategy adaptations of the highly sensitive search strategy designed by Cochrane for identifying randomised controlled trials (as described in the *Cochrane Handbook for Systematic Reviews of Interventions* Version 6.5, 2024, Chapter 4 [69]).

We present the search strategies used for the four bibliographic databases in Supplementary material 1. The search strategy used in the previous review versions is documented in Supplementary material 7.

### Searching other resources

- The Cochrane Hypertension Information Specialist searched the Hypertension Specialised Register segment (which includes searches in MEDLINE, Embase and Epistemonikos for systematic reviews) to retrieve existing reviews relevant to this systematic review, so that we could scan their reference lists for additional trials. The Specialised Register also includes searches for controlled trials in the Allied and Complementary Medicine Database (AMED), CAB Abstracts & Global Health, CINAHL, ProQuest Dissertations & Theses and Web of Science.
- We checked the bibliographies of the included studies and any relevant systematic reviews we identified for further references to relevant trials.
- Where necessary, we contacted authors of key papers and abstracts to request additional information about their trials (see Supplementary material 8).
- We contacted manufacturers of weight-reducing drugs to request information about potentially relevant trials (see Supplementary material 8).
- We checked the included studies for retractions and errata via PubMed (www.ncbi.nlm.nih.gov/pubmed) and the Retraction Watch Database (http://retractiondatabase.org) and reported the search dates in the review.
- We checked potentially relevant studies for additional publications on hypertensive subgroups via PubMed.

## Data collection and analysis

### Selection of studies

The Cochrane Hypertension's Information Specialist used the Known Assessments and RCT classifier components of Cochrane's Screen4Me (S4Me) workflow to automatically exclude non-RCTs prior to screening. Following this, two review authors (USK, NP) independently screened the title and abstract of each remaining reference identified by the search and applied the inclusion criteria. We retrieved potentially relevant studies in full, and again two review authors independently decided whether they met the inclusion criteria. In cases of disagreement, we also obtained the full article and the two review authors inspected it independently, resolving differences in opinion by recourse to a third review author (TS). If a resolution of the disagreement was not possible, we classified the article as 'awaiting assessment' and contacted the authors of the study for clarification. We re-assessed the articles after receiving the study authors' replies.

### Data extraction and management

Two review authors (USK, NP) independently extracted data using a predefined data extraction form. We resolved differences in data extraction by consensus, referring back to the original article. We sought information from the authors of the primary studies if necessary. We extracted, checked and recorded the following data.

- General information: all publications of a single trial, sponsor of the trial (specified, known or unknown), and country in which the trial was conducted.

- Methods section: characteristics of the trial, participants, and interventions, and outcome measures used and reported in the publication.
  - Characteristics of the trial: design and duration of the trial, randomisation (and method), allocation concealment (and method), blinding (participants, people administering treatment, outcome assessors), and checking of blinding.
  - Characteristics of participants: number of participants in each group, how the participants were selected (random, convenience), exclusion criteria used, and their demographic characteristics (e.g. age, sex/gender, nationality, ethnicity). We extracted disease-related information about the duration of hypertension. We checked the similarity of groups at baseline, as well as reports about withdrawals and losses to follow-up (reasons/description), and described them in the risk of bias tables in Supplementary material 2. If hypertensive subpopulations were analysed, we noted and reported the reasons and methods.
  - Characteristics of interventions: duration of the intervention, length of follow-up (in months), the type of long-term weight management drug (orlistat, phentermine/topiramate, naltrexone/bupropion, liraglutide, semaglutide or tirzepatide), dose and administration route.
  - Characteristics of outcome measures: measures as reported in the outcome section of the studies: time points, longest follow-up for critical outcomes, 6-month, 12-month and longest follow-up for important outcomes.
- Data from clinical trial registers: if data from included trials were available as study results in clinical trial registers such as ClinicalTrials.gov or similar sources, we made full use of this information and extracted the data. If there was also a full publication of the trial, we collated and critically appraised all available data. If an included trial was marked as a completed study in a clinical trial register but no additional information (study results, publication or both) was available, we contacted the study authors for further information. If no data were submitted, we labelled the trial as awaiting classification (see Supplementary material 4).

### Risk of bias assessment in included studies

Two review authors (USK, NP) independently assessed trials fulfilling the inclusion criteria in order to evaluate methodological quality, resolving any differences in opinion through discussion with a third review author (TS). We assessed all trials using the Cochrane RoB 1 assessment tool [70], and we judged the risk of bias criteria as having low, high or unclear risk. We evaluated individual bias items as described in the *Cochrane Handbook for Systematic Reviews of Interventions* according to the categories: adequate sequence generation, allocation concealment, blinding, incomplete outcome data, selective reporting, and other potential biases [70]. We evaluated the risk of bias for the categories of blinding and incomplete outcome data separately for each outcome [71]. We noted whether endpoints were self-reported, investigator-assessed or adjudicated outcome measures.

### Measures of treatment effect

When at least two included trials were available for a comparison and a given outcome, we tried to express dichotomous data as a risk ratio (RR) with a 95% confidence interval (CI). For continuous outcomes (blood pressure and body weight), we estimated the intervention effect using the mean difference (MD) with a 95% CI of the differences of change between baseline and end of study.

The position of the participant during a blood pressure measurement may affect the blood pressure-lowering effect. When measurements were reported for more than one position, our order of preference was: 1) sitting, 2) standing and 3) supine [72].

### Unit of analysis issues

For included studies with multiple arms, we addressed any potential unit-of-analysis issues in line with the guidance in Chapter 10.4 of the *Cochrane Handbook for Systematic Reviews of Interventions* (Version 6.4) [73]. The same approach would have been considered for studies that used cluster-randomised or cross-over designs; however, none of the included studies used such designs. This guidance covers the appropriate handling of clustering, within-person correlation, multiple time points and multiple intervention groups to avoid double-counting and ensure valid synthesis [73].

### Dealing with missing data

We obtained relevant missing data from authors and from the Institute for Quality and Efficiency in Health Care report [74], which is a systematic review that provides previously unpublished data for some of the included studies, including data derived from author requests. We evaluated important numerical data such as screened, eligible and randomised participants, as well as intention-to-treat (ITT) and per-protocol (PP) populations. We investigated attrition rates, for example, dropouts, losses to follow-up and withdrawals. We critically appraised issues of missing data, ITT and PP and, if available, compared them to the specification of primary outcome parameters and power calculations.

If necessary, we contacted authors of trials reporting incomplete information to provide the missing information.

### Reporting bias assessment

If we included 10 or more trials that investigated a particular outcome, we assessed publication bias and small-study effects using funnel plots. Several explanations may account for funnel plot asymmetry, including true heterogeneity of effect with respect to trial size, poor methodological design (and hence bias of small trials) and publication bias. We therefore interpreted the results cautiously [75].

### Synthesis methods

We summarised data statistically when it was available, sufficiently similar, and of sufficient quality. We performed analyses separately for each drug. In order to ensure a uniform data basis, we used the results from the most comparable follow-up periods for the meta-analyses. We performed data synthesis and analysis using Cochrane Review Manager software [76]. We performed statistical analysis according to the statistical guidelines referenced in the current version of the *Cochrane Handbook for Systematic Reviews of Interventions* [73]. We undertook both fixed-effect and random-effects meta-analysis, and we presented the results of the random-effects meta-analysis. If the standard deviation of the mean change was not explicitly given in the study, we calculated it from confidence intervals and the standard error of the mean, or estimated it from P values.

#### Investigation of heterogeneity and subgroup analysis

Heterogeneity amongst participants could be related to a range of factors. According to the original protocol, we planned to perform sensitivity or subgroup analyses for the following items: sex, age, body mass index, concomitant diseases, ethnicity, blood pressure at baseline, blood pressure goals, concomitant antihypertensive therapy and socioeconomic status. Due to limited data, subgroup analyses were not possible in the previous or current version of the review. However, we dealt with the issue of increased heterogeneity during the discussion section.

##### *Equity-related assessment*

We did not examine any equity-related characteristics in this review.

#### Sensitivity analysis

According to the original protocol, we planned to test the robustness of our results where appropriate, using several sensitivity analyses:

- studies with high overall risk of bias
- PP versus ITT analyses
- studies with large dropout rates and losses to follow-up.

Due to limited data, sensitivity analyses were not possible in the previous or the current version of the review.

#### Certainty of the evidence assessment

We used the GRADE approach to assess the certainty of the evidence (TS) [77]. We present the main results of the review, including a summary of the data, the magnitude of the effect and the overall certainty of the evidence, for each type of weight-reducing drug separately in Summary of findings 1, Summary of findings 2, Summary of findings 3, Summary of findings 4 and Summary of findings 5.

We include all critical outcomes in the summary of findings tables of the main article, listed according to priority:

- all-cause mortality;
- cardiovascular morbidity;
- serious adverse events;
- all adverse events

Important outcomes (change in systolic and diastolic blood pressure, and change in body weight) were included in additional summary of findings tables in Supplementary material 10.

### Consumer involvement

Consumers were not involved in this review.

## RESULTS

### Description of studies

Please see the characteristics of included studies in Supplementary material 2, overview of included studies and synthesis in Table 1, overview of study populations in Table 2, and population baseline characteristics in Table 3 and Supplementary material 3. For the characteristics of studies awaiting classification, see Supplementary material 4.

#### Results of the search

In the previous version of the review, six RCTs (33 records) were included in the analyses. Our search of the electronic databases for this fourth update up to 22 April 2024 resulted in 3169 records. Two publications on one potentially relevant trial were identified by checking potentially relevant studies for publications on hypertensive subpopulations via PubMed. We removed 1726 duplicate records, leaving 1445 unique records. Of these, 70 records were marked as ineligible by automation tools (Known Assessments and RCT classifier components of Cochrane's S4Me workflow), and 1318 were judged as not relevant to the question under study based on their title and abstract by the review authors (USK, NP). We identified no further studies from the reference lists of the included trials nor from relevant systematic reviews and meta-analyses. We therefore identified 57 publications for further examination.

After screening the full texts of these publications, we excluded 53 of these publications by consensus, leaving two additional trials (four publications) as relevant for our review [78, 79, 80, 81]. Together with the 33 publications (six studies) from the previous version of the review [63], we include 37 publications describing eight completed studies in this fourth update (see Figure 1 for details of the PRISMA statement [66]).

**Figure 1. Flow diagram**

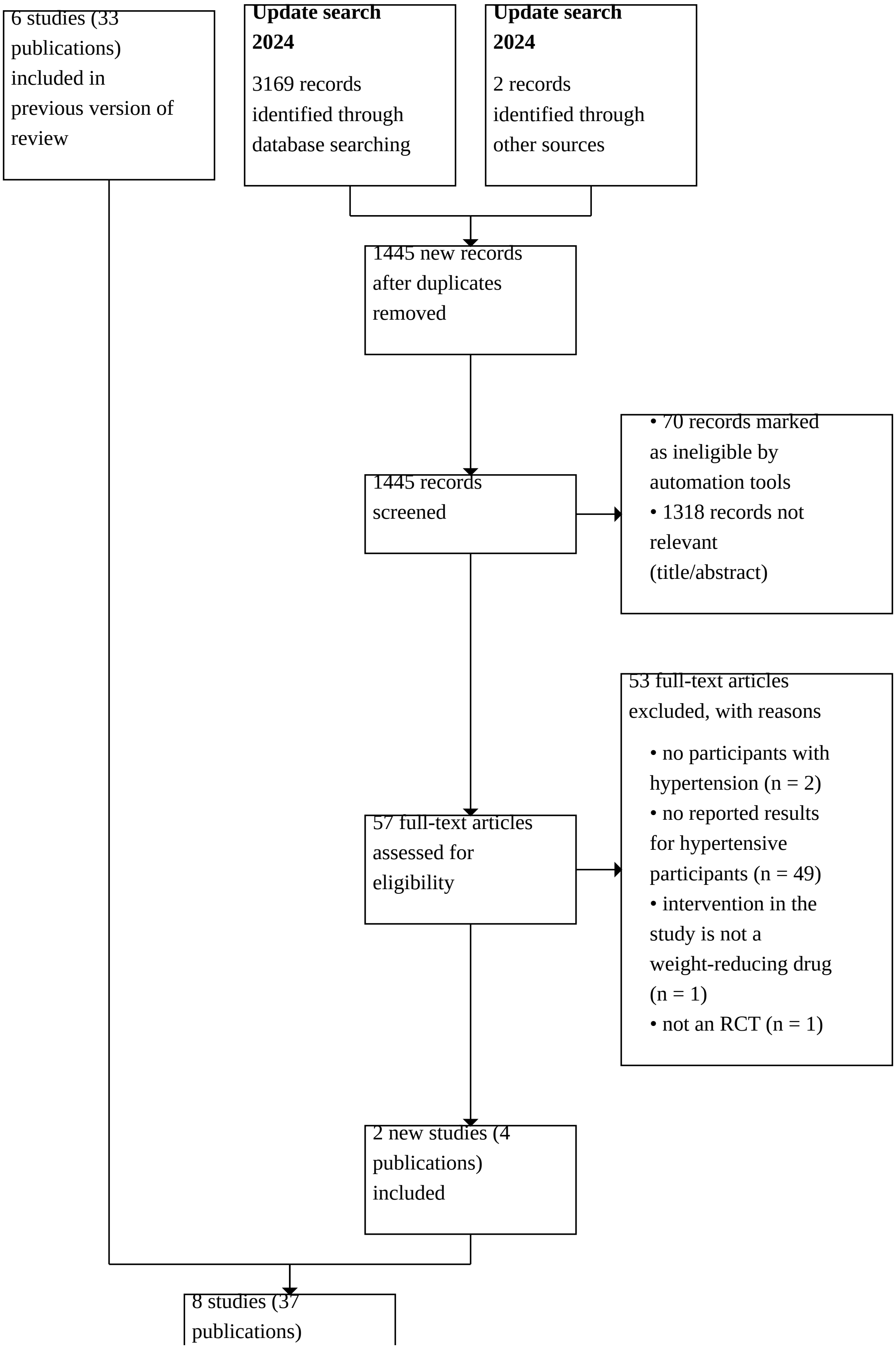
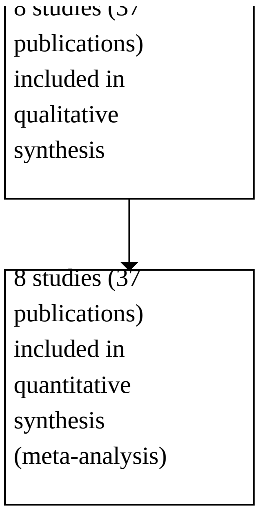


**Figure 1. (Continued)**




All relevant studies were published after the year 2000 and were written in English. For the initial review in 2009 [82], unpublished data for some of the included studies on orlistat could be obtained from a report published in 2006 by the German Institute for Quality and Efficiency in Health Care (IQWiG) [74], a systematic review only published in German language. For this report, the authors of the IQWiG report made requests to the authors of all studies investigating orlistat. In addition, the manufacturer sponsoring the XENDOS 2001-2006 [83, 84] was contacted. Data were only provided by the authors of Cocco 2005 [85] and the manufacturer that sponsored XENDOS 2001-2006 (see Supplementary material 8). For all other drugs approved after 2009, we conducted author and/or manufacturer enquiries during the review updates for all included studies, as well as for potentially relevant studies with mixed study populations (normotensive/hypertensive). Only the authors of the Nissen 2016 [86, 87, 88, 89, 90, 91] study provided data [88]. For details on these author and manufacturer requests, see Supplementary material 8.

### Included studies

We have provided details of the comparisons and studies included in this update below. Please see also the characteristics of included studies in Supplementary material 2, overview of included studies and synthesis in Table 1, overview of study populations in Table 2, and participant baseline characteristics in Table 3 and in Supplementary material 9.

In this update to the review, we identified two new trials that reported results for participants with hypertension. One RCT compared semaglutide with placebo (STEP 1 2023 [78, 79]), and the second RCT compared tirzepatide with placebo (SURMOUNT-1 2024 [80, 81]). From the previous versions of the review, we brought forward four RCTs comparing orlistat to placebo (Bakris 2002 [92]; Cocco 2005; Guy-Grand 2004 [93]; XENDOS 2001-2006), one RCT comparing naltrexone/bupropion to placebo (Nissen 2016) and one RCT comparing phentermine/topiramate to placebo (CONQUER 2013 [94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115]). No additional RCTs reporting results for participants with hypertension could be identified for these drugs in the update search. We are still unable to include any results for liraglutide in our review, since no separate results for participants with hypertension have been made available from RCTs investigating a mixed population of hypertensive and normotensive participants.

#### *Orlistat versus placebo*

##### Study design and duration

The four included studies that evaluated this comparison had a parallel, double-blind design (Bakris 2002; Cocco 2005; Guy-Grand 2004; XENDOS 2001-2006). Only Cocco 2005 did not mention any industry sponsorship, and it was also the only study that was a single-centre trial.

The mean treatment duration was six to 48 months.

##### Participants

The four studies involved a total of 3132 hypertensive participants with a mean age of 46 to 55 years, a mean baseline systolic blood pressure (BP) of 142 to 154 mm Hg and a mean baseline diastolic BP of 85 to 98 mm Hg. All participants in Bakris 2002 and Cocco 2005 were overweight and hypertensive. In Cocco 2005, all participants also had diabetes mellitus type 2. One trial included overweight people with diabetes mellitus type 2, hypercholesterolaemia or hypertension, and reported results for the predefined subpopulation of participants with hypertension (Guy-Grand 2004). XENDOS 2001-2006 included both normotensive and hypertensive obese participants. Upon request, data for two predefined subpopulations of hypertensive participants were provided by the manufacturer of orlistat, Hoffmann-La Roche (first subpopulation: diastolic BP at baseline ≥ 90 mm Hg; second subpopulation: systolic BP at baseline ≥ 140 mm Hg). Participants in one subpopulation may also be included in the other subpopulation. Therefore, the data could not be summarised. We present the data for the two subpopulations separately in this review. For meta-analyses, we used the results of the subpopulation with participants having a diastolic BP of ≥ 90 mm Hg at baseline.

**Interventions**

Participants received either 120 mg orlistat three times daily or placebo in all studies. In Bakris 2002, Cocco 2005 and XENDOS 2001-2006, participants in all study groups were also encouraged to modify their lifestyle and increase physical activity.

**Outcomes**

**Critical outcomes**

No study included mortality or cardiovascular morbidity as predefined outcomes, but results were reported in the context of adverse events: mortality in three RCTs (Bakris 2002; Cocco 2005; XENDOS 2001-2006) and cardiovascular morbidity in two RCTs (Bakris 2002; XENDOS 2001-2006). All four studies reported adverse events.

**Important outcomes**

All studies described the mean change in systolic and diastolic BP and the mean change in body weight between baseline and end of study.

#### *Phentermine/topiramate versus placebo*

**Study design and duration**

The one included study that evaluated this comparison was multicentre, with 93 study sites in the USA (CONQUER 2013). The study had a parallel, double-blind design. Industry sponsorship was mentioned.

The treatment duration was 56 weeks.

**Participants**

The study involved 2487 obese or overweight people with two or more comorbidities (hypertension, dyslipidaemia, diabetes or prediabetes, or abdominal obesity). The hypertensive subpopulation included 1305 participants with a mean age of 53 years, a mean baseline systolic BP of 134 mm Hg and a mean baseline diastolic BP of 84 mm Hg. Of these 1305 participants, about 90% were taking antihypertensive medication. In the hypertensive subpopulation, 216 participants had uncontrolled hypertension at baseline.

**Interventions**

The study compared two different dose regimens of phentermine/topiramate once daily versus placebo. All participants in the active groups received an initial dose of 7.5 mg phentermine and 23 mg topiramate. During an initial four-week titration period, doses were increased weekly (3.75 mg phentermine and 23 mg topiramate) until the assigned dosages of 7.5 mg phentermine/46.0 mg topiramate (low dose (LD) group) or 15 mg phentermine/92.0 mg topiramate (high dose (HD) group) were achieved. The assigned dosages were maintained for 52 weeks. In addition, all participants received standardised counselling for diet (to reduce caloric intake by 500 kcal/day) and lifestyle modification.

**Outcomes**

**Critical outcomes**

The study did not include mortality or cardiovascular morbidity as predefined outcomes, but results were reported. Adverse events were also reported.

**Important outcomes**

The study described the mean change in systolic and diastolic BP and the mean percentage change in body weight between baseline and end of study.

#### *Naltrexone/bupropion versus placebo*

**Study design and duration**

The one included study that evaluated this comparison was a parallel, double-blind placebo-controlled multicentre trial with 266 study sites in the USA (Nissen 2016). The study was industry funded.

In order to attain the calculated number of 378 primary events, the necessary duration of the randomised treatment was estimated by the study authors to be between two and four years. After the unplanned public release of confidential interim data by the sponsor, the study was terminated prematurely (in May 2015). All results were based on the 50% interim analysis of the study.

The median duration of follow-up was 121 weeks (interquartile range, 114 to 128 weeks).

**Participants**

The study involved 8910 obese or overweight people aged at least 45 years (men) or 50 years (women), with an increased risk of adverse cardiovascular events.

Results for the subpopulation of participants with hypertension were provided by the authors upon request. The hypertensive subpopulation included 8287 participants (93% of the total study population), with a mean age of 62 years, a mean baseline systolic BP of 126 mm Hg and a mean baseline diastolic BP of 75 mm Hg. About 97% of the included participants were taking antihypertensive medication. In the hypertensive subpopulation, about 87% of the participants had diabetes mellitus type 2 and about 30% had cardiovascular disease.

**Interventions**

The participants received either a fixed-dose combination of naltrexone (8 mg) and bupropion (90 mg) or placebo once or twice a day. During the initial four-week titration period, the number of tablets was increased weekly (one tablet a day during the first week, two tablets a day during week two, three tablets a day during week three, and four tablets a day during week four and thereafter). All participants also received an internet-based weight management programme focused on healthy eating and physical activity.

**Outcomes**

**Critical outcomes**

All-cause mortality was reported as an additional outcome. Cardiovascular death, fatal or nonfatal stroke, and fatal or nonfatal myocardial infarction were reported as primary and secondary outcomes. Adverse events were also reported.

**Important outcomes**

The study described the mean change in systolic and diastolic BP and the mean change in body weight between baseline and end of study.

***Semaglutide versus placebo***

**Study design and duration**

The one included study that evaluated this comparison was a double-blind, placebo-controlled, multinational trial at 129 sites in 16 countries (STEP 1 2023). The study was industry-sponsored.

The treatment duration was 68 weeks.

**Participants**

The study involved 1961 normotensive or hypertensive adults with a body mass index (BMI) of ≥ 30 $kg/m^2$ and who did not have diabetes mellitus. In a post hoc analysis by the study authors [79], data were presented for the subpopulation of participants with uncontrolled hypertension at baseline, defined as an average systolic BP ≥ 140 mmHg or an average diastolic BP ≥ 90 mmHg at baseline. This subpopulation included 274 participants (14% of the total study population). No further information on mean age and baseline systolic and diastolic BP for participants of the subpopulation was provided by the study authors.

**Interventions**

Participants received semaglutide 2.4 mg or placebo once weekly subcutaneously, with a lifestyle intervention for 68 weeks.

**Outcomes**

**Critical outcomes**

For the subpopulation of participants with hypertension, the study did not report results on mortality, cardiovascular morbidity or adverse events.

**Important outcomes**

For the subpopulation of participants with hypertension, results on the mean change in systolic and diastolic BP between baseline and end of study were available. As explained above, we could only use the results from the limited number of participants with uncontrolled hypertension at baseline (which was 14% of the total sample size). The study did not report results on the mean percentage change in body weight for this subpopulation of patients with hypertension.

***Tirzepatide versus placebo***

**Study design and duration**

The one included study that evaluated this comparison was a double-blind, placebo-controlled, multinational trial conducted at 118 sites in 10 countries (SURMOUNT-1 2024). The study was industry-sponsored.

The treatment duration was 72 weeks.

**Participants**

The study included 2539 normotensive or hypertensive adults with a BMI ≥ 30 or a BMI ≥ 27 $kg/m^2$ and at least one weight-related complication, excluding diabetes. For the whole study population, the mean age at baseline was 44.9 years and baseline systolic BP was 123.3 mm Hg and diastolic BP was 79.5 mm Hg. In a post hoc analysis by the study authors [81], data were presented for three subpopulations of hypertensive participants. The first subpopulation included 195 participants with a systolic BP ≥ 140 mm Hg at baseline (7.7% of the total study population), the second subpopulation included 207 participants with a diastolic BP ≥ 90 mm Hg at baseline (8.2% of the total study population), and the third subpopulation included 635 participants taking antihypertensive medication at baseline (25% of the total study population). Participants in one subpopulation may also be included in the other subpopulation. We present the data for the three subpopulations separately in this review. For these subpopulations, no further information on mean age and baseline systolic and diastolic BP was provided by the authors.

**Interventions**

Participants were assigned to tirzepatide (5 mg, 10 mg or 15 mg) or placebo once weekly subcutaneously for 72 weeks, as an adjunct to lifestyle intervention.

**Outcomes**

**Critical outcomes**

For the subpopulations of participants with hypertension, the study did not report results on mortality, cardiovascular morbidity or adverse events.

**Important outcomes**

The study described the least-square means change in systolic and diastolic BP at week 72 from baseline for the subpopulation of participants taking antihypertensive medication at baseline, and the change in systolic BP for the subpopulation of participants with systolic BP ≥ 140 mm Hg at baseline, and the change in diastolic BP for the subpopulation of participants with diastolic BP ≥ 90 at baseline. Change in body weight was not reported for these three subpopulations of participants with hypertension.

### Excluded studies

We excluded 53 records at the full-text evaluation stage of our study selection process for this review update. The main reason for exclusion was a lack of data for the hypertensive subpopulation in studies including normotensive as well as hypertensive participants. Other reasons that we excluded a record was that it did not describe a randomised controlled trial, did not include participants with essential hypertension, had a duration of intervention of less than 24 weeks or included different accompanying antihypertensive therapies in the study groups. We provide our reasons for excluding each trial in the Supplementary material 3 table.

In the previous version of the review, one completed study that we identified in the trial registries was put in the 'awaiting classification' section (see Supplementary material 4). This was due to a lack of information and the absence of published results. There are still no data available for this study (CTRI/2017/05/008655 [116]).

## Risk of bias in included studies

Our judgements of the risk of bias for all included studies are shown in summary figures (Figure 2; Figure 3). We provide a brief overview of these below. For further details, see the risk of bias tables in Supplementary material 2.

**Figure 2. Risk of bias graph: review authors' judgements about each risk of bias item presented as percentages across all included studies**

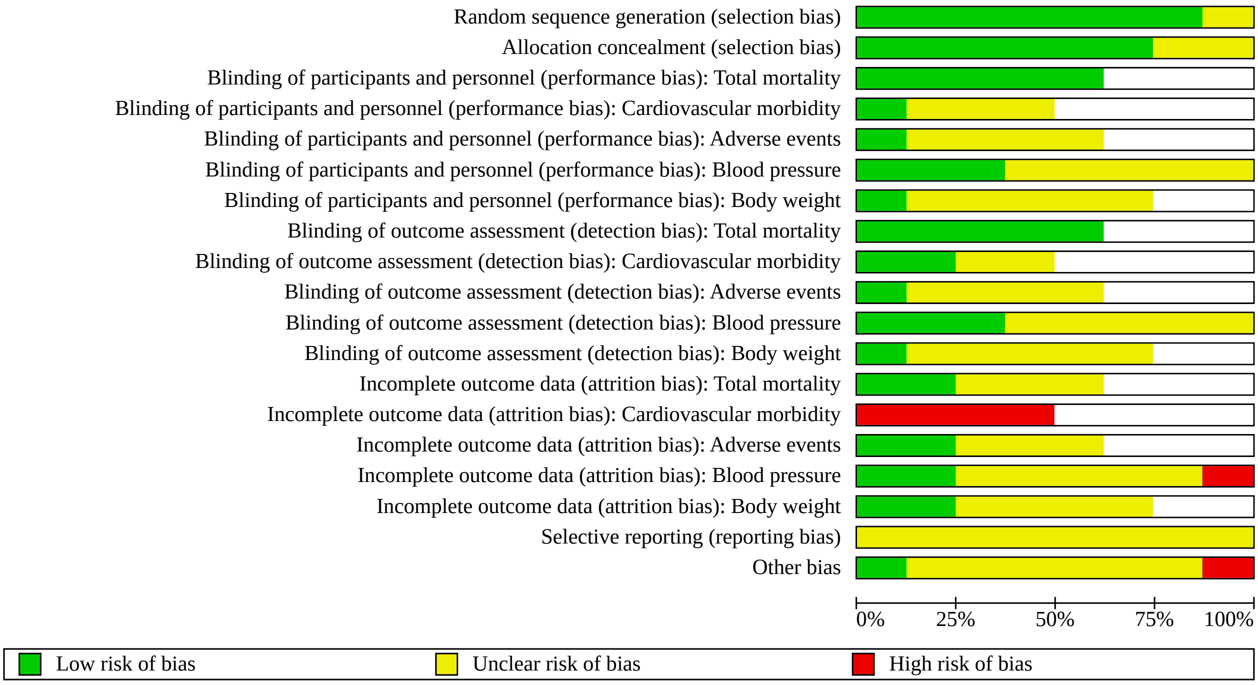
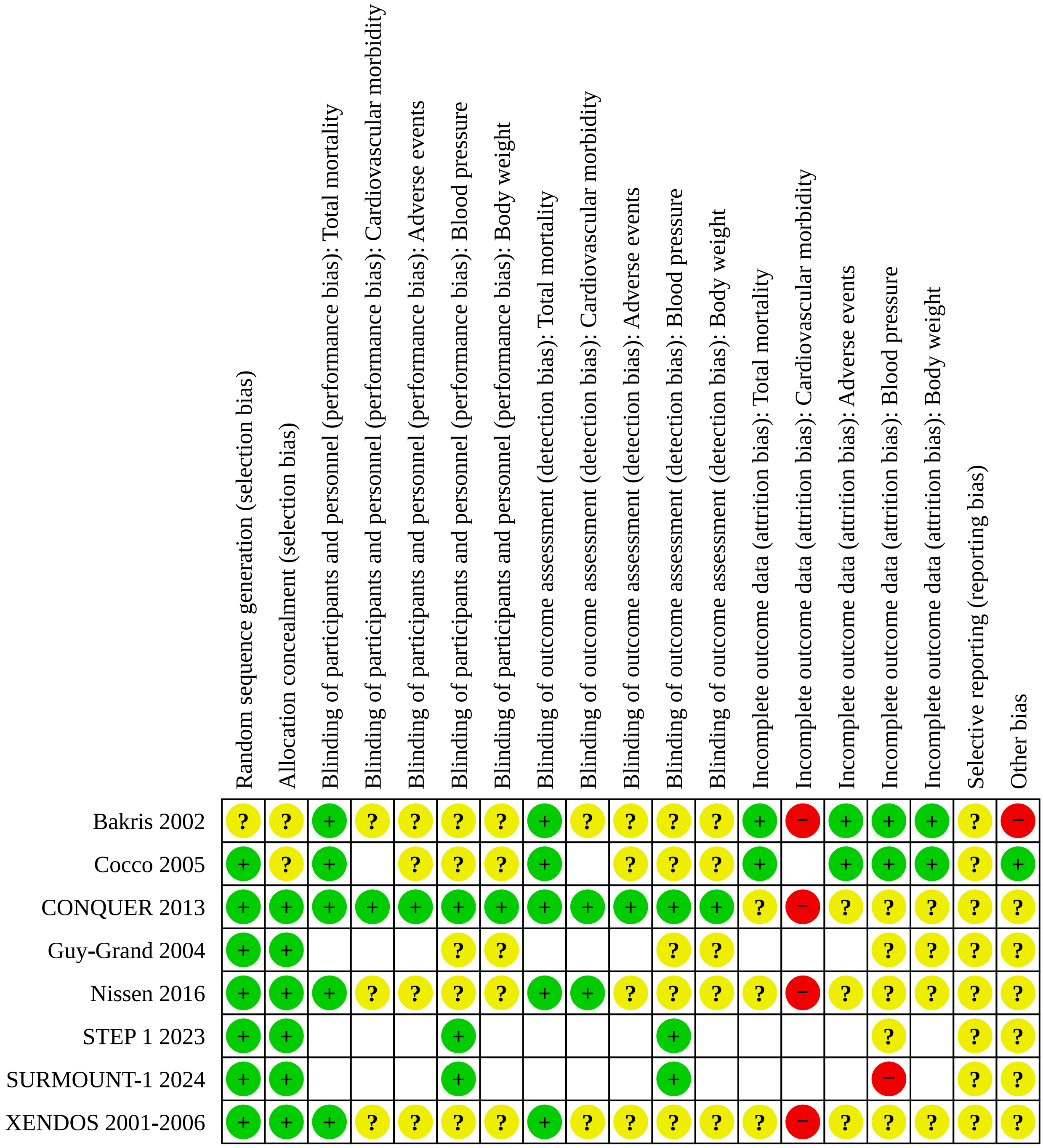


**Figure 3. Risk of bias summary: review authors' judgements about each risk of bias item for each included study**




### Random sequence generation and allocation concealment

#### *Orlistat versus placebo*

All included studies were randomised controlled trials that randomised individuals. For the method of randomisation, we judged three trials (Cocco 2005; Guy-Grand 2004; XENDOS 2001-2006) to have a low risk of bias based on the information from journal publications or the IQWIG report [74]. In one study, the method of randomisation was not reported and was therefore judged 'unclear' (Bakris 2002).

The method of allocation concealment was adequate in two studies (Guy-Grand 2004; XENDOS 2001-2006), while it was unclear in the other two studies (Bakris 2002; Cocco 2005).

***Phentermine/topiramate versus placebo***

The included study evaluating this comparison, CONQUER 2013, adequately described the method of randomisation and allocation concealment.

***Naltrexone/bupropion versus placebo***

The included study evaluating this comparison, Nissen 2016, adequately described the method of randomisation and allocation concealment.

***Semaglutide versus placebo***

The included study evaluating this comparison, STEP 1 2023, adequately described the method of randomisation and allocation concealment.

***Tirzepatide versus placebo***

The included study evaluating this comparison, SURMOUNT-1 2024, adequately described the method of randomisation and allocation concealment.

**Blinding**

***Orlistat versus placebo***

Although all included trials were described as double-blind, three trials provided too little information, and the blinding of participants and key study personnel was uncertain (Bakris 2002; Guy-Grand 2004; XENDOS 2001-2006). Based on the authors' information, we can only assume that blinding took place throughout the duration of one study (Cocco 2005).

***Phentermine/topiramate versus placebo***

CONQUER 2013 was described as double-blind. The investigators, participants, and study sponsors were masked to treatment assignment, and all study drugs were administered as capsules that were identical in size and appearance.

***Naltrexone/bupropion versus placebo***

Nissen 2016 was described as double-blind. The investigators and participants were masked for a treatment assignment. A blinded independent clinical events committee adjudicated clinical outcomes, including cardiovascular death, nonfatal myocardial infarction, nonfatal stroke, hospitalisation for unstable angina and all-cause mortality.

***Semaglutide versus placebo***

STEP 1 2023 was described as a randomised, double-blind, placebo-controlled trial. The investigators and participants were masked to treatment assignment. A committee of independent external event adjudicators reviewed selected adverse events and deaths. All standard assays were performed in a central laboratory. The investigators were responsible for data collection, while the study sponsor undertook site monitoring, data collation and data analysis.

***Tirzepatide versus placebo***

SURMOUNT-1 2024 was described as double-blind. Investigators, site staff, clinical monitors and participants stayed blinded to the treatment assignments until the study was complete.

**Incomplete outcome data**

***Orlistat versus placebo***

In two studies, the outcome data description was complete for all-cause mortality, adverse events, change in blood pressure and change in body weight: Cocco 2005 had no losses to follow-up, and Bakris 2002 described all reasons for withdrawals and losses to follow-up. Cardiovascular morbidity was reported only in the context of adverse events in Bakris 2002 and XENDOS 2001-2006, so the completeness of the information is unknown. For this endpoint, we judged the risk of bias to be high. In Guy-Grand 2004, withdrawals were only reported for the whole study population and not for the hypertensive subpopulation, and in XENDOS 2001-2006, the reasons for withdrawals were incompletely reported; therefore, we rated the risk of bias in these two studies as unclear for all reported outcomes.

***Phentermine/topiramate versus placebo***

The total number of withdrawals was only reported for the whole study population and not for the hypertensive subpopulation of the study (CONQUER 2013). For the hypertensive subpopulation, only the number of withdrawals due to adverse events was reported, so the reasons for withdrawals were incompletely reported. We therefore judged the risk of bias to be unclear for all reported outcomes other than cardiovascular morbidity. Cardiovascular morbidity was reported only in the context of treatment-emergent adverse events, so the completeness of the information is unknown. For this endpoint, we judged the risk of bias to be high.

***Naltrexone/bupropion versus placebo***

In Nissen 2016, all primary analyses were performed on the ITT population, defined as participants who underwent randomisation into the treatment period and were dispensed study medication. No imputation was performed for missing data. The number of withdrawals and reasons were only reported for the whole study population, but not for the subset of participants with hypertension. We therefore judged the risk of bias to be unclear for all reported outcomes.

***Semaglutide versus placebo***

In STEP 1 2023, all primary analyses were performed on the ITT population. This includes all randomised participants regardless of adherence to treatment or rescue interventions (other anti-obesity medication or bariatric surgery). The total number of withdrawals was only reported for the whole study population and not for the hypertensive subpopulation of the study. We therefore judged the risk of bias to be unclear for all reported outcomes.

***Tirzepatide versus placebo***

In SURMOUNT-1 2024, randomised participants who took at least one dose of the study drug were included in the modified ITT population (mITT). All analyses were conducted based on the efficacy analysis set, which included data obtained during the treatment period from mITT, excluding data after discontinuation of the study drug. Only 84.8% (N = 1607) of participants in the intervention group and 73.6% (N = 473) of participants of the control group completed the study period on the study drug and were therefore included in the analysis. The number of withdrawals and reasons were only reported for the whole study population, but not for the subset of participants with hypertension. We therefore judged the risk of bias to be high for all reported outcomes.

#### Selective reporting

##### *Orlistat versus placebo*

We classified the risk of bias for selective reporting as unclear, as either no study protocol was provided (Bakris 2002; Cocco 2005; Guy-Grand 2004), more outcomes were reported than were prespecified (Bakris 2002) or no full publication was obtainable (XENDOS 2001-2006).

##### *Phentermine/topiramate versus placebo*

No study protocol was available for CONQUER 2013. We therefore classified the risk of bias for selective reporting as unclear.

##### *Naltrexone/bupropion versus placebo*

A study protocol for Nissen 2016 was provided, and all prespecified outcomes were reported. The study was terminated prematurely after the unplanned release of the 25% interim data, which we judged as compromising the scientific integrity of the ongoing study. Outcome measure data based on the 50% interim analysis were designated as the primary analysis. Safety data were based on all available data at the time of database lock, which occurred after the 50% interim analysis. We therefore classified the risk of bias for selective reporting as unclear.

##### *Semaglutide versus placebo*

Post-hoc analyses in STEP 1 2023 include the change of systolic and diastolic BP in participants with uncontrolled hypertension at baseline, regardless of antihypertensive medication. No outcome data for the change of body weight in the hypertensive subpopulation were reported. No safety data were reported for this subpopulation of participants with uncontrolled hypertension. We therefore classified the risk of bias for selective reporting as unclear.

##### *Tirzepatide versus placebo*

Post-hoc analyses of SURMOUNT-1 2024 include the change of systolic and diastolic BP in participants with stage 2 hypertension at baseline, or participants on antihypertensive medication at baseline, respectively. No outcome data for the change of body weight in the hypertensive subpopulations were reported. No safety data were reported for these subpopulations of participants with hypertension. We therefore classified the risk of bias for selective reporting as unclear.

#### Other bias

##### *Orlistat versus placebo*

Three studies were commercially sponsored, and no funding information was available for one study. No study included in the review reported any significant differences between groups in the main characteristics of participants at baseline. However, in Bakris 2002 and XENDOS 2001-2006, we judged there to be a high risk of bias due to the combination of a high withdrawal rate and the unknown length of participants' involvement in the trial, even when the last observation carried forward (LOCF) analysis was used. In addition, in Bakris 2002, there were inconsistencies between the text and flowchart for the numbers of participants who finished the study. In the remaining two trials, we judged the risk of bias as unclear, as the possibility of sponsor influence cannot be ruled out.

##### *Phentermine/topiramate versus placebo*

We judged the risk of other bias to be unclear in CONQUER 2013, as the possibility of sponsor influence cannot be ruled out.

##### *Naltrexone/bupropion versus placebo*

We judged the risk of other bias to be unclear in Nissen 2016, as the possibility of sponsor influence cannot be ruled out.

##### *Semaglutide versus placebo*

We judged the risk of bias to be unclear in STEP 1 2023, as the possibility of sponsor influence cannot be ruled out.

##### *Tirzepatide versus placebo*

We judged the risk of bias to be unclear in SURMOUNT-1 2024, as the possibility of sponsor influence cannot be ruled out.

### Synthesis of results

#### Orlistat versus placebo

For critical outcomes, see Summary of findings 1; for important outcomes, see additional summary of findings table in Supplementary material 10.

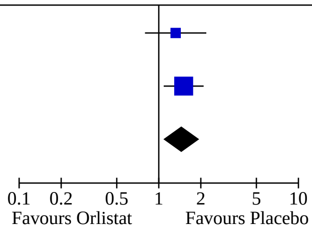
In all the meta-analyses presented, we used the results of the subpopulation of participants with diastolic BP ≥ 90 mm Hg at baseline. In addition, we calculated meta-analyses, taking into account the subpopulation of participants in XENDOS 2001-2006 who had systolic BP ≥ 140 mm Hg at baseline. The results of these meta-analyses with the random-effects model correspond to the results presented in Figure 4 to Figure 5.

**Figure 4. Forest plot of comparison: 1 Orlistat versus placebo, outcome: 1.1 Serious adverse events.**

| Study or Subgroup | Orlistat Events | Orlistat Total | Placebo Events | Placebo Total | Weight | Risk Ratio IV, Random, 95% CI | Risk of Bias A | B | C | D |
|---|---|---|---|---|---|---|---|---|---|---|
| Bakris 2002 | 31 | 268 | 24 | 274 | 29.8% | 1.32 [0.80 , 2.19] | ? | ? | ? | − |
| Cocco 2005 | 0 | 45 | 0 | 45 | | Not estimable | + | ? | ? | + |
| XENDOS 2001-2006[a] | 73 | 407 | 52 | 437 | 70.2% | 1.51 [1.08 , 2.09] | + | + | ? | ? |
| **Total (Wald[b])** | | **720** | | **756** | **100.0%** | **1.45 [1.10 , 1.91]** | | | | |
| Total events: | 104 | | 76 | | | | | | | |

Test for overall effect: Z = 2.64 (P = 0.008)

Heterogeneity: Tau² (DL[c]) = 0.00; Chi² = 0.18, df = 1 (P = 0.67); I² = 0%




**Footnotes**
[a]Participants with diastolic BP ≥ 90 mm Hg at baseline.
[b]CI calculated by Wald-type method.
[c]Tau² calculated by DerSimonian and Laird method.

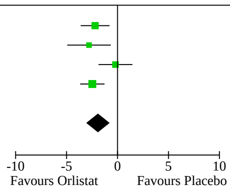
**Risk of bias legend**
(A) Random sequence generation (selection bias)
(B) Allocation concealment (selection bias)
(C) Selective reporting (reporting bias)
(D) Other bias

**Figure 5. Forest plot of comparison: 1 Orlistat versus placebo, outcome: 1.5 Change in diastolic blood pressure from baseline to endpoint (6 to 12 months follow-up) [mm Hg].**

| Study or Subgroup | Orlistat Mean [mm Hg] | Orlistat SD [mm Hg] | Orlistat Total | Placebo Mean [mm Hg] | Placebo SD [mm Hg] | Placebo Total | Weight | Mean Difference IV, Random, 95% CI [mm Hg] | Risk of Bias A | B | C | D |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Bakris 2002 | -11.4 | 8.3 | 267 | -9.2 | 8.4 | 265 | 27.6% | -2.20 [-3.62 , -0.78] | ? | ? | ? | − |
| Cocco 2005 | -3.6 | 5.18 | 45 | -0.8 | 5.18 | 45 | 17.0% | -2.80 [-4.94 , -0.66] | + | ? | ? | + |
| Guy-Grand 2004 | -7.5 | 10.46 | 304 | -7.3 | 10.56 | 310 | 23.3% | -0.20 [-1.86 , 1.46] | + | + | ? | ? |
| XENDOS 2001-2006[a] | -9.12 | 7.85 | 399 | -6.65 | 9.59 | 423 | 32.1% | -2.47 [-3.67 , -1.27] | + | + | ? | ? |
| **Total (Wald[b])** | | | **1015** | | | **1043** | **100.0%** | **-1.92 [-2.99 , -0.85]** | | | | |

Test for overall effect: Z = 3.52 (P = 0.0004)

Heterogeneity: Tau² (DL[c]) = 0.56; Chi² = 5.70, df = 3 (P = 0.13); I² = 47%




**Footnotes**
[a]Participants with diastolic BP ≥ 90 mm Hg at baseline.
[b]CI calculated by Wald-type method.
[c]Tau² calculated by DerSimonian and Laird method.

**Risk of bias legend**
(A) Random sequence generation (selection bias)
(B) Allocation concealment (selection bias)
(C) Selective reporting (reporting bias)
(D) Other bias

Although we present random-effects meta-analyses in the review, we also calculated meta-analyses using the fixed-effect model. The outcomes of these analyses are similar to those obtained from the random-effects model.

### *Critical outcomes*

For details on critical outcome data, see Table 4 and Supplementary material 9.

#### All-cause mortality

Three studies including 1488 participants reported on mortality. No deaths were reported in either Bakris 2002 or Cocco 2005. In XENDOS 2001-2006, there were two deaths in the orlistat-treated group in the first subpopulation analysis (diastolic BP ≥ 90 mm Hg) and one death in the orlistat group in the second subpopulation analysis (systolic BP ≥ 140 mm Hg); no deaths occurred in the placebo group. In summary, orlistat may have little to no effect on all-cause mortality, but the evidence is very uncertain.

#### Cardiovascular morbidity

Two studies (1721 participants) presented data on cardiovascular morbidity in the context of adverse events. In Bakris 2002, five participants (1.9%) in the orlistat group reported cardiovascular adverse events. Two suffered a myocardial infarction, two had chest pain, and one had an episode of atrial fibrillation. In the placebo group, four participants (0.7%) had a cardiovascular adverse event. One suffered a myocardial infarction, one had worsening atherosclerotic coronary artery disease, and two had an episode of chest pain. In XENDOS 2001-2006, the percentage of participants with vascular complications was reported. In both subpopulations (diastolic BP ≥ 90 mm Hg and systolic BP ≥ 140 mm Hg), 17% of the participants treated with orlistat and 19% treated with placebo reported a vascular adverse event. In summary, orlistat may have little to no effect on cardiovascular morbidity, but the evidence is very uncertain.

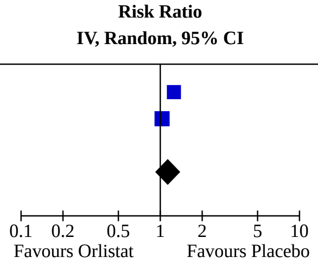
**Adverse events**

Data on serious adverse events (SAEs) were available from three trials (Bakris 2002; Cocco 2005; XENDOS 2001-2006) with orlistat probably increasing the risk of SAEs compared to placebo (RR 1.45, 95% CI 1.10 to 1.91; P = 0.008, $I^2$ = 0%; 3 trials, 1476 participants; moderate-certainty evidence; Analysis 1.1; Figure 4). All adverse events (AEs) were reported in two RCTs (Bakris 2002; XENDOS 2001-2006). There may be little or no difference between the orlistat and the placebo group for all AEs, but the evidence is very uncertain (RR 1.13, 95% CI 0.93 to 1.37; P = 0.21, $I^2$ = 95%; 2 trials, 1386 participants; very low-certainty evidence; Analysis 1.2; Figure 6).

**Figure 6. Forest plot of comparison: 1 Orlistat versus placebo, outcome: 1.2 All adverse events.**

| | Orlistat | | Placebo | | | Risk Ratio | Risk of Bias | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Study or Subgroup | Events | Total | Events | Total | Weight | IV, Random, 95% CI | A | B | C | D |
| Bakris 2002 | 239 | 268 | 195 | 274 | 47.6% | 1.25 [1.15 , 1.37] | ? | ? | ? | − |
| XENDOS 2001-2006[a] | 403 | 407 | 420 | 437 | 52.4% | 1.03 [1.01 , 1.05] | + | + | ? | ? |
| **Total (Wald[b])** | | **675** | | **711** | **100.0%** | **1.13 [0.93 , 1.37]** | | | | |
| Total events: | 642 | | 615 | | | | | | | |

Test for overall effect: Z = 1.26 (P = 0.21)




Heterogeneity: Tau² (DL[c]) = 0.02; Chi² = 18.72, df = 1 (P < 0.0001); I² = 95%

**Footnotes**
[a]Participants with diastolic BP ≥ 90 mm Hg at baseline.
[b]CI calculated by Wald-type method.
[c]Tau² calculated by DerSimonian and Laird method.

**Risk of bias legend**
(A) Random sequence generation (selection bias)
(B) Allocation concealment (selection bias)
(C) Selective reporting (reporting bias)
(D) Other bias

In Bakris 2002, at least one AE was reported by more participants in the orlistat-treated group (89%) than in the placebo-treated group (71%). Of those, 7% of participants in the orlistat group versus 7% of participants in the placebo group withdrew. There was evidence that gastrointestinal AEs were higher in the orlistat-treated group than in the placebo group (73% versus 44%); 8% of those participants in the orlistat group and 5% in the placebo group stopped taking the medication for this reason. Musculoskeletal AEs were also reported more often in the orlistat than the placebo group (23% versus 16%).

In Cocco 2005, as reported by the study authors, AEs were generally mild, but event rates on all AEs were not provided. Gastrointestinal AEs were the most common adverse events and were described for 24% of the placebo group within the first three weeks and for 36% of the orlistat group within the first four weeks.

In XENDOS 2001-2006's first subpopulation (diastolic BP ≥ 90 mm Hg), AEs were reported in 99% of participants in the orlistat and 96% of participants in the placebo group. Gastrointestinal side effects were more common in the orlistat (93%) versus the placebo group (70%). Musculoskeletal, nervous, dermatological and vascular events were comparable in both treatment groups. Nine per cent withdrew due to AEs in the orlistat versus 4% in the placebo group. It is not clear whether the reported AEs were study-drug-related.

In XENDOS 2001-2006's second subpopulation (systolic BP ≥ 140 mm Hg), AEs were reported in 99% of participants in the orlistat and 97% of participants in the placebo group. Gastrointestinal side effects were more common in the orlistat (93%) versus the placebo group (71%). Musculoskeletal, nervous, dermatological and vascular events were comparable in both treatment groups. Nine per cent withdrew due to AEs in the orlistat versus 4% in the placebo group. It is not clear whether the reported AEs were study-drug-related.

In Guy-Grand 2004, data were presented only for the whole study group, with no information provided on the hypertensive subpopulation.

For details on AEs, see Table 4 and Supplementary material 9.

***Important outcomes***

For details on important outcome data, see tables in Supplementary material 9.

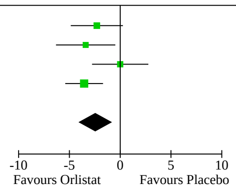
In Bakris 2002, the study duration was 52 weeks, while in Cocco 2005 and Guy-Grand 2004, it was six months. Results from XENDOS 2001-2006 were available after 12 and 48 months. In order to ensure a uniform data basis, we used the results from XENDOS 2001-2006 after a study duration of 12 months for meta-analyses of important outcomes.

#### Change in systolic blood pressure

We could include all four studies in the meta-analysis investigating the effects of orlistat on systolic BP. After a follow-up of six to 12 months, orlistat probably reduces systolic BP compared to placebo (mean difference (MD) −2.46, 95% CI −4.01 to −0.90; P = 0.002, $I^2$ = 36%; 4 trials, 2058 participants; moderate-certainty evidence; Analysis 1.4, Figure 7). As the risk of bias is high in all included studies, the increased statistical heterogeneity cannot be explained by differences in study quality. We could not deduce any plausible explanation for heterogeneity from differences in study design, study duration, sample sizes, interventions or characteristics of included participants. In XENDOS 2001-2006, additional results after 48 months of follow-up were reported for the subpopulations of participants with diastolic BP ≥ 90 mm Hg at baseline and systolic BP ≥ 140 mm Hg, with a mean reduction in systolic BP with orlistat compared to placebo of −2.4 mm Hg (95% CI −4.4 to −0.4) and −2.9 mm Hg (95% CI −4.7 to −1.1), respectively.

### Figure 7. Forest plot of comparison: 1 Orlistat versus placebo, outcome: 1.4 Change in systolic blood pressure from baseline to endpoint (6 to 12 months follow-up) [mm Hg].

| | Orlistat | | | Placebo | | | | Mean Difference | Risk of Bias | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Study or Subgroup | Mean [mm Hg] | SD [mm Hg] | Total | Mean [mm Hg] | SD [mm Hg] | Total | Weight | IV, Random, 95% CI [mm Hg] | A | B | C | D | E |
| Bakris 2002 | -13.3 | 15.2 | 267 | -11 | 15 | 265 | 23.9% | -2.30 [-4.87 , 0.27] | ? | ? | − | ? | − |
| Cocco 2005 | -4.3 | 7.07 | 45 | -0.9 | 7.07 | 45 | 20.0% | -3.40 [-6.32 , -0.48] | + | ? | | ? | + |
| Guy-Grand 2004 | -9.8 | 17.43 | 304 | -9.8 | 17.61 | 310 | 21.5% | 0.00 [-2.77 , 2.77] | + | + | | ? | ? |
| XENDOS 2001-2006[a] | -11.21 | 13.45 | 399 | -7.66 | 13.75 | 423 | 34.6% | -3.55 [-5.41 , -1.69] | + | + | − | ? | ? |
| **Total (Wald[b])** | | | **1015** | | | **1043** | **100.0%** | **-2.46 [-4.01 , -0.90]** | | | | | |




Test for overall effect: Z = 3.10 (P = 0.002)

Heterogeneity: $Tau^2$ ($DL_c$) = 0.92; $Chi^2$ = 4.72, df = 3 (P = 0.19); $I^2$ = 36%

**Footnotes**
[a]Participants with diastolic BP ≥ 90 mm Hg at baseline.
[b]CI calculated by Wald-type method.
[c]$Tau^2$ calculated by DerSimonian and Laird method.

**Risk of bias legend**
(A) Random sequence generation (selection bias)
(B) Allocation concealment (selection bias)
(C) Incomplete outcome data (attrition bias): Cardiovascular morbidity
(D) Selective reporting (reporting bias)
(E) Other bias

#### Change in diastolic blood pressure

We could include all four studies in the meta-analysis investigating the effects of orlistat on diastolic BP. After six to 12 months, orlistat probably reduces diastolic BP compared to placebo (MD −1.92 mm Hg, 95% CI −2.99 to −0.85; P < 0.001, $I^2$ = 47%; 4 trials, 2058 participants; moderate-certainty evidence; Analysis 1.5; Figure 5). As the risk of bias is high in all included studies, the increased statistical heterogeneity cannot be explained by differences in study quality. We could not deduce any plausible explanation for heterogeneity from differences in study design, study duration, sample sizes, interventions or characteristics of included participants. In XENDOS 2001-2006, additional results after 48 months of follow-up were reported for the subpopulations of participants with diastolic BP ≥ 90 mm Hg at baseline and systolic BP ≥ 140 mm Hg, with a mean reduction in diastolic BP with orlistat compared to placebo of −1.9 mm Hg (95% CI −3.2 to −0.6) and −2.0 mm Hg (95% CI −3.3 to −0.7), respectively.

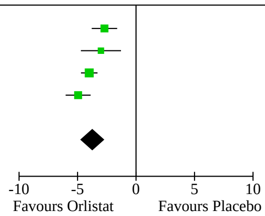
#### Change in body weight

We could include all four studies in the meta-analysis investigating the effects of orlistat on body weight. After six to 12 months, orlistat may reduce body weight compared to placebo (MD −3.74 kg, 95% CI −4.70 to −2.78; P < 0.001, $I^2$ = 68%; 4 trials, 2080 participants; low-certainty evidence; Analysis 1.3; Figure 8). As the risk of bias is high in all included studies, the increased statistical heterogeneity cannot be explained by differences in study quality. We could not deduce any plausible explanation for heterogeneity from differences in study design, study duration, sample sizes, interventions or characteristics of included participants. In XENDOS 2001-2006, additional results after 48 months of follow-up were reported for the subpopulations of participants with diastolic BP ≥ 90 mm Hg at baseline and systolic BP ≥ 140 mm Hg, with a mean reduction in body weight with orlistat compared to placebo of −2.8 kg (95% CI −3.9 to −1.7) and −3.6 kg (95% CI −4.6 to −2.6), respectively.

**Figure 8. Forest plot of comparison: 1 Orlistat versus placebo, outcome: 1.3 Change in body weight from baseline to endpoint (6 to 12 months follow-up) [kg].**

| | Orlistat | | | Placebo | | | | Mean Difference | Risk of Bias | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Study or Subgroup | Mean [kg] | SD [kg] | Total | Mean [kg] | SD [kg] | Total | Weight | IV, Random, 95% CI [kg] | A | B | C | D |
| Bakris 2002 | -5.4 | 6.4 | 267 | -2.7 | 6.4 | 265 | 25.5% | -2.70 [-3.79 , -1.61] | ? | ? | ? | − |
| Cocco 2005 | -5.6 | 4.15 | 45 | -2.6 | 4.15 | 45 | 17.1% | -3.00 [-4.71 , -1.29] | + | ? | ? | + |
| Guy-Grand 2004 | -5.8 | 5.23 | 304 | -1.8 | 3.52 | 310 | 31.5% | -4.00 [-4.71 , -3.29] | + | + | ? | ? |
| XENDOS 2001-2006[a] | -11.88 | 8.22 | 407 | -6.93 | 7.57 | 437 | 25.8% | -4.95 [-6.02 , -3.88] | + | + | ? | ? |
| **Total (Wald[b])** | | | **1023** | | | **1057** | **100.0%** | **-3.74 [-4.70 , -2.78]** | | | | |

Mean Difference
IV, Random, 95% CI [kg]




Test for overall effect: Z = 7.65 (P < 0.00001)

Heterogeneity: Tau² (DL[c]) = 0.63; Chi² = 9.48, df = 3 (P = 0.02); I² = 68%

**Footnotes**
[a]Participants with diastolic BP ≥ 90 mm Hg at baseline.
[b]CI calculated by Wald-type method.
[c]Tau² calculated by DerSimonian and Laird method.

**Risk of bias legend**
(A) Random sequence generation (selection bias)
(B) Allocation concealment (selection bias)
(C) Selective reporting (reporting bias)
(D) Other bias

### Phentermine/topiramate versus placebo

For critical outcomes, see Summary of findings 2; for important outcomes, see the additional summary of findings table in Supplementary material 10.

#### *Critical outcomes*

##### All-cause mortality

Since there were no deaths in the hypertensive subpopulation during the study, phentermine/topiramate may have little to no effect on all-cause mortality, but the evidence is very uncertain.

##### Cardiovascular morbidity

In CONQUER 2013, 2.3% of the hypertensive participants in the LD group, 3.7% in the HD group, and 1.7% of the placebo group experienced treatment-emergent cardiac AEs. There was little or no difference between the LD group and the control group (RR 1.51, 95% CI 0.43 to 5.27; 1 trial, 785 participants; very low-certainty evidence) and between the HD group and the control group (RR 2.39, 95% CI 0.82 to 6.69; 1 trial, 1044 participants; very low-certainty evidence). Palpitations occurred in 0.8%, 1.2% and 0.6% of the hypertensive participants in the LD, HD and placebo groups, respectively. Serious adverse cardiac events occurred in six hypertensive participants and serious adverse vascular events in two hypertensive participants, but the study did not report to which treatment groups these participants belonged. In summary, phentermine/topiramate may have little to no effect on cardiovascular morbidity, but the evidence is very uncertain.

##### Adverse events

In the hypertensive subpopulation, SAEs occurred in 3.4% (LD group), 3.7% (HD group) and in 4.2% of participants in the placebo group. There may be little or no difference in SAEs between phentermine/topiramate and placebo, but the evidence is very uncertain (RR 0.85, 95% CI 0.49 to 1.48; 1 study, 1305 participants; very low-certainty evidence; Analysis 2.1). In terms of all AEs, there are probably more in participants with phentermine/topiramate compared to participants with placebo (RR 1.13, 95% CI 1.08 to 1.20; 1 study, 1305 participants; moderate-certainty evidence; Analysis 2.2). The most common treatment-emergent AEs in the phentermine/topiramate groups were dry mouth (14.2% (LD group) and 22.7% (HD group)) and paraesthesia (14.2% (LD group) and 22.3% (HD group)) when compared to placebo (2.3% for each outcome).

For details on AEs, see Table 4 and Supplementary material 9.

#### *Important outcomes*

For details on important outcome data, see tables in Supplementary material 9.

##### Change in systolic blood pressure

In hypertensive participants, there are probably greater reductions in systolic BP in the phentermine/topiramate groups than in the placebo group after 13 months: LD group versus placebo (MD −2.00 mm Hg, 95% CI −3.97 to −0.03; 1 trial, 772 participants; moderate-certainty evidence; Analysis 2.4) and HD group versus placebo (MD −4.20 mm Hg, 95% CI −5.85 to −2.55; 1 trial, 1030 participants; moderate-certainty evidence; Analysis 2.4).

Compared with placebo, there is probably a greater percentage of participants with uncontrolled hypertension at baseline (≥ 140/90 mm Hg) treated in the HD group who achieved the BP goal of 140/90 mm Hg by week 56 (RR 1.42, 95% CI 1.13 to 1.78; 1 trial, 176 participants; moderate-certainty evidence). There is probably little or no difference between the LD group and placebo (RR 1.18, 95% CI 0.87 to 1.60; 1 trial, 144 participants; moderate-certainty evidence).

##### Change in diastolic blood pressure

With both doses of phentermine/topiramate, there are probably greater reductions in diastolic BP compared to placebo after 13 months in hypertensive participants: LD group versus placebo (MD −1.30 mm Hg, 95% CI −2.60 to −0.00; 1 trial, 772 participants; moderate-certainty evidence; Analysis 2.5); HD group versus placebo (MD -1.90 mm Hg, 95% CI −2.88 to −0.92; 1 trial, 1030 participants; moderate-certainty evidence; Analysis 2.5).

#### Change in body weight

Compared to placebo, phentermine/topiramate probably produces a greater percentage weight loss after 13 months in participants with hypertension at baseline: LD group versus placebo (MD −6.30 kg, 95% CI −7.37 to −5.23; 1 trial, 772 participants; moderate-certainty evidence; Analysis 2.3) and HD group versus placebo (MD −8.20 kg, 95% CI −9.09 to −7.31; 1 trial, 1030 participants; moderate-certainty evidence; Analysis 2.3). In addition, more participants in the hypertensive subpopulation achieved weight loss of ≥ 5%, ≥ 10% and ≥ 15% with phentermine/topiramate compared with placebo.

### Naltrexone/bupropion versus placebo

For critical outcomes, see Summary of findings 3; for important outcomes, see additional summary of findings table in Supplementary material 10.

#### *Critical outcomes*

#### All-cause mortality

Naltrexone/bupropion may have little to no effect on all-cause mortality, but the evidence is very uncertain (RR 0.99, 95% CI 0.70 to 1.40; 1 trial, 8283 participants; very low-certainty evidence; Analysis 3.1). During the study period, there were 63 deaths in the naltrexone/bupropion group (1.51%) and 63 deaths in the placebo group (1.53%) reported for the subpopulation of participants with hypertension. The number of cardiovascular deaths was 26 (0.62%) in the naltrexone/bupropion group and 37 (0.90%) in the placebo group (RR 0.80, 95% CI 0.62 to 1.04; 1 trial, 8283 participants).

#### Cardiovascular morbidity

Naltrexone/bupropion may have little to no effect on cardiovascular morbidity, but the evidence is very uncertain (RR 1.11, 95% CI 0.87 to 1.41; 1 trial, 8283 participants; very low-certainty evidence; Analysis 3.2). There were 132 participants with hypertension in the naltrexone/bupropion group and 118 in the placebo group who experienced a cardiovascular event. Sixty-two participants had a non-fatal myocardial infarction (1.54%), 28 had a non-fatal stroke (0.67%) and 49 participants were hospitalised for unstable angina (1.18%) in the naltrexone/bupropion group. In the placebo group, the respective numbers were 61 non-fatal myocardial infarctions (1.48%), 20 non-fatal strokes (0.49%) and 45 hospitalisations for unstable angina (1.09%).

#### Adverse events

In the hypertensive subpopulation, there is probably little or no difference in SAEs between naltrexone/bupropion and placebo (RR 1.05, 95% CI 0.96 to 1.14; 1 trial, 8283 participants; moderate-certainty evidence; Analysis 3.3). All AEs probably occur more frequently in participants treated with naltrexone/bupropion compared to those receiving placebo (RR 1.69, 95% CI 1.58 to 1.80; 1 trial, 8283 participants; moderate-certainty evidence; Analysis 3.4). Gastrointestinal AEs were higher in the naltrexone/bupropion group (17.2%) than in the placebo group (3.4%). Nervous system AEs (7.5% versus 2.7%) and dermatological AEs (1.1% versus 0.5%) were also reported more often in the naltrexone/bupropion versus the placebo group. In addition, AEs leading to withdrawal of study medication occurred more often in the naltrexone/bupropion group (30.6%) than in the placebo group (9.2%) (RR 3.33, 95% CI 2.99 to 3.70; 1 trial, 8283 participants).

For details on AEs, see Table 4 and Supplementary material 9.

#### *Important outcomes*

For details on important outcome data, see tables in Supplementary material 9.

#### Change in systolic blood pressure

In hypertensive participants, there was an increase in systolic BP in both study groups during the study period, with probably little to no difference between the naltrexone/bupropion group and the placebo group after 28 months (MD 0.00, 95% CI −0.60 to 0.60; 1 trial, 8283 participants; moderate-certainty evidence; Analysis 3.6).

#### Change in diastolic blood pressure

In hypertensive participants, there was an increase in diastolic BP in both study groups during the study period, with probably little to no difference between the naltrexone/bupropion group and the placebo group after 28 months (MD 0.30, 95% CI −0.08 to 0.68; 1 trial, 8283 participants; moderate-certainty evidence; Analysis 3.7).

#### Change in body weight

Compared to placebo, naltrexone/bupropion probably leads to a greater reduction in body weight after 28 months in participants with hypertension (MD −1.90, 95% CI −2.07 to −1.73; 1 trial, 8283 participants; moderate-certainty evidence; Analysis 3.5).

### Semaglutide versus placebo

For critical outcomes, see Summary of findings 4; for important outcomes, see the additional summary of findings table in Supplementary material 10.

#### *Critical outcomes*

#### All-cause mortality

This outcome was not reported for the hypertensive subpopulation.

#### Cardiovascular morbidity

This outcome was not reported for the hypertensive subpopulation.

#### Adverse events

This outcome was not reported for the hypertensive subpopulation.

#### *Important outcomes*

For details on important outcome data, see tables in Supplementary material 9.

#### Change in systolic blood pressure

In participants with uncontrolled hypertension, there was a decrease in systolic BP in the intervention group and an increase in the placebo group during the study period. After 15 months, there may be a reduction in systolic BP in favour of semaglutide compared to placebo (MD −6.26 mm Hg, 95% CI −6.57 to −5.95; 1 trial, 274 participants; low-certainty evidence; Analysis 4.1).

#### Change in diastolic blood pressure

In participants with uncontrolled hypertension, there was a decrease in diastolic BP in the intervention group and an increase in the placebo group during the study period. After 15 months, there may be a reduction in diastolic BP in favour of semaglutide compared to placebo (MD −3.74 mm Hg, 95% CI −3.95 to −3.53; 1 trial, 274 participants; low-certainty evidence; Analysis 4.2).

#### Change in body weight

This outcome was not reported for the hypertensive subpopulation.

### Tirzepatide versus placebo

For critical outcomes, see Summary of findings 5; for important outcomes, see the additional summary of findings table in Supplementary material 10.

#### *Critical outcomes*

#### All-cause mortality

This outcome was not reported for the hypertensive subpopulation.

#### Cardiovascular morbidity

This outcome was not reported for the hypertensive subpopulation.

#### Adverse events

This outcome was not reported for the hypertensive subpopulation.

#### *Important outcomes*

For details on important outcome data, see tables in Supplementary material 9.

#### Change in systolic blood pressure

In the subpopulation of participants with systolic BP ≥ 140 mm Hg at baseline, there was a decrease in systolic BP in both study groups during the study period. After 16 months, there may be a reduction in systolic BP in favour of tirzepatide compared to placebo (MD −6.10, 95% CI −10.40 to −1.80; 1 trial, 195 participants; low-certainty evidence; Analysis 5.1).

In the subpopulation of participants using antihypertensive medication at baseline, there was a decrease in systolic BP in both study groups during the study period. After 16 months, there may be a reduction in systolic BP in favour of tirzepatide compared to placebo (MD −3.10, 95% CI −5.31 to −0.89; 1 trial, 635 participants; low-certainty evidence; Analysis 5.1).

#### Change in diastolic blood pressure

In the subpopulation of participants with diastolic BP ≥ 90 mm Hg at baseline, there was a decrease in diastolic BP in both study groups during the study period. After 16 months, there may be a reduction in diastolic BP in favour of tirzepatide compared to placebo (MD −5.20, 95% CI −8.20 to −2.20; 1 trial, 207 participants; low-certainty evidence; Analysis 5.2).

In the subpopulation of participants using antihypertensive medication at baseline, there was a decrease in systolic BP in both study groups during the study period. After 16 months, there may be a reduction in diastolic BP in favour of tirzepatide compared to placebo (MD −1.60, 95% CI −3.10 to −0.10; 1 trial, 635 participants; low-certainty evidence; Analysis 5.2).

#### Change in body weight

This outcome was not reported for the hypertensive subpopulation.

### Subgroup analyses

Due to lack of data, subgroup analyses could not be performed for any of the included drugs.

### Sensitivity analyses

Due to lack of data, sensitivity analyses could not be performed for any of the included drugs.

### Reporting biases

We provide our assessment of the risk of selective outcome reporting for each of the included studies in the Risk of bias in included studies section. We did not utilise funnel plots to investigate publication bias, as our analyses contained fewer than 10 studies.

# DISCUSSION

Since our last update in 2020, we have added to the review two new randomised controlled trials (RCTs) that reported results for participants with hypertension. The trials evaluated recently approved drugs: semaglutide and tirzepatide. We found no new trials for the other drugs currently approved for long-term weight management. Most of the trials conducted recently have recruited people with and without hypertension, but none of them have made separate data available for the subpopulation of participants with hypertension.

## Summary of main results

Our systematic review attempted to determine the long-term effects of approved drugs for long-term weight management on patient-relevant endpoints, namely death and cardiovascular complications, in the antihypertensive therapy of people with essential hypertension. We found no RCTs that were specifically designed to answer this particular question, although these outcomes were sometimes reported under the category of adverse events. Our search revealed studies that focused mainly on the evaluation of effects on body weight and blood pressure.

We included four relevant trials investigating orlistat in our review. Only one RCT each could be included for the other drugs: phentermine/topiramate, naltrexone/bupropion, semaglutide, and tirzepatide. For liraglutide, no separate results for participants with hypertension have been published or made available by the drug manufacturer. Therefore, we were unable to include any study examining liraglutide in this review update. In addition, for this review update, we excluded studies investigating sibutramine, rimonabant or lorcaserin, since these three products lost their market approval and are therefore no longer relevant for long-term weight loss. Results of trials investigating these three products were discussed in the previous version of this review [63].

In people with hypertension, there may be little to no difference in all-cause mortality between orlistat and placebo, phentermine/topiramate and placebo, and naltrexone/bupropion and placebo, but the evidence is very uncertain for all three comparisons. Treatment with orlistat, phentermine/topiramate or naltrexone/bupropion may also result in little to no difference in cardiovascular morbidity compared to placebo, but again the evidence is very uncertain.

Serious adverse events (SAEs) are probably more common with orlistat than placebo, while there is probably little to no difference between naltrexone/bupropion and placebo. There may be little or no difference between phentermine/topiramate and placebo, but the evidence is very uncertain. When considering all adverse events (AEs), these are probably more common with naltrexone/bupropion and with phentermine/topiramate, than either compared to placebo. The evidence is very uncertain for all AEs for orlistat, although gastrointestinal AEs are reported more frequently with orlistat than with placebo.

We could not assess all-cause mortality, cardiovascular morbidity and AEs in people with hypertension for semaglutide or tirzepatide.

Naltrexone/bupropion and phentermine/topiramate probably lead to a reduction in body weight compared to placebo, as may orlistat. No data were available for semaglutide and tirzepatide. Furthermore, in people with hypertension, orlistat and phentermine/topiramate probably result in a reduction in both systolic and diastolic BP, while semaglutide and tirzepatide may result in a reduction of systolic and diastolic BP. With naltrexone/bupropion, on the other hand, there is probably little or no difference in systolic and diastolic BP compared to placebo.

To strengthen the evidence regarding patient-relevant outcomes for pharmacological weight management in people with hypertension, improved access to data from hypertensive subpopulations in existing studies is needed, as are higher-quality studies with large samples that measure long-term cardiovascular outcomes in people with hypertension.

## Limitations of the evidence included in the review

For five of the included RCTs, no protocol was available, and one RCT was terminated prematurely. For the remaining two trials, only limited results were available for the hypertensive subpopulation based on post-hoc analyses. This resulted in some concerns about selective reporting for all included trials. Two trials lacked sufficient information on the method of randomisation and/or allocation concealment. In five trials, information on blinding was insufficient. All but one trial had some concerns about bias or even high risk of bias due to incomplete outcome data for at least one of the reported outcomes. Seven of the eight included RCTs were commercially funded. There were some concerns about bias, since the possibility of sponsor influence could not be ruled out. Therefore, there were at least some concerns regarding the overall risk of bias in all the included studies. As a result, we downgraded the certainty of evidence in all comparisons for most of the outcomes of interest.

Overall, only a few trials could be included in the comparisons of individual drugs against placebo. Other than for orlistat, which was investigated in four RCTs, the results for each weight-reducing drug were from a single trial. Furthermore, the available data were sparse, particularly for critical endpoints such as mortality or cardiovascular morbidity. Results were mostly reported in the context of AEs with low event rates. This reduced our certainty about the evidence for many outcomes due to imprecision. For semaglutide and tirzepatide, only results regarding change in blood pressure were available. No studies on liraglutide could be included in our review.

Focusing on the effects of weight-reducing drugs on patient-relevant outcomes, we included only trials with a follow-up period of at least 24 weeks in our review. Nevertheless, to reach firm conclusions about long-term effects on mortality or cardiovascular complications, trials with follow-up periods spanning several years are necessary. The follow-up periods in those trials that reported on critical outcomes ranged from six to 48 months. Consequently, statements about long-term effects would be possible in principle. However, only in Nissen 2016, investigating naltrexone/bupropion, was the prespecified primary endpoint a clinical outcome, namely MACE, defined as cardiovascular death, nonfatal stroke or nonfatal myocardial infarction. Therefore, only in this study was the power calculation based on a clinical endpoint. The other trials included in the review focused mainly on the evaluation of effects on body weight and BP. These trials, not being powered for clinical events, are at high risk of bias due to a low number of events for clinical outcomes.

## Limitations of the review processes

We searched four electronic databases up to April 2024, and the clinical trials registries ClinicalTrials.gov and WHO ICTRP up to March 2020. In our 2024 update search, we searched ClinicalTrials.gov and WHO ICTRP within Cochrane CENTRAL. We examined the reference lists of included trials and relevant systematic reviews and meta-analyses. We contacted authors of studies and manufacturers of the investigated drugs with mixed study populations (normotensive and hypertensive participants) or the manufacturers of the investigated drugs (or both) for further information about the subset of people with hypertension. We assessed each study's quality and summarised the results. Although no ongoing studies were identified, the possibility that relevant publications have appeared since the search in April 2024 cannot be entirely excluded.

Our review focuses on the narrow question of the effects of drugs approved for long-term weight management in people with essential hypertension. Furthermore, we used the WHO definition for hypertension [1]. Since 2017, clinical guidelines in the USA have used a different definition for hypertension, namely ≥ 130/80 mmHg [6]. As a consequence, we did not include studies or subpopulations that met the American hypertension criteria. The results of our review are therefore not generalisable to all people with hypertension. In addition, our review does not provide any results for pregnant women with hypertension, since we explicitly excluded this group.

Another limitation of the review is that we were unable to examine publication bias using funnel plot analysis because there were too few studies available. Publication bias may therefore influence the results of the review.

Some meta-analyses showed high heterogeneity. According to our Methods, we intended to perform subgroup and sensitivity analyses. However, this was not possible due to the limited data available. Therefore, the heterogeneity remains largely unexplored.

One major limitation of this review is that information available from most of the included publications was insufficient to draw reliable conclusions. Furthermore, a substantial body of research, encompassing numerous large cardiovascular outcome trials, has been conducted to investigate the impact of drugs for long-term weight management on individuals irrespective of their hypertension status. However, the publication of results specifically for the subpopulation of people with hypertension from these studies is rare. Although we conducted authors and sponsors of all potentially relevant RCTs investigating mixed study populations, only occasionally was further information provided. As a consequence, only limited or no data, especially on outcomes such as mortality, cardiovascular morbidity and AEs were available from the included trials.

It was also difficult to judge the risk of bias in most RCTs, so this was often judged unclear. Conversely, it seems that there are unpublished data from many trials on subpopulations of people with hypertension, which unfortunately were not made available to us. These data could significantly enhance the evidence base for the research questions addressed in our review. Therefore, in consideration of the extant data, we cannot draw reliable conclusions regarding the effects of pharmaceutical weight-loss interventions on long-term patient-relevant outcomes in people with essential hypertension.

## Agreements and disagreements with other studies or reviews

Our review focuses on the narrow question of the efficacy and safety of drugs for long-term weight management in people with essential hypertension. Unfortunately, only very limited data are available on this topic. However, there are numerous RCTs that have investigated the approved drugs in a mixed population of normotensive and hypertensive individuals. The results of these studies are briefly presented below.

### Other studies

For the combination therapy of phentermine/topiramate, we could include only one trial involving a hypertensive subpopulation in our analysis (CONQUER 2013). An additional 56-week trial (EQUIP 2012) compared phentermine/topiramate to placebo in obese participants with blood pressure below 140/90 mm Hg [117]. In both trials, the mean weight loss with phentermine/topiramate was about 6 to 8 kg, which is higher than in the case of orlistat, while the antihypertensive effect of phentermine/topiramate is comparable to orlistat. After completion of the CONQUER 2013 trial, participants could take part in a further 52-week extension trial (SEQUEL 2014 [118]). Although there was significant sustained weight loss in the phentermine/topiramate groups compared to placebo, the reduction in blood pressure did not differ significantly between the groups after two years.

Only one trial could be included for the combination of naltrexone/bupropion in our analysis, since the author provided results for the subpopulation of participants with hypertension upon request (Nissen 2016). In addition to this RCT, naltrexone/bupropion was also assessed in four 56-week trials of the Contrave Obesity Research (COR) study programme, including 4536 overweight or obese people with or without hypertension, dyslipidaemia, or diabetes mellitus type 2. Results for the subpopulation of participants with hypertension were not available. In two of the trials, the percentage of participants with hypertension at baseline was about 20% (COR-I 2010 [119]; COR-II 2013 [120]; in the other two trials, it was not reported (COR-BMOD 2011[121]; COR-Diabetes 2013 [122]). A statistically significant weight reduction of about 4 to 5 kg was reported for naltrexone/bupropion compared to placebo in addition to diet and exercise counselling (COR-Diabetes 2013 [122]; COR-I 2010 [119]; COR-II 2013 [120]), or to an intensive behaviour modification programme (COR-BMOD 2011 [121]). Blood pressure remained unchanged or was slightly reduced in the naltrexone/bupropion groups, but there was a larger reduction in the placebo groups. The group differences were significant for systolic BP in the COR-I 2010 [119], COR-II 2013 [120], and COR-BMOD 2011 [121] trials and for diastolic BP in the COR-I 2010 [119] and COR-BMOD 2011 [121] trials. None of these studies investigated the effects on patient-related outcomes. The dropout rates in the four trials ranged from 42% to 50%. Nausea, dizziness, constipation and tinnitus were the most common adverse events in the naltrexone/bupropion groups.

For liraglutide, we could not include any study results in our review. One Cochrane review of glucagon-like peptide 1 (GLP-1) receptor agonists for the treatment of type 2 diabetes mellitus showed a significant weight reduction of 1.33 kg ($P < 0.0001$) and a marginal reduction in systolic BP of −2.42 mm Hg ($P = 0.05$) for a daily dose of 1.8 mg liraglutide compared to placebo after 24 weeks [123]. No difference in diastolic BP was reported. The most common AEs of liraglutide were gastrointestinal, such as nausea or vomiting [123]. There were also more cases of pancreatitis amongst liraglutide users compared with comparators, and in 2011, the FDA issued a warning that liraglutide may cause pancreatitis and thyroid carcinoma [124].

In 2016, the Liraglutide Effect and Action in Diabetes: Evaluation of cardiovascular outcome Results (LEADER) trial, a double-blind, placebo-controlled randomised controlled cardiovascular outcome trial including 9340 participants with type 2 diabetes mellitus at high risk of cardiovascular disease, investigated the long-term effects of 1.8 mg liraglutide in comparison to placebo on cardiovascular outcomes as well as on neoplasms and other adverse events [125]. After a mean follow-up of 3.8 years, the primary outcome, a composite endpoint including the first occurrence of death from cardiovascular causes, nonfatal myocardial infarction or nonfatal stroke, occurred in significantly fewer participants in the liraglutide group compared to the placebo group. On the other hand, significantly more participants in the liraglutide group discontinued the study due to AEs, and the rate of pancreatic carcinoma was twice as high in the liraglutide group as in the placebo group. However, the extent to which the results of these studies can be extrapolated to liraglutide in the context of weight management remains uncertain. This is due to the fact that the approved dose for this indication (3.0 mg per day) is almost double the dose used in the treatment of diabetes mellitus (maximum 1.8 mg).

For weight management, 3.0 mg liraglutide was assessed in the SCALE study programme, in which 5358 obese or overweight people with or without weight-related comorbidities participated (SCALE diabetes 2014 [126]; SCALE maintenance 2013 [127]; SCALE obesity and prediabetes 2014 [128]; SCALE sleep apnoea 2014 [129]). Although the percentage of hypertensive participants was as high as 70% in these trials, no results for a hypertensive subpopulation were reported. At the end of the studies (32 to 56 weeks), mean body weight was significantly reduced amongst liraglutide users compared to placebo (−3 to −6 kg). Systolic BP was also significantly reduced in all four trials (−3 mm Hg), while for diastolic BP only one trial showed a significant difference between liraglutide and placebo (−1 mm Hg). The dropout rates in the studies were about 23% to 34% and therefore lower than in other weight-management trials. As in the glycaemic control trials, the most common AEs were nausea, diarrhoea and vomiting.

Data from only one trial could be included for semaglutide (STEP 1 [78]), which included an additional post-hoc analysis on systolic and diastolic BP in participants with uncontrolled hypertension (i.e. average systolic BP ≥ 140 mm Hg or average diastolic BP ≥ 90 mm Hg) at baseline, regardless of antihypertensive medication [79]. The trial also reported that semaglutide may reduce the need for antihypertensive medication in overweight or obese adults without diabetes. However, these potential benefits were not sustained after treatment discontinuation. In addition to STEP 1, semaglutide was assessed in seven more trials of the STEP study programme [130, 131, 132, 133, 134, 135, 136], including 6102 people with a BMI greater than or equal to 30.0 kg/m$^2$ or greater than or equal to 27.0 kg/m$^2$ with the presence of at least one of the following weight-related comorbidities (treated or untreated): hypertension, dyslipidaemia, obstructive sleep apnoea or cardiovascular disease and history of at least one self-reported unsuccessful dietary effort to lose body weight. The results in these RCTs showed a significant weight reduction with semaglutide compared to placebo of 10 to 15 kg, and a significant decrease in systolic and diastolic BP. While there were no differences in the numbers of deaths or cardiovascular events between the semaglutide and placebo groups, participants in the semaglutide group experienced significantly more gastrointestinal AEs.

In addition to semaglutide for weight management, it was assessed in a number of trials for the treatment of diabetes (SUSTAIN programme [137]). This includes a large cardiovascular outcome study that evaluated the effects of 0.5 mg or 1.0 mg semaglutide compared to placebo in patients with type 2 diabetes who were at high cardiovascular risk (SUSTAIN-6). The trial enroled 3297 patients and primarily assessed major adverse cardiovascular events (MACE). After a median follow-up of 2.1 years, the rate of MACE was significantly lower amongst participants receiving semaglutide than amongst those receiving placebo. Rates of cardiovascular death were comparable between the groups [138].

Another cardiovascular outcome trial (SELECT) assessed semaglutide in the context of weight-management in 2023. The trial enroled 17,604 participants with pre-existing cardiovascular disease and a BMI ≥ 27 kg/m$^2$. About 82% of the participants had hypertension at baseline, defined as a BP ≥ 130/80 mm Hg. After a mean duration of follow-up of 40 months, semaglutide was reported to be superior to placebo in reducing the incidence of cardiovascular mortality, nonfatal myocardial infarction, or nonfatal stroke [139].

An individual patient data (IPD) meta-analysis of three RCTs of the STEP programme was published in August 2024, which was after the search for our review update had been carried out [140]. This IPD analysis goes beyond a conventional systematic review by enabling numerical comparisons of blood pressure effects across subgroups defined by baseline hypertension status. While the Kennedy analysis primarily included participants with baseline blood pressure in the normotensive range, it also reported subgroup estimates for different subpopulations with hypertension. In participants with systolic BP > 140 at randomisation (397 participants), the adjusted MD in systolic BP change was −4.09 mm Hg (95% CI −7.12 to −1.06, P = 0.005) in favour of semaglutide compared to placebo. The magnitude of blood pressure reduction was comparable to the overall pooled effect reported in that analysis and is consistent with the reductions observed in our hypertensive population. However, these effects may have been partly influenced by concurrent reductions in antihypertensive medication, as de-escalation of antihypertensive medication was more frequent in participants in the semaglutide groups compared to those in the placebo groups of the STEP trials (OR 2.61, 95% CI 0.92 to 9.24, P = 0.094 for the subpopulation of patients with systolic BP > 140). In another pooled post hoc analysis of the STEP trials [141], treatment with semaglutide 2.4 mg compared to placebo was associated with a higher proportion of participants who discontinued all antihypertensive medications (pooled results of STEP 1, 3, 6 and 8: 17.7% versus 9.0%; pooled results of STEP 2 and 6: 9.8% versus 7.3%) or had a reduction of antihypertensive treatment intensity (pooled results of STEP 1, 3, 6 and 8: 16.5% versus 4.8%; pooled results of STEP 2 and 6: 14.7% versus 5.7%). In addition, fewer participants in the semaglutide group required intensification of antihypertensive therapy by the end of treatment compared to those in the placebo group (pooled results of STEP 1, 3, 6 and 8: 8.3% versus 4.2%; pooled results of STEP 2 and 6: 7.9% versus 11.5%). Amongst participants receiving antihypertensive medication at randomisation, these changes were accompanied by greater mean reductions of the initial body weight with semaglutide versus placebo (pooled results of STEP 1, 3, 6 and 8: 14.8% (8.7%) versus 3.8% (6.9%); pooled results of STEP 2 and 6: 11.0% (8.2%) versus 3.2% (5.8%).

Within the SURMOUNT-1 2024 trial, one post hoc analysis investigated the effect of tirzepatide on body weight and BP reduction in the subpopulation of participants with hypertension [81]. It was observed that the BP reduction with tirzepatide was more pronounced amongst those hypertensive study participants not on antihypertensive medication at baseline. The study did not collect robust data on medication changes, and the dosage and number of BP medications may have been reduced in response to the decline in BP associated with tirzepatide. The data for the subpopulation of hypertensive patients not on antihypertensive medications may therefore provide the clearest and least confounded indication of the effect of tirzepatide on BP. In addition, tirzepatide was assessed in three more RCTs of the SURMOUNT study programme [142, 143, 144], including 4839 people with or without hypertension. The results in these studies showed a significant weight reduction with tirzepatide compared to placebo of 12 to 25 kg, as well as a significant reduction in systolic and diastolic BP. Blood pressure-related treatment-emergent AEs were reported more frequently with tirzepatide compared to placebo. The most commonly reported AEs with tirzepatide were diarrhoea, nausea and vomiting. Like liraglutide and semaglutide, tirzepatide was also assessed in a number of trials for the treatment of diabetes (SURPASS programme [145]). In addition, there is currently one ongoing cardiovascular outcome trial investigating tirzepatide for weight management. The SURMOUNT MNO trial assesses the impact on morbidity and mortality with once-weekly 2.4 mg tirzepatide compared with placebo in non-diabetic adults with obesity and with, or at risk of, cardiovascular disease [146].

These results indicate that further studies of the hypertensive subpopulation are required to more precisely determine the extent of differences between this group and the normotensive group.

### Systematic reviews

Systematic reviews on the long-term effects of weight-management drugs in people with hypertension are rare. One systematic review dealing with the long-term effects of weight loss on hypertension outcome measures reached the same conclusion as our review [147], that only short-term trials were available. The authors also warned "that extrapolation of short-term blood pressure changes with weight loss to the longer term is potentially misleading. The weight/hypertension relationship is complex and needs well-conducted studies with long-term follow-up to examine the effects of weight loss on hypertension outcomes". Systematic reviews on long-term effects of weight-management drugs in a mixed population of overweight or obese people with or without hypertension reported blood pressure-reducing effects for orlistat, liraglutide, lorcaserin and phentermine/topiramate, but not for naltrexone/bupropion [56, 57]. Although LeBlanc 2018 could include more than 30 trials that addressed medications for weight loss, no reliable conclusions could be drawn about clinical outcomes [57]. A systematic review investigated the effect of pharmacological weight reduction versus placebo on all-cause mortality and cardiovascular risk factors [148]. The meta-analyses, including up to seven RCTs with normotensive and hypertensive participants, reported that there is evidence of a benefit for the pharmacological weight-loss interventions in cardiovascular mortality, but no effect on all-cause mortality. However, these meta-analyses combine all different weight loss drugs regardless of their different mechanisms of action and dosage forms. This represents a high degree of heterogeneity amongst the included studies, which is why we consider the different drugs separately in our review. Unlike our review, the review by Kane 2019 does not include any evaluations of hypertensive subpopulations [148]. The reported results are therefore not directly transferable to patients with hypertension. In addition, only data on CVD outcomes, including myocardial infarction, stroke and/or mortality, were analysed, while our current review also analysed data on (serious) adverse events.

A recently published systematic review investigated different GLP-1 receptor agonists for weight loss in patients without diabetes [58]. This review by Moiz and colleagues included six RCTs with normotensive and hypertensive participants for semaglutide and one RCT for tirzepatide. The analyses showed reductions in BP and body weight for semaglutide and tirzepatide. Most AEs with GLP-1 receptor agonists were gastrointestinal disorders.

We are aware of all major and relevant systematic reviews and studies, and can say with confidence that our findings are in good agreement with recently-published reviews and studies.

## AUTHORS' CONCLUSIONS

### Implications for practice

There have been multiple developments in the domain of medications for long-term weight management since the original publication of this review in 2009. A number of pharmaceutical products have been removed from the market due to safety concerns, whilst new drugs have been introduced. Although the drugs currently available on the market for long-term weight management probably can, for the most part, help to lower blood pressure, they are probably also associated with high rates of side effects. Our critical outcomes of death and cardiovascular complications were not the focus of the trials included in this review but were sometimes presented as part of an 'adverse events' outcome, with the resultant data providing only very low certainty evidence. Overall, the evidence for people with hypertension remains insufficient to draw conclusions regarding the benefits of pharmacological weight loss in terms of reducing the risk of mortality or cardiovascular morbidity. Further large, high-quality trials are needed. Moreover, separate results for participants with hypertension should be made available from any completed trials involving both normotensive and hypertensive participants.

### Implications for research

Since there is little data on the hypertensive population, especially for patient-relevant outcomes, long-term trials assessing the effects of the approved drugs used for weight management on mortality and morbidity in this subpopulation are very much needed. It is also important to make existing data for this hypertensive subpopulation from already completed cardiovascular outcome trials publicly available, particularly since long-term trials assessing the effects of sibutramine, rimonabant and lorcaserin on mortality and morbidity have confirmed concerns about the potentially severe side effects that led to marketing withdrawal of these drugs throughout the world.

## SUPPLEMENTARY MATERIALS

Supplementary materials are available with the online version of this article: 10.1002/14651858.CD007654.pub6.

**Supplementary material 1** Search strategies

**Supplementary material 2** Characteristics of included studies

**Supplementary material 3** Characteristics of excluded studies

**Supplementary material 4** Characteristics of studies awaiting classification

**Supplementary material 5** Analyses

**Supplementary material 6** Data package

**Supplementary material 7** Search strategies used in the previous versions of the review

**Supplementary material 8** Survey of trial investigators and manufacturers of the investgated drugs providing information on included trials

**Supplementary material 9** Additional tables

**Supplementary material 10** Additional summary of findings tables for important outcomes

## ADDITIONAL INFORMATION

### Acknowledgements

Cochrane Hypertension Group supported the authors in the development of this review. The review authors would like to thank Douglas Salzwedel for assisting in updating the literature search for the 2016, 2020 and 2024 updates of this review. The review authors would also like to thank Ulrich Siering for assisting in the development of the original review.

#### Editorial and peer-reviewer contributions

Cochrane Hypertension supported the authors in the development of this updated review.

The following people conducted the editorial process for this article.

- Sign-off Editor (final editorial decision): Juan VA Franco, Cochrane Evidence Synthesis Unit Germany, Sub-Unit Düsseldorf, Institute of General Practice, Centre for Health and Society, Medical Faculty, Heinrich Heine University Düsseldorf
- Managing Editor (selected peer reviewers, provided editorial guidance to authors, edited the article): Ben Ridley, Cochrane Editorial Services
- Editorial Assistant (conducted editorial policy checks, collated peer-reviewer comments and supported editorial team): Andrew Savage, Cochrane Editorial Services
- Copy Editor (copy editing and production): Laura MacDonald, Cochrane Central Production Service
- Peer reviewers (provided comments and recommended an editorial decision): Anshula Ambasta, Assistant Professor at University of British Columbia (clinical/content review); Guillaume Grenet, Department of Anesthesiology, Pharmacology, and Therapeutics, Faculty of Medicine, University of British Columbia, Vancouver, BC, Canada (clinical/content review); Devarajulu Reddy S, Scientist-D(M), ICMR-NIRT (patient and public review); Nuala Livingstone, Cochrane Evidence Production and Methods Directorate (methods review); Jo Platt, Central Editorial Information Specialist (search review).

## Contributions of authors

Ulrike Spary-Kainz: selection of studies, development of fourth review update.

Nicole Posch: selection of studies, development of second and fourth review updates, corresponding author.

Andrea Siebenhofer: protocol development, quality assessment of trials, data extraction, development of initial version of the review and first review update.

Klaus Jeitler: protocol development, searching for trials, quality assessment of trials, data extraction, development of review updates.

Karl Horvath: protocol development, quality assessment of trials, data extraction, development of the initial version of the review and first review update.

Andrea Berghold: statistical analysis, development of initial version of the review.

Sebastian Winterholer: selection of studies, quality assessment of trials, data extraction in the third review update.

Cornelia Krenn: selection of studies, quality assessment of trials, data extraction, development of third review update.

Thomas Semlitsch: searching for trials, selection of studies, quality assessment of trials, data extraction, development of review updates.

All authors approved the review update for publication.

Two review authors (Anne Stich and Eva Matyas) did not contribute to the 2013 update of this review and were removed from the list of authors.

Thomas Semlitsch joined the team of review authors for the 2013 version of this review and provided substantive intellectual contributions that justify his inclusion as author.

Ulrich Siering did not contribute to the 2015 update of this review and was removed from the list of authors.

Jutta Meschik joined the team of review authors for the 2015 version of this review and provided substantive intellectual contributions that justify her inclusion as author.

Nicole Posch and Jutta Meschik did not contribute to the 2020 update of this review and were removed from the list of authors.

Sebastian Winterholer and Cornelia Krenn joined the team of review authors for the 2020 version of this review and provided substantive intellectual contributions that justify their inclusion as authors.

Ulrike Spary-Kainz and Christina Radl-Karimi joined the team of review authors for the 2026 version of this review and provided substantive intellectual contributions that justify their inclusion as authors.

Nicole Posch rejoined the team of authors for the 2026 update and provided substantive intellectual contributions that justify her inclusion as author.

## Declarations of interest

Nicole Posch: no known conflicts of interest

Ulrike Spary-Kainz: no known conflicts of interest

Christina Radl-Karimi: no known conflicts of interest.

Andrea Siebenhofer, Klaus Jeitler, and Karl Horvath were involved in the preparation of a report on the evaluation of the benefits and harms of non-drug treatment strategies in people with essential hypertension: weight reduction for the Institute for Quality and Efficiency in Health Care (iqwig.de/).

Andrea Berghold: Andrea Berghold is involved in Data and Safety Monitoring Boards for hematological and oncological diseases at ROCHE.

Cornelia Krenn: no known conflicts of interest.

Sebastian Winterholer: no known conflicts of interest.

Thomas Semlitsch: no known conflicts of interest.

## Sources of support

### Internal sources

- Institute of General Practice, Goethe University Frankfurt, Germany

  Salary
- Medical University of Graz, Austria

  Salary, office space, computer support, library resources

**External sources**

- None, Other

  No sources of support supplied

### Registration and protocol

The protocol for this Cochrane review was published in 2009: Siebenhofer A, Horvath K, Jeitler K, Berghold A, Stich AK, Matyas E, et al. Long-term effects of weight-reducing drugs in hypertensive patients [Protocol]. Cochrane Database of Systematic Reviews 2009, Issue 1. Art. No.: CD007654. DOI: 10.1002/14651858.CD007654

Original review: Long-term effects of weight-reducing drugs in people with hypertension DOI: 10.1002/14651858.CD007654.pub2

Update 2013: DOI: 10.1002/14651858.CD007654.pub3

Update 2016: DOI: 10.1002/14651858.CD007654.pub4

Update 2021: DOI: 10.1002/14651858.CD007654.pub5

### Data, code and other materials

As part of the published Cochrane review, the following is made available for download for users of the Cochrane Library: full search strategies for each database (Supplementary material 1); full citations of each unique report for all included studies (Supplementary material 2) or studies excluded from the full-text screen (Supplementary material 3), in the final review, and analysis data (Supplementary material 5); analyses and data management were conducted within Cochrane's authoring tool, RevMan, using the inbuilt computation methods (Supplementary material 6). Template data extraction forms from Excel are available from the authors on reasonable request.

## What's new

| Date | Event | Description |
|---|---|---|
| 19 June 2026 | New citation required but conclusions have not changed | Update published with changed authors, updated search and two new studies, but conclusions unchanged. |
| 19 June 2026 | New search has been performed | We updated the search for new studies in April 2024. We identified two new studies that met the inclusion criteria for this review and added Background information. Studies investigating lorcaserin are no longer included in this review update, since this product lost its market approval in 2020 and is therefore no longer relevant for long-term weight loss. |

## History

Protocol first published: Issue 1, 2009
Review first published: Issue 3, 2009

| Date | Event | Description |
|---|---|---|
| 4 January 2021 | New search has been performed | We updated the electronic search to March 2020. We found one additional study investigating weight-reducing drugs in people with hypertension. Studies investigating sibutramine or rimonabant are no longer included in this review update, since these products lost their market approval nearly 10 years ago and are therefore no longer relevant for long-term weight loss. |
| 4 January 2021 | New citation required but conclusions have not changed | Update published with changed authors, additional Background information and one additional study investigating weight-reducing drugs in people with hypertension. |
| 26 February 2016 | New search has been performed | The electronic search for new studies was extended to include four new weight-reducing drugs and updated to April 2015. We identified one additional study investigating phentermine/topiramate versus placebo that met the inclusion criteria of this review. |
| 26 February 2016 | New citation required and conclusions have changed | Update published with changed authors, additional background information, and one additional study included. |

| Date | Event | Description |
|---|---|---|
| 15 January 2013 | New search has been performed | The electronic search for new studies was updated to August 2012. No new studies were identified that met the inclusion criteria of this review. |

# REFERENCES


**1.** World Health Organization. Global report on hypertension. The race against a silent killer. 2023. Available from: https://www.who.int/publications/i/item/9789240081062 (accessed 18 November 2024). [ISBN: 978-92-4-008106-2]

**2.** Turnbull F, Neal B, Ninomiya T, Algert C, Arima H, Barzi F, et al. Effects of different regimens to lower blood pressure on major cardiovascular events in older and younger adults: meta-analysis of randomised trials. *BMJ* 2008;**336**(7653):1121-3. [DOI: 10.1136/bmj.39548.738368.BE]

**3.** Beunza JJ, Martínez-González MA, Ebrahim S, Bes-Rastrollo M, Núñez J, Martínez JA, et al. Sedentary behaviors and the risk of incident hypertension: the SUN Cohort. *American Journal of Hypertension* 2007;**20**(11):1156-62. [DOI: 10.1016/j.amjhyper.2007.06.007] [PMID: 17954361]

**4.** Li Z, Zhong W, Gao J, Zhang X, Lin G, Qi C, et al. Association between leisure sedentary behaviors and hypertension risk: z prospective cohort study and two-sample Mendelian randomization analysis in Europeans. *Preventive Medicine* 2024;**181**:107915. [DOI: 10.1016/j.ypmed.2024.107915] [PMID: 38408649]

**5.** Rabi D, McBrien K, Sapir-Pichhadze R, Nakhla M, Ahmed S, Dumanski S, et al. Hypertension Canada's 2020 comprehensive guidelines for the prevention, diagnosis, risk assessment, and treatment of hypertension in adults and children. *Canadian Journal of Cardiology* 2020;**36**(5):596-624. [DOI: 10.1016/j.cjca.2020.02.086]

**6.** Whelton PK, Carey RM, Aronow WS, Casey DE Jr, Collins KJ, Dennison Himmelfarb C, et al. 2017 ACC/AHA/AAPA/ABC/ACPM/AGS/APhA/ASH/ASPC/NMA/PCNA Guideline for the Prevention, Detection, Evaluation, and Management of High Blood Pressure in Adults: Executive Summary: a report of the American College of Cardiology/American Heart Association Task Force on Clinical Practice Guidelines. *Hypertension* 2018;**71**(6):1269-324. [PMID: 29133354]

**7.** McEvoy J, McCarthy C, Bruno R, Brouwers S, Canavan M, Ceconi C, et al. 2024 ESC Guidelines for the management of elevated blood pressure and hypertension. European Heart Journal 2024;**45**(38):3912-4018. [DOI: doi.org/10.1093/eurheartj/ehae178]

**8.** National Institute for Health and Care Excellence (NICE). Hypertension in adults: diagnosis and management. Available from: https://www.nice.org.uk/guidance/ng136 28 August 2019 (last updated 26 February 2026).

**9.** Horvath K, Jeitler K, Siering U, Stich AK, Skipka G, Gratzer TW, et al. Long-term effects of weight-reducing interventions in hypertensive patients: systematic review and meta-analysis. *Archives of Internal Medicine* 2008;**168**(6):571-80. [PMID: 18362248]

**10.** James WP, Caterson ID, Coutinho W, Finer N, Van Gaal LF, Maggioni AP, et al. Effect of sibutramine on cardiovascular outcomes in overweight and obese subjects. *New England Journal of Medicine* 2010;**363(10)**:905-17. [DOI: 10.1056/NEJMoa1003114 ] [PMID: 20818901]

**11.** European Medicines Agency (EMA). Press release - European Medicines Agency recommends suspension of marketing authorisations for sibutramine. Available from: https://www.ema.europa.eu/system/files/documents/press-release/wc500069995_en.pdf (updated 21 January 2010; accessed 2 June 2026).

**12.** European Medicines Agency (EMA). Questions and answers on the suspension of medicines containing sibutramine. Available from: https://www.ema.europa.eu/system/files/documents/referral/wc500094238_en.pdf (updated 6 August 2010; accessed 2 June 2026).

**13.** US Food and Drug Administration (FDA). Recommendation on a regulatory decision for Meridia (sibutramine). http://wayback.archive-it.org/7993/20170113112219/http:/www.fda.gov/downloads/Drugs/DrugSafety/UCM228795.pdf (updated 7 October 2010; accessed 18 December 2012).

**14.** Abbott Germany Theobald F. Press release: [Abbott setzt Vertrieb seiner Medikamente mit dem Wirkstoff Sibutramin zur Behandlung von Adipositas in den Ländern der Europäischen Union aus.] 2010. Available from: https://www.abbott.de/press/show/e7340/e19695/e18095/Sibutramin_210110_Theobald_de.pdf (accessed 14 January 2013).

**15.** US Food and Drug Administration (FDA). FDA news release: Abbott Laboratories agrees to withdraw its obesity drug Meridia. Available from: https://wayback.archive-it.org/7993/20170112215610/http:/www.fda.gov/NewsEvents/Newsroom/PressAnnouncements/2010/ucm228812.htm (updated 8 October 2010; cited 18 December 2012).

**16.** US Food and Drug Administration Egan AG, Colman EG. FDA rimonabant briefing document. Endocrine and Metabolic Drugs Advisory Committee. 2007. Available from: https://wayback.archive-it.org/7993/20170405052424/https://www.fda.gov/ohrms/dockets/ac/07/briefing/2007-4306b1-fda-backgrounder.pdf (updated 13 June 2007; accessed 18 December 2012).

**17.** European Medicines Agency (EMA). Questions and answers on the recommendation to suspend the marketing authorisation of Acomplia (rimonabant). Available from: https://www.ema.europa.eu/en/documents/medicine-qa/questions-and-answers-recommendation-suspend-marketing-authorisation-acomplia-rimonabant_en.pdf (updated 23 October 2008; accessed 17 December 2012).

**18.** European Medicines Agency (EMA). Press release - the European Medicines Agency recommends suspension of the marketing authorisation of Acomplia. Available from: https://www.ema.europa.eu/en/documents/press-release/european-medicines-agency-recommends-suspension-marketing-authorisation-acomplia_en.pdf (updated 23 October 2008; accessed 1 June 2026).

**19.** European Medicines Agency (EMA). Public statement on Acomplia (rimonabant) - withdrawal of the marketing authorisation in the European Union. Available from: https://www.ema.europa.eu/en/documents/public-statement/public-statement-acomplia-withdrawal-marketing-authorisation-european-union_en.pdf (updated 30 January 2009; accessed 2 June 2026).

**20.** US Food and Drug Administration (FDA). Qsymia (phentermine and topiramate extended-release) - approval letter. Available from: https://www.accessdata.fda.gov/drugsatfda_docs/nda/2012/022580Orig1s000Approv.pdf (updated 17 July 2012; accessed 20 July 2015).

**21.** US Food and Drug Administration (FDA). Belviq (lorcaserin hydrochloride) - approval letter. https://www.accessdata.fda.gov/drugsatfda_docs/appletter/2012/022529Orig1s000ltr.pdf (updated 27 June 2012; accessed 20 July 2015).

**22.** US Food and Drug Administration (FDA). Contrave (naltrexone hydrochloride/bupropion hydrochloride) - approval letter. Available from: https://www.accessdata.fda.gov/drugsatfda_docs/nda/2014/200063Orig1s000Approv.pdf (updated 10 September 2014; accessed 20 July 2015).

**23.** US Food and Drug Administration (FDA). Saxenda (liraglutide) - approval letter. Available from: https://www.accessdata.fda.gov/drugsatfda_docs/appletter/2014/206321Orig1s000ltr.pdf (updated 23 December 2014; accessed 20 July 2015).

**24.** Apovian CM, Aronne LJ, Bessesen DH, McDonnell ME, Murad MH, Pagotto U, et al. Pharmacological management of obesity: an Endocrine Society clinical practice guideline. *Journal of Clinical Endocrinology and Metabolism* 2015;**100**(2):342-62. [PMID: 25590212]

**25.** European Medicines Agency (EMA). Mysimba (naltrexone/bupropion) - EPAR summary for the public. Available from: https://www.ema.europa.eu/en/documents/overview/mysimba-epar-summary-public_en.pdf (updated 15 April 2015; accessed 29 November 2024).

**26.** European Medicines Agency (EMA). Saxenda (liraglutide) - EPAR summary for the public. Available from: https://www.ema.europa.eu/en/documents/overview/saxenda-epar-summary-public_en.pdf (updated August 2023; accessed 29 November 2024).

**27.** Government of Canada. Health Canada. Drug Product Database – Contrave (DIN 02472945). https://health-products.canada.ca/dpd-bdpp/info?lang=eng&code=96230 16 March 2018.

**28.** Government of Canada. Approval of saxenda (liraglutide) for chronic weight management in adults with overweight or obesity. https://health-products.canada.ca/dpd-bdpp/info?lang=eng&code=92226 27 May 2015.

**29.** Australian Government, Department of Health, Disability and Ageing, Therapeutic Goods Administration. Approval of Liraglutide. Available from: https://www.tga.gov.au/resources/artg/225804 24 December 2015-cited October 2025.

**30.** Australian Government, Department of Health, Disability and Ageing, Therapeutic Goods Administration. Approval naltrexone/bupropion ARTG. https://www.tga.gov.au/resources/prescription-medicines-registrations/contrave-inova-pharmaceuticals-australia-pty-ltd accessed October 2025.

**31.** Korean Ministry of Food and Drug Safety. Approval liraglutide. https://nedrug.mfds.go.kr/CEAAA21?sort=&sortOrder=&searchYn=true&ExcelRowdata=&page=1&searchDivision=
21 July 2017.

**32.** Korean Ministry of Food and Drug Safety. Naltrexone/bupropion - approval information. https://nedrug.mfds.go.kr/CEAAA21?sort=&sortOrder=&searchYn=true&ExcelRowdata=&page=1&searchDivision=
%2Fbupropion&itemSeq=&stdrCodeName=&atcCodeName=&indutyClassC
29 April 2016.

**33.** European Medicines Agency (EMA). Questions and answers on the withdrawal of the marketing authorisation application for Belviq (lorcaserin). Available from: https://www.ema.europa.eu/en/documents/medicine-qa/questions-and-answers-withdrawal-marketing-authorisation-application-belviq_en.pdf (updated 30 May 2013; accessed 2 June 2026).

**34.** Eisai Incorporation. Eisai to Voluntarily Withdraw BELVIQ®/BELVIQ XR® in the U.S.. Available from: https://eisai.mediaroom.com/2020-02-13-Eisai-to-Voluntarily-Withdraw-BELVIQ-R-BELVIQ-XR-R-in-the-U-S (13 February 2020; accessed 3 December 2024).

**35.** US Food and Drug Administration (FDA). Drug safety communication - FDA requests the withdrawal of the weight-loss drug Belviq, Belviq XR (lorcaserin) from the market. Available from: https://www.fda.gov/media/135189/download (updated 13 February 2020; accessed 30 March 2020).

**36.** European Medicines Agency (EMA). Questions and answers on the refusal of the marketing authorisation for Qsiva (phentermine/topiramate). Available from: https://www.ema.europa.eu/en/documents/smop-initial/questions-and-answers-refusal-marketing-authorisation-qsiva-phentermine-topiramate_en.pdf (updated 21 February 2013; accessed 29 November 2024).

**37.** European Medicines Agency (EMA). Scientific conclusions and grounds for refusal of the marketing authorisation for Qsiva (13 June 2013). Available from: https://www.ema.europa.eu/en/documents/other/qsiva-epar-scientific-conclusions-and-grounds-refusal-marketing-authorisation-qsiva_en.pdf (updated 13 June 2013; accessed 2 June 2026).

**38.** US Food and Drug Administration (FDA). Wegovy (semaglutide) - approval letter. Available from: https://www.accessdata.fda.gov/drugsatfda_docs/appletter/2021/215256Orig1s000ltr.pdf (updated 4 June 2021; accessed 9 December 2024).

**39.** US Food and Drug Administration (FDA). Zepbound (tirzepatide) - approval letter. Available from: Approval Letter_217806Orig1s000ltr.pdf (updated 08 November 2023; accessed 9 December 2024).

**40.** European Medicines Agency (EMA). Wegovy (semaglutide) - an overview of Wegovy and why it is authorised in the EU. Available from: https://www.ema.europa.eu/en/documents/

overview/wegovy-epar-medicine-overview_en.pdf (updated April 2023; accessed 9 December 2024).

**41.** European Medicines Agency (EMA). Mounjaro (tirzepatide): EPAR - medicine overview. Available from: https://www.ema.europa.eu/en/documents/overview/mounjaro-epar-medicine-overview_en.pdf (last updated 21 February 2024; accessed 29 November 2024).

**42.** Government of Canada. Approval of semaglutide. https://health-products.canada.ca/dpd-bdpp/info?lang=eng&code=101774 28 March 2024.

**43.** Government of Canada. Approval of tirzepatide. https://health-products.canada.ca/dpd-bdpp/info?lang=eng&code=104066 10 December 2024.

**44.** Australian Government, Department of Health, Disability and Ageing, Therapeutic Goods Administration. WEGOVY 1.7 mg semaglutide 2.27 mg/mL solution for injection pre-filled pen with pre-assembled needle (356287) - approval information. https://www.tga.gov.au/resources/artg/356287 accessed October 2025.

**45.** Australian Government, Department of Health, Disability and Ageing, Therapeutic Goods Administration. Approval of tirzepatide. https://www.tga.gov.au/resources/artg/379331 accessed October 2025.

**46.** Korean Ministry of Food and Drug Safety. Approval semaglutide. https://nedrug.mfds.go.kr/CEAAA21?sort=&sortOrder=&searchYn=true&ExcelRowdata=&page=1&searchDivision=detail&itemEngName=&entpEngName=&ingrEngName=semaglutide& 27 April 2023.

**47.** Korean Ministry of Food and Drug Safety. Approval tirzepatide. https://nedrug.mfds.go.kr/CEAAA21?sort=&sortOrder=&searchYn=true&ExcelRowdata=&page=1&searchDivision=detail&itemEngName=&entpEngName=&ingrEngName=tirzepatide&it 28 June 2023.

**48.** Padwal RS, Majumdar SR. Drug treatments for obesity: orlistat, sibutramine, and rimonabant. *Lancet* 2007;**369**(9555):71-7. [PMID: 17208644]

**49.** Aronne LJ, Wadden TA, Peterson C, Winslow D, Odeh S, Gadde KM. Evaluation of phentermine and topiramate versus phentermine/topiramate extended-release in obese adults. *Obesity (Silver Spring, Md.)* 2013;**21**(11):2163-71. [PMID: 24136928]

**50.** Caixas A, Albert L, Capel I, Rigla M. Naltrexone sustained-release/bupropion sustained-release for the management of obesity: review of the data to date. *Drug Design, Development and Therapy* 2014;**8**:1419-27. [PMID: 25258511]

**51.** Russell-Jones D. Molecular, pharmacological and clinical aspects of liraglutide, a once-daily human GLP-1 analogue. *Molecular and Cellular Endocrinology* 2009;**297**(1-2):137-40. [PMID: 19041364]

**52.** Friedrichsen M, Breitschaft A, Tadayon S, Wizert A, Skovgaard D. The effect of semaglutide 2.4 mg once weekly on energy intake, appetite, control of eating, and gastric emptying in adults with obesity. *Diabetes, Obesity and Metabolism* 2020;**23**(3):754-62. [DOI: 10.1111/dom.14280 ] [PMID: 33269530]

**53.** Forzano I, Varzideh F, Avvisato R, Jankauskas S, Mone P, Santulli G. Tirzepatide: a systematic update. *International Journal of Molecular Sciences* 2022;**23**(23):1-16. [DOI: 10.3390/ijms232314631] [PMID: 36498958]

**54.** Kaneko S. Tirzepatide: a novel, once-weekly dual GIP and GLP-1 receptor agonist for the treatment of type 2 diabetes. *TouchREVIEWS in Endocrinology* 2022;**18**(1):10-19. [DOI: 10.17925/EE.2022.18.1.10] [PMID: 35949358]

**55.** Chan EW, He Y, Chui CS, Wong AY, Lau WC, Wong IC. Efficacy and safety of lorcaserin in obese adults: a meta-analysis of 1-year randomized controlled trials (RCTs) and narrative review on short-term RCTs. *Obesity Reviews: an Official Journal of the International Association for the Study of Obesity* 2013;**14**(5):383-92. [PMID: 23331711]

**56.** Khera R, Pandey A, Chandar AK, Murad MH, Prokop LJ, Neeland IJ, et al. Effects of weight-loss medications on cardiometabolic risk profiles: a systematic review and network meta-analysis. *Gastroenterology* 2018;**154**(5):1309-19.e1307. [DOI: 10.1053/j.gastro.2017.12.024] [PMID: 29305933]

**57.** LeBlanc EL, Patnode CD, Webber EM, Redmond N, Rushkin M, O'Connor EA. Behavioral and pharmacotherapy weight loss interventions to prevent obesity-related morbidity and mortality in adults: an updated systematic review for the U.S. Preventive Services Task Force. Agency for Healthcare Research and Quality (US). U.S. Preventive Services Task Force evidence syntheses, formerly systematic evidence reviews, report no.: 18-05239-EF-1 2018. [PMID: 30354042]

**58.** Moiz A, Filion KB, Toutounchi H, Tsoukas MA, Yu OHY, Peters TM, et al. Efficacy and safety of glucagon-like peptide-1 receptor agonists for weight loss among adults without diabetes: a systematic review of randomized controlled trials. *Annals of Internal Medicine* 2025;**178**:Epub ahead of print. [PMID: 39761578]

**59.** Zhang F, Tong Y, Su N, Li Y, Tang L, Huang L, et al. Weight loss effect of glucagon-like peptide-1 mimetics on obese/overweight adults without diabetes: a systematic review and meta-analysis of randomized controlled trials. *Journal of Diabetes* 2015;**7**(3):329-39. [DOI: 10.1111/1753-0407.12198] [PMID: 25043423]

**60.** Flint AC, Conell C, Ren X, Banki NM, Chan SL, Rao VA, et al. Effect of systolic and diastolic blood pressure on cardiovascular outcomes. *New England Journal of Medicine* 2019;**381**(3):243-51. [DOI: 10.1056/NEJMoa1803180 ] [PMID: 31314968]

**61.** Rapsomaniki E, Timmis A, George J, Pujades-Rodriguez M, Shah AD, Denaxas S, et al. Blood pressure and incidence of twelve cardiovascular diseases: lifetime risks, healthy life-years lost, and age-specific associations in 1·25 million people. *Lancet* 2014;**383**(9932):1899-911. [DOI: 10.1016/S0140-6736(14)60685-1 ] [PMID: 24881994]

**62.** Gepner Y, Goldstein N, Shelef I, Schwarzfuchs D, Zelicha H, Meir AY, et al. Dissociation between long-term weight loss intervention and blood pressure: an 18-month randomized controlled trial. *Journal of General Internal Medicine* 2021;**36**(8):2300-6. [DOI: 10.1007/s11606-021-06655-2 ] [PMID: 33634382]

**63.** Siebenhofer A, Winterholer S, Jeitler K, Horvath K, Berghold A, Krenn C, et al. Long-term effects of weight-reducing drugs in people with hypertension. *Cochrane Database of Systematic Reviews* 2021, Issue 1. Art. No: CD007654. [DOI: 10.1002/14651858.CD007654.pub5] [PMID: 33454957]

**64.** Siebenhofer A, Horvath K, Jeitler K, Berghold A, Stich AK, Matyas E, et al. Long-term effects of weight-reducing drugs in hypertensive patients. *Cochrane Database of Systematic Reviews* 2009, Issue 1. Art. No: CD007654. [DOI: 10.1002/14651858.CD007654]

**65.** Higgins JPT, Lasserson T, Thomas J, Flemyng E, Churchill R. Methodological Expectations of Cochrane Intervention Reviews. Cochrane: London Version August 2023.

**66.** Page MJ, McKenzie JE, Bossuyt PM, Boutron I, HoJmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. *BMJ (Clinical Research Ed.)* 2021;**371**(4):n71. [DOI: 10.1136/bmj.n71] [PMID: 33782057]

**67.** Siebenhofer A, Jeitler K, Horvath K, Berghold A, Posch N, Meschik J, Semlitsch T. Long-term effects of weight-reducing drugs in people with hypertension. *Cochrane Database of Systematic Reviews* 2016, Issue 3. Art. No: CD007654. [DOI: 10.1002/14651858.CD007654.pub4] [PMID: 26934640]

**68.** Pedersen SD, Manjoo P, Dash S, Jain A, Pearce N, Poddar M. Pharmacotherapy for obesity management in adults: 2025 clinical practice guideline update. *Canadian Medical Association Journal* 2025;**197**(27):E797-E809. [DOI: 10.1503/cmaj.250502] [PMID: 40789597]

**69.** Lefebvre C, Glanville J, Briscoe S, Featherstone R, Littlewood A, Metzendorf M-I, et al. Chapter 4: Searching for and selecting studies. In: Higgins JP, Thomas J, Chandler J, Cumpston M, Li T, Page MJ, et al (editors). *Cochrane Handbook for Systematic Reviews of Interventions* version 6.5 The Cochrane Collaboration, 2024. Available from https://www.cochrane.org/handbook.

**70.** Higgins JP, Savović J, Page MJ, Elbers RG, Sterne JA. Chapter 8: Assessing risk of bias in a randomized trial. In: Higgins JP, Thomas J, Chandler J, Cumpston M, Li T, Page MJ, et al (editors). *Cochrane Handbook for Systematic Reviews of Interventions* version 6.1 (updated September 2020). The Cochrane Collaboration, 2020. Available from https://www.cochrane.org/handbook.

**71.** Hróbjartsson A, Thomsen AS, Emanuelsson F, Tendal B, Hilden J, Boutron I, et al. Observer bias in randomized clinical trials with measurement scale outcomes: a systematic review of trials with both blinded and nonblinded assessors. *Canadian Medical Association Journal* 2013;**185**(4):E201-11. [PMID: 23359047]

**72.** Musini VM, Wright JM. Factors affecting blood pressure variability: lessons learned from two systematic reviews of randomized controlled trials. *PloS One* 2009;**4**(5):e5673.

**73.** Deeks JJ, Higgins JP, Altman DG (editors). Chapter 10: Analysing data and undertaking meta-analyses. In: Higgins JP, Thomas J, Chandler J, Cumpston M, Li T, Page MJ, et al (editors). *Cochrane Handbook for Systematic Reviews of Interventions* version 6.4 (updated August 2023). The Cochrane Collaboration, 2023. Available from https://www.cochrane.org/handbook.

**74.** Institut für Qualität und Wirtschaftlichkeit im Gesundheitswesen (IQWiG). Evaluation of the benefits and harms of non-drug weight-reducing treatment strategies in patients with essential hypertension [Nutzenbewertung nichtmedikamentöser Behandlungsstrategien bei Patienten mit Bluthochdruck: A-Gewichtsreduktion]. Available from: iqwig.de/download/A05-21A_Abschlussbericht_Gewichtsreduktion_bei_Bluthochdruck_neu.pd 200619-119.

**75.** Sterne JA, Sutton AJ, Ioannidis JP, Terrin N, Jones DR, Lau J, et al. Recommendations for examining and interpreting funnel plot asymmetry in meta-analyses of randomised controlled trials. *BMJ* 2011;**343**:d4002. [PMID: 21784880]

**76.** Review Manager (RevMan). Version 8.7.0. The Cochrane Collaboration, 2024. Available at https://revman.cochrane.org.

**77.** Guyatt G, Oxman AD, Akl EA, Kunz R, Vist G, Brozek J, et al. GRADE guidelines: 1. Introduction-GRADE evidence profiles and summary of findings tables. *Journal of Clinical Epidemiology* 2011;**64**(4):383-94. [PMID: 21195583]

**78.** Wilding JP, Batterham RL, Calanna S, Davies M, Van Gaal LF, Lingvay I, et al. Once-weekly semaglutide in adults with overweight or obesity. *New England Journal of Medicine* 2021;**384**(11):989-1002. [DOI: 10.1056/NEJMoa2032183] [PMID: 33567185]

**79.** Kosiborod MN, Bhatta M, Davies M, Deanfield JE, Garvey WT, Khalid U, et al. Semaglutide improves cardiometabolic risk factors in adults with overweight or obesity: STEP 1 and 4 exploratory analyses. *Diabetes, Obesity and Metabolism* 2022;**25**(2):468-78. [DOI: 10.1111/dom.14890] [PMID: 36200477]

**80.** Jastreboff AM, Aronne LJ, Ahmad NN, Wharton S, Connery L, Alves B, et al. Tirzepatide once weekly for the treatment of obesity. *New England Journal of Medicine* 2022;**387**(3):205-16. [DOI: 10.1056/NEJMoa2206038] [PMID: 35658024]

**81.** Krumholz HM, De Lemos JA, Sattar N, Linetzky B, Sharma P, Mast CJ, et al. Tirzepatide and blood pressure reduction: stratified analyses of the SURMOUNT-1 randomised controlled trial. *Heart (British Cardiac Society)* 2024;**110**(39):1165-71. [DOI: 10.1136/heartjnl-2024-324170] [PMID: 39084707]

**82.** Siebenhofer A, Horvath K, Jeitler K, Berghold A, Stich A K, Matyas E, et al. Long-term effects of weight-reducing drugs in hypertensive patients. *Cochrane Database of Systematic Reviews* 2009, Issue 3. Art. No: CD007654. [DOI: 10.1002/14651858.CD007654.pub2] [PMID: 19588440]

**83.** Torgerson JS, Hauptman J, Boldrin MN, Sjöström L. XENical in the prevention of diabetes in obese subjects (XENDOS) study: a randomized study of orlistat as an adjunct to lifestyle changes for the prevention of type 2 diabetes in obese patients. *Diabetes Care* 2004;**27**(1):155-61.

**84.** Torgerson JS, Arlinger K, Kappi M, Sjöström L. Principles for enhanced recruitment of subjects in a large clinical trial. the XENDOS (XENical in the prevention of Diabetes in Obese Subjects) study experience. *Controlled Clinical Trials* 2001;**22**(5):515-25.

**85.** Cocco G, Pandolfi S, Rousson V. Sufficient weight reduction decreases cardiovascular complications in diabetic patients with the metabolic syndrome: a randomized study of orlistat as an adjunct to lifestyle changes (diet and exercise). *Heart Drug* 2005;**5**(2):68-74.

**86.** Nissen SE, Wolski KE, Prcela L, Wadden T, Buse JB, Bakris G, et al. Effect of naltrexone-bupropion on major adverse cardiovascular events in overweight and obese patients with cardiovascular risk factors: a randomized clinical trial. *JAMA* 2016;**315**(10):990-1004. [PMID: 26954408]

**87.** Buse JB, Smith SR, Gilder K, Shan K, Halseth A. Effect of naltrexone/bupropion on cardiovascular events in overweight and obese participants with type 2 diabetes and cardiovascular risk factors in a large randomized, double-blind study. In: Diabetes. Vol. 66. 2017:A557. [EMBASE: 8024951]

**88.** Wolski K. Light study - data on hypertensive subgroup [personal communication]. Email to: T Semlitsch 23 April 2020.

**89.** Halseth A, Gilder K, Shan K, Acevedo L, Buse J. Effect on body weight of naltrexone/bupropion in overweight and obese participants with cardiovascular risk factors in a large randomized double-blind study. In: Obesity Facts. Vol. 10. 2017:46. [EMBASE: 8024954]

**90.** Halseth A, Gilder K, Shan K, Acevedo L, Smith S. Naltrexone/bupropion is well tolerated and had no effect on serious adverse events in participants receiving antidepressant medication, including SSRIs, in a large randomized double-blind study. In: Obesity Facts. Vol. 10. 2017:46. [EMBASE: 6521721]

**91.** Smith SR, Buse JB, Gilder K, Shan K, Halseth A. Effect on body weight of naltrexone/bupropion in overweight and obese participants with type 2 diabetes and cardiovascular risk factors in a large, randomized, double-blind study. In: Diabetes. Vol. 66. 2017:A556. [EMBASE: 8024952]

**92.** Bakris G, Calhoun D, Egan B, Hellmann C, Dolker M, Kingma I, et al. Orlistat improves blood pressure control in obese subjects with treated but inadequately controlled hypertension. *Journal of Hypertension* 2002;**20**(11):2257-67.

**93.** Guy-Grand B, Drouin P, Eschwege E, Gin H, Joubert JM, Valensi P. Effects of orlistat on obesity-related diseases - a six-month randomized trial. *Diabetes, Obesity & Metabolism* 2004;**6**(5):375-83.

**94.** Davidson MH, Tonstad S, Oparil S, Schwiers M, Day WW, Bowden CH. Changes in cardiovascular risk associated with phentermine and topiramate extended-release in participants with comorbidities and a body mass index >27 kg/m(2). *American Journal of Cardiology* 2013;**111**(8):1131-8. [PMID: 23375187]

**95.** Ryan DH, Garvey WT, Day WW. Achievement of composite of recommended goals in obese subjects with type 2 diabetes mellitus using extended-release phentermine/topiramate in a 56-week weight loss intervention. In: Endocrine Reviews. Vol. 33. 2012:(3 Meeting Abstracts). [EMBASE: 70832584]

**96.** Astrup AV, Oppert JM, Peterson CA. Weight loss and improvements in cardiometabolic outcomes with extended-release phentermine/topiramate treatment in overweight/obese subjects with type 2 diabetes. In: Diabetologia. Vol. 55. 2012:S284-S285. [EMBASE: 70888716]

**97.** Cheskin LJ, Bowden CH. Reduction of cardiovascular risk factors by magnitude of weight loss in obese and overweight subjects with >2 comorbidities. In: Journal of General Internal Medicine. Vol. 28. 2013:S168. [EMBASE: 71292986]

**98.** Garvey WT, Henry RR, Peterson CA, Bowden CH. Weight loss with controlled-release phentermine/topiramate (PHEN/TPM CR) reverses metabolic syndrome (MetS) and improves associated traits. In: Diabetes. Vol. 60. 2011:A506. [EMBASE: 70629626]

**99.** Kahan S, Karnawat S. Effect of weight loss (Wl) on concomitant medication costs in obese/overweight individuals with metabolic syndrome (METS) receiving phentermine/topiramate extended-release (PHEN/TPM ER). In: Value in Health. Vol. 18. 2015:A297. [EMBASE: 1819490]

**100.** Cheskin LJ, Peterson CA. Improved cardiometabolic risk factors in overweight/obese subjects receiving controlled-release phentermine/topiramate (PHEN/TPM CR). In: Diabetes. Vol. 61. 2012:A710. [EMBASE: 70799295]

**101.** Davidson M, Bowden CH, Day WW. Weight loss and cardiovascular risk reduction over 2 years with controlled-release phentermine-topiramate. In: Journal of the American College of Cardiology. Vol. 57. 2011:E545. [EMBASE: 70400222]

**102.** Finer N, Jordan J, Dvorak R. Weight loss and improvements in blood pressure (BP) status associated with phentermine and topiramate extended-release (PHEN/TPM ER) treatment in obese and overweight adults. In: Obesity Facts. Vol. 6. 2013:45. [EMBASE: 71300090]

**103.** Gadde K, Najarian T, Day WW. Magnitude of weight loss experienced with a low-dose, controlled-release formulation of phentermine/topiramate (PHEN/TPM) may drive degree of cardiometabolic benefit. In: Diabetes. Vol. Conference Publication. 2010:(var. pagings). [EMBASE: 71602039]

**104.** Gadde K, Peterson C, Troupin B, Day WW. 12-month weight loss and antihypertensive benefits with PHEN/TPM in overweight and obese subjects with hypertension. In: European Journal of Cardiovascular Prevention and Rehabilitation. Vol. 17. 2010:S2. [EMBASE: 70349685]

**105.** Gadde KM, Allison DB, Ryan DH, Peterson CA, Troupin B, Schwiers ML, et al. Effects of low-dose, controlled-release, phentermine plus topiramate combination on weight and associated comorbidities in overweight and obese adults (CONQUER): a randomised, placebo-controlled, phase 3 trial. [Erratum appears in Lancet. 2011 Apr 30;377(9776):1494]. *Lancet* 2011;**377**(9774):1341-52. [PMID: 21481449]

**106.** Gadde KM, Day WW. Low-dose, controlled-release phentermine/topiramate for reduction of weight, related risks in overweight/obese adults with >2 comorbidities. In: Obesity Reviews. Vol. 11. 2010:42-3. [EMBASE: 70226591]

**107.** Garvey WT, Gadde K, Tam P, Peterson C. Changes in insulin sensitivity in overweight/obese patients treated with low-dose, controlled-release phentermine/topiramate (PHEN/TPM). In: Diabetes. Vol. Conference Publication. 2010:(var. pagings). [EMBASE: 71602034]

**108.** Garvey WT, Gadde K, Wilson LF, Peterson C. Improvements in dyslipidemia and other cardiometabolic disease risk factors with low-dose, controlled-release phentermine/topiramate (PHEN/TPM). In: Diabetes. Vol. Conference Publication. 2010: (var. pagings). [EMBASE: 71601293]

**109.** Gupta AK, Church T, Day WW. Obesity treatment-induced weight loss improves resting blood pressure and reduces progression from prehypertension to hypertension in overweight/obese adults. In: Circulation. Vol. 127. 2013:(12 Meeting Abstracts). [EMBASE: 71026610]

**110.** Kushner RF, Varghese ST. Weight loss with phentermine and topiramate extended-release in obese and overweight subjects over 56 weeks. In: Journal of General Internal Medicine. Vol. 28. 2013:S241. [EMBASE: 71293155]

**111.** Rossner SK, Sharma AM, Troupin B, Padwal RS. Effects of extended-release phentermine/topiramate on weight loss and blood pressure in obese subjects with type 2 diabetes mellitus. In: Diabetologia. Vol. 55. 2012:S285. [EMBASE: 70888717]

**112.** Finkelstein EA, Kruger E, Karnawat S. Cost-effectiveness analysis of Qsymia for weight loss. *PharmacoEconomics* 2015;**33**(7):699-706. [PMID: 24986038]

**113.** Garvey WT, Troupin B, Day WW. Once-daily, low-dose, controlled-release phentermine/topiramate results in significant clinical improvements in overweight/obese patients with type 2 diabetes. In: Diabetologia. Vol. 53. 2010:S308. [EMBASE: 70262814]

**114.** Troupin B, Hankin CS. Low-dose controlled-release phentermine/topiramate (PHEN/TPM CR) results in significant weight loss and improves atherosclerotic biomarkers in overweight/obese patients with obesity-related comorbidities. In: Clinical Pharmacology and Therapeutics. Vol. 89. 2011:S39. [EMBASE: 70343842]

**115.** Troupin B, Hankin CS. Magnitude of weight loss conferred by controlled-release low-dose phentermine/topiramate (PHEN/TPM CR) determines extent of cardiometabolic benefit. In: Clinical Pharmacology and Therapeutics. Vol. 89. 2011:S38-S39. [EMBASE: 4272365]

**116.** CTRI/2017/05/008655. A clinical study to study the effect of lorcaserin tablets in the treatment of obesity. https://trialsearch.who.int/Trial2.aspx?TrialID=CTRI/2017/05/008655 (first received 25 May 2017).

**117.** Allison D, Gadde K, Garvey W, Peterson C, Schwiers M, Najarian T, et al. Controlled-release phentermine/topiramate in severely obese adults: a randomized controlled trial (EQUIP). *Obesity* 2012;**20**(2):330-42. [PMID: 22051941]

**118.** Garvey WT, Ryan DH, Look M, Gadde KM, Allison DB, Peterson CA, et al. Two-year sustained weight loss and metabolicbenefits with controlled-release phentermine/topiramate in obese and overweight adults (SEQUEL): a randomized, placebo-controlled, phase 3 extension study. *American Journal of Clinical Nutrition* 2012;**95**:297-308. [EMBASE: 2012064424]

**119.** Greenway FL, Fujioka K, Plodkowski RA, Mudaliar S, Guttadauria M, Erickson J, et al. Effect of naltrexone plus bupropionon weight loss in overweight and obese adults (COR-I): a multicentre, randomised, double-blind, placebo-controlled, phase 3 trial. *Lancet* 2010;**376**(9741):595-605. [PMID: 20673995]

**120.** Apovian CM, Aronne L, Rubino D, Still C, Wyatt H, Burns C, et al. A randomized, phase 3 trial of naltrexone SR/bupropionSR on weight and obesity-related risk factors (COR-II). *Obesity* 2013;**21**(5):935-43. [PMID: 23408728]

**121.** Wadden TA, Foreyt JP, Foster GD, Hill JO, Klein S, O'Neil PM, et al. Weight loss with naltrexone SR/bupropion SR combination therapy as an adjunct to behavior modification: the COR-BMOD trial. *Obesity* 2011;**19**(1):110-20. [EMBASE: 2011434989]

**122.** Hollander P, Gupta AK, Plodkowski R, Greenway F, Bay H, Burns C, et al. Eff ects of naltrexone sustained-release/bupropion sustained-release combination therapy on body weight and glycemic parameters in overweight and obese patients with type 2 diabetes. *Diabetes Care* 2013;**36**:4022-9. [PMID: 24144653]

**123.** Shyangdan DS, Royle P, Clar C, Sharma P, Waugh N, Snaith A. Glucagon-like peptide analogues for type 2 diabetes mellitus. *Cochrane Database of Systematic Reviews* 2011, Issue 10. Art. No: CD006423. [DOI: 10.1002/14651858.CD006423.pub2] [PMID: 21975753]

**124.** NEJM Journal Watch. FDA reminds clinicians of liraglutide's risks for pancreatitis and cancer. Available from: https://clinician.nejm.org/fda-reminds-clinicians-liraglutides-risks-pancreatitis-cancer-FW201106140000002 14 June 2011.

**125.** Marso SP, Poulter NR, Nissen SE, Nauck MA, Zinman B, Daniels GH, et al. Design of the liraglutide eff ect and action indiabetes: evaluation of cardiovascular outcome results (LEADER) trial. *American Heart Journal* 2013;**166**(5):823-30. [PMID: 24176437]

**126.** Davies MJ, Bergenstal R, Bode B, Kushner RF, Lewin A, Skjoth TV, et al. Efficacy of liraglutide for weight loss among patients with type 2 diabetes: the SCALE diabetes randomized clinical trial. *JAMA* 2015;**314**(7):687-99. [DOI: 10.1001/jama.2015.9676]

**127.** Wadden TA, Hollander P, Klein S, Niswender K, Woo V, Hale PM, et al. Weight maintenance and additional weight loss with liraglutide after low-calorie-diet-induced weight loss: the SCALE maintenance randomized study. *International Journal of Obesity (London)* 2013;**37**(11):1443-51. [DOI: 10.1038/ijo.2013.120]

**128.** Le Roux CW, Astrup A, Fujioka K, Greenway F, Lau DC, Van Gaal L, et al. 3 years of liraglutide versus placebo for type 2 diabetes risk reduction and weight management in individuals with prediabetes: a randomised, double-blind trial. *Lancet* 2017;**389**(10077):1399-409. [PMID: 28237263]

**129.** Blackman A, Foster GD, Zammit G, Rosenberg R, Aronne L, Wadden T, et al. Effect of liraglutide 3.0mg in individuals with obesity and moderate or severe obstructive sleep apnea: the SCALE sleep apnea randomized clinical trial. *International Journal of Obesity* 2016;**40**(8):1310-9. [PMID: 27005405]

**130.** Kadowaki T, Isendahl J, Khalid U, Lee SY, Nishida T, Ogawa W, et al. Semaglutide once a week in adults with overweight or obesity, with or without type 2 diabetes in an east Asian population (STEP 6): a randomised, double-blind, double-dummy, placebo-controlled, phase 3a trial. *Lancet Diabetes & Endocrinology* 2022;**10**(3):193-206. [DOI: 10.1016/S2213-8587(22)00008-0]

**131.** Rubino D, Abrahamsson N, Davies M, Hesse D, Greenway FL, Jensen C, et al. Effect of continued weekly subcutaneous semaglutide vs pacebo one weight loss maintenance in adults with overweight or obesity: the STEP 4 randomized clinical trial. *JAMA* 2021;**325**(14):1414-25. [DOI: 10.1001/jama.2021.3224]

**132.** Davies M, Faerch L, Jeppesen OK, Pakseresht A, Pedersen SD, Perreault L, et al. Semaglutide 2.4 mg once a week in adults with overweight or obesity, and type 2 diabetes (STEP 2): a randomised, double-blind, double-dummy, placebo-controlled, phase 3 trial. *Lancet* 2021;**397**(10278):971-84. [DOI: 10.1016/S0140-6736(21)00213-0]

**133.** Garvey WT, Batterham RL, Bhatta M, Buscemi S, Christensen LN, Frias JP, et al. Two-year effects of semaglutide in adults with overweight or obesity: the STEP 5 trial. *Nature Medicine* 2022;**28**(10):2083-91. [DOI: 10.1038/s41591-022-02026-4]

**134.** Mu Y, Bao X, Eliaschewitz FG, Hansen MR, Kim BT, Koroleva A, et al. Efficacy and safety of once weekly semaglutide 2.4 mg for weight management in a predominantly east Asian population with overweight or obesity (STEP 7): a double-blind, multicentre, randomised controlled trial. *Lancet Diabetes & Endocrinology* 2024;**12**(3):184-95. [DOI: 10.1016/S2213-8587(23)00388-1]

**135.** Rubino DM, Greenway FL, Khalid U, O'Neil PM, Rosenstock J, Sorrig R, et al. Effect of weekly subcutaneous semaglutide vs daily liraglutide on body weight in adults with overweight or obesity without diabetes: the STEP 8 randomized clinical trial. *JAMA* 2022;**327**(2):138-50. [DOI: 10.1001/jama.2021.23619]

**136.** Wadden TA, Bailey TS, Billings LK, Davies M, Frias JP, Koroleva A, et al. Effect of subcutaneous semaglutide vs placebo as an adjunct to intensive behavioral therapy on body weight in adults with overweight or obesity: the STEP 3 randomized clinical trial. *JAMA* 2021;**325**(14):1403-13. [DOI: 10.1001/jama.2021.1831]

**137.** Aroda VR, Ahmann A, Cariou B, Chow F, Davies MJ, Jodar E, et al. Comparative efficacy, safety, and cardiovascular outcomes with once-weekly subcutaneous semaglutide in the treatment of type 2 diabetes: insights from the SUSTAIN 1-7 trials. *Diabetes & Metabolism* 2019;**45**(5):409-18. [DOI: 10.1016/j.diabet.2018.12.001]

**138.** Marso SP, Bain SC, Consoli A, Eliaschewitz FG, Jodar E, Leiter LA, et al. Semaglutide and cardiovascular outcomes in patients with type 2 diabetes. *New England Journal of Medicine* 2016;**375**(19):1834-44. [DOI: 10.1056/NEJMoa1607141]

**139.** Lincoff AM, Brown-Frandsen K, Colhoun HM, Deanfield J, Emerson SS, Esbjerg S, et al. Semaglutide and cardiovascular outcomes in obesity without diabetes. *New England Journal of Medicine* 2023;**389**(24):2221-32. [DOI: 10.1056/NEJMoa2307563]

**140.** Kennedy C, Hayes P, Cicero A, Dobner S, Le Roux C, McEvoy J, et al. Semaglutide and blood pressure: an individual patient data meta-analysis. *European Heart Journal* 2024;**45**(38):4124-34. [DOI: https://doi.org/10.1093/eurheartj/ehae564]

**141.** Tchang B, Knight M, Adelborg K, Clements J, Iversen A, Traina A. Effect of semaglutide 2.4 mg on use of antihypertensive and lipid-lowering treatment in five randomized controlled STEP trials. *Obesity* 2024;**33**:267-77. [DOI: 10.1002/oby.24202]

**142.** Aronne LJ, Sattar N, Horn DB, Bays HE, Wharton S, Lin WY, et al. Continued treatment with tirzepatide for maintenance of weight reduction in adults with obesity: the SURMOUNT-4 randomized clinical trial. *JAMA* 2024;**331**(1):38-48. [DOI: 10.1001/jama.2023.24945]

**143.** Garvey WT, Frias JP, Jastreboff AM, Le Roux CW, Sattar N, Aizenberg D, et al. Tirzepatide once weekly for the treatment of obesity in people with type 2 diabetes (SURMOUNT-2): a double-blind, randomised, multicentre, placebo-controlled, phase 3 trial. *Lancet* 2023;**402**(10402):613-26. [DOI: 10.1016/S0140-6736(23)01200-X]

**144.** Wadden TA, Chao AM, Machineni S, Kushner R, Ard J, Srivastava G, et al. Tirzepatide after intensive lifestyle intervention in adults with overweight or obesity: the SURMOUNT-3 phase 3 trial. *Nature Medicine* 2023;**29**(11):2909-18. [DOI: 10.1038/s41591-023-02597-w]

**145.** Kushner P, Anderson JE, Simon J, Boye KS, Ranta K, Torcello-Gomez A, et al. Efficacy and safety of tirzepatide in adults with type 2 diabetes: a perspective for primary care providers. *Clinical Diabetes* 2023;**41**(2):258-72. [DOI: 10.2337/cd22-0029]

**146.** Lam CSP, Rodriguez A, Aminian A, Ferrannini E, Heerspink HJL, Jastreboff AM, et al. Tirzepatide for reduction of morbidity and mortality in adults with obesity: rationale and design of the SURMOUNT-MMO trial. *Obesity (Silver Spring)* 2025;**33**(9):1645-56. [DOI: 10.1002/oby.24332]

**147.** Aucott L, Poobalan A, Smith WC, Avenell A, Jung R, Broom J. Effects of weight loss in overweight/obese individuals and long-term hypertension outcomes: a systematic review. *Hypertension* 2005;**45**(6):1035-41. [PMID: 15897373]

**148.** Kane JA, Mehmood T, Munir I, Kamran H, Kariyanna PT, Zhyvotovska A, et al. Cardiovascular risk reduction associated with pharmacological weight loss: a meta-analysis. *International Journal of Clinical Research Trials* 2019;**4**(1):1-5. [DOI: 10.15344/2456-8007/2019/131]

# ADDITIONAL TABLES

## Table 1. Overview of included studies and synthesis

| Study name year; country | Study design | Intervention details | Control details | Population (sample size: interventions/control) | Outcomes with available data (synthesis method/ metric) | Outcome measures | Outcome time point | Method of synthesis |
|---|---|---|---|---|---|---|---|---|
| Intervention type: orlistat | | | | | | | | |
| Bakris 2002; USA | RCT, parallel, double-blind | Orlistat 120 mg; 3 times a day | Placebo | 278/276 | All-cause mortality (events); cardiovascular morbidity (events); adverse events (MA); body weight (MA); blood pressure (MA) | All-cause mortality | 12 months | summary |
| | | | | | | Cardiovascular morbidity | | summary |
| | | | | | | SAEs | | MA |
| | | | | | | All AEs | | MA |
| | | | | | | AEs leading to withdrawal | | summary |
| | | | | | | Change in body weight | | MA |
| | | | | | | Change in SBP | | MA |
| | | | | | | Change in DBP | | MA |
| Cocco 2005; Switzerland | RCT, parallel, double-blind | Orlistat 120 mg; 3 times a day | Placebo | 45/45 | All-cause mortality (events); cardiovascular morbidity (rate); adverse events (MA); body weight (MA); blood pressure (MA) | All-cause mortality | 6 months | summary |
| | | | | | | Cardiovascular morbidity | | summary |
| | | | | | | SAEs | | MA |
| | | | | | | Change in body weight | | MA |
| | | | | | | Change in SBP | | MA |
| | | | | | | Change in DBP | | MA |
| Guy-Grand 2004; France | RCT, parallel, double-blind | Orlistat 120 mg; 3 times a day | Placebo | 304/310[a] | Body weight (MA); blood pressure (MA) | Change in body weight | 6 months | MA |
| | | | | | | Change in SBP | | MA |

**Table 1. Overview of included studies and synthesis** *(Continued)*

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | Change in SBP | | MA |
| XENDOS 2001-2006; Sweden | RCT, parallel, double-blind | Orlistat 120 mg; 3 times a day | Placebo | 924/950[a] | All-cause mortality (events); cardiovascular morbidity (rate); adverse events (MA); body weight (MA); blood pressure (MA) | All-cause mortality | 48 months | summary |
| | | | | | | Cardiovascular morbidity | | summary |
| | | | | | | SAEs | | MA |
| | | | | | | All AEs | | MA |
| | | | | | | AEs leading to withdrawal | | summary |
| | | | | | | Change in body weight | | MA |
| | | | | | | Change in SBP | | MA |
| | | | | | | Change in DBP | | MA |
| Intervention: phentermine/topiramate | | | | | | | | |
| CONQUER 2013; USA | RCT, parallel, double-blind | Phentermine (7.5 mg)/topiramate (46 mg) or phentermine (15 mg)/topiramate (92 mg); once daily | Placebo | 261/520/524[a] | All-cause mortality (events); cardiovascular morbidity (rate); adverse events (events); body weight (MD); blood pressure (MD) | All-cause mortality | 56 weeks | summary |
| | | | | | | Cardiovascular morbidity | | summary |
| | | | | | | SAEs | | summary |
| | | | | | | All AEs | | summary |
| | | | | | | AEs leading to withdrawal | | summary |
| | | | | | | Change in body weight | | summary |
| | | | | | | Change in SBP | | summary |
| | | | | | | Change in DBP | | summary |
| Intervention: naltrexone/bupropion | | | | | | | | |
| Nissen 2016; USA | RCT, parallel, double-blind | Naltrexone (8 mg)/ bupropion (90 mg); 1 to 4 times per day | Placebo | 4164/4123[a] | All-cause mortality (events); cardiovascular morbidity (events); adverse events (events); body weight (MD); blood pressure (MD) | All-cause mortality | 121 weeks | summary |
| | | | | | | Cardiovascular morbidity | | summary |

**Table 1. Overview of included studies and synthesis** *(Continued)*

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | SAEs | | summary |
| | | | | | | All AEs | | summary |
| | | | | | | AEs leading to withdrawals | | summary |
| | | | | | | Change in body weight | | summary |
| | | | | | | Change in SBP | | summary |
| | | | | | | Change in DBP | | summary |
| Intervention: semaglutide | | | | | | | | |
| STEP 1 2023; USA | RCT, parallel, double-blind | Semaglutide (dose escalation 0.25 to 2.4 mg); once weekly | Placebo | 177/97[b] | Blood pressure (MD) | Change in SBP | 68 weeks | summary |
| | | | | | | Change in DBP | | summary |
| Intervention: tirzepatide | | | | | | | | |
| SURMOUNT-1 2024; USA | RCT, parallel, double-blind | Tirzepatide (5, 10 or 15 mg); once weekly | Placebo | 153/42[c]<br>168/39[d]<br>500/135[e] | Blood pressure (MD) | Change in SBP | 72 weeks | summary |
| | | | | | | Change in DBP | | summary |
| AEs: adverse events; DBP: diastolic blood pressure; MA: meta-analysis of standardised effect sizes; MD: mean difference; RCT: randomised controlled trial; SAEs: serious adverse events; SBP: systolic blood pressure | | | | | | | | |

[a]Hypertensive subgroup
[b]Subgroup with uncontrolled hypertension (i.e. SBP ≥ 140 mmHg and/or DBP ≥ 90 mmHg)
[c]Subgroup with SBP ≥ 140 mmHg
[d]Subgroup with DBP ≥ 90 mmHg
[e]Subgroup on antihypertensive medication

**Table 2. Overview of study populations**

| Study | Intervention(s) | Description of power and sample size calculation | Randomised (N) | Safety (N) | ITT (N) | Finishing trial | Randomised | Follow-up |
|---|---|---|---|---|---|---|---|---|

**Table 2. Overview of study populations** *(Continued)*

| | and comparator(s) | | | | | | (N) | finishing trial (%) | (extended follow-up) |
|---|---|---|---|---|---|---|---|---|---|
| **Orlistat vs placebo** | | | | | | | | | |
| Bakris 2002 | Orlistat | "The sample size determination for the present trial was based on a 2-sample t-test (2-tailed). Since there were 2 primary efficacy parameters in this trial, body weight reduction and sitting diastolic BP reduction, Holm's sequential rejection procedure was used to project the overall type I error rate 0.05, and α = 0.025 was chosen for the calculation of sample size for each parameter. A mean body weight change of 2.1 kg and a within-group standard deviation of 6.1 kg would require 161 participants per group to provide a power of 0.8 at α = 0.025. A mean change of 3.0 mm Hg in sitting diastolic BP with a within-group standard deviation of 8.5 mm Hg would require 153 participants per group to provide power of 0.8 with α = 0.025. Based on these calculations, and assuming a dropout rate of 35%, 496 participants (248 participants per group) had to be enroled to ensure an adequate statistical power of at least 80% in either of the 2 primary efficacy parameters." | 278 | | nr | 267 | 162 | 58 | 24 weeks |
| | Placebo | | 276 | | nr | 265 | 108 | 39 | |
| Cocco 2005 | Orlistat | nr | 45 | | 45 | 45 | 45 | 100 | 24 weeks |
| | Placebo | | 45 | | 45 | 45 | 45 | 100 | |
| Guy-Grand 2004 | Orlistat | "Power calculations indicated that, with a power of 80% at the 0.05 significance level, 408 participants were needed to detect a 2.5 mm Hg difference in diastolic BP, 152 participants were needed to detect a 0.05% difference in HbA1c. 140 participants were needed to detect a 0.35 mmol/l difference in LDL-cholesterol." | 499 (HT: 304) | | nr | 499 (HT: 304) | 458 (HT: nr) | 91.6% (HT: nr) | 24 weeks |
| | Placebo | | 505 (HT: 310) | | nr | 505 (HT: 310) | 458 (HT: nr) | 90.7% (HT: nr) | |
| XENDOS 2001-2006 | Orlistat | "A 2-sided log-rank test would require a minimum of ~ 95 primary cases of type 2 diabetes in both study groups combined to have 90% | 1650 | (HT: 408) | nr | 1640 (HT: nr) | (HT: 386) | 94.6 | 208 weeks |

**Table 2. Overview of study populations** *(Continued)*

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | (DBP ≥ 90 mm Hg) (OD) | power of detecting a significant outcome at α = 0.05. With this event-based design, 3305 participants were randomised and followed until sufficient events occurred. As a consequence of the design, study power would be unaffected by dropout rate." | | | | | | | |
| | Orlistat<br>(SBP ≥ 140 mm Hg) (OS) | | | (HT: 516) | nr | | (HT: 491) | 95.2 | |
| | Placebo<br>(DBP ≥ 90 mm Hg) (PD) | | 1655 | (HT: 441) | nr | 1637 (HT: nr) | (HT: 421) | 95.5 | |
| | Placebo<br>(SBP ≥ 140 mm Hg) (PS) | | | (HT: 509) | nr | | (HT: 487) | 95.7 | |
| **Phentermine/topiramate vs placebo** | | | | | | | | | |
| CONQUER 2013 | Phen/Top (LD) | "Power analysis based on data from a previous study suggested that 250 participants in each group would provide > 95% power to detect a difference of 4.4% in weight loss between placebo and active treatments at a significance level of 0.05. To enhance the power for detecting differences in safety outcomes, we planned to enrol about 2500 participants." | 498 (HT: 261) | | 398 (HT: 261) | 488 (HT: 261) | 374 (HT: 256) | 75.1 (HT: 98.1) | 56 weeks |
| | Phen/Top (HD) | | 995 (HT: 520) | | 994 (HT: 520) | 981 (HT: 520) | 733 (HT: 514) | 73.7 (HT: 98.8) | |
| | Placebo | | 994 (HT: 524) | | 993 (HT: 524) | 976 (HT: 524) | 616 (HT: 516) | 61.9 (HT: 98.5) | |
| Naltrexone/bupropion vs placebo | | | | | | | | | |
| Nissen 2016 | Nal/Bup | "The trial was designed to provide 90% power to rule out the 1.4 margin (i.e. the upper limit of the confidence interval would not exceed 1.4) when the true HR is 1.0, which required 378 primary events. The early pre-approval analysis to rule out the 2.0 margin required 87 primary events to provide 90% power when the true HR is 1.0. In both settings, a 1-sided type I error (α) of 2.5% was used. To obtain | 4456 (HT: 4164) | | 4455 (HT: 4164) | 4455 (HT: 4164) | 705 (HT: nr) | 15.8% (HT: nr) | 121 weeks |
| | Placebo | | 4454 (HT: 4123) | | 4450 (HT: 4119) | 4450 (HT: 4119) | 275 (HT: nr) | 6.2% (HT: nr) | |

**Table 2. Overview of study populations** *(Continued)*

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | | sample sizes, an annualised rate of primary events of 1.5% in the placebo group was assumed. The recruitment was assumed to take 1.5 years, with maximum participant follow-up of 4 years. It was assumed that 7% of the study population would discontinue during the lead-in period, with a loss–to–follow-up rate of 1.2% annually." | | | | | | |
| Semaglutide vs placebo | | | | | | | | |
| STEP 1 2023 | Semaglutide | "A sample size of 1950 participants provided an effective power of 99% for the coprimary and confirmatory secondary end points, tested in a prespecified hierarchical order. Efficacy end points were analyzed in the full analysis population (all randomly assigned participants according to the intention-to-treat principle); safety end points were analyzed in the safety analysis population (all randomly assigned participants exposed to at least one dose of semaglutide or placebo). Observation periods included the in-trial period (the time from random assignment to last contact with a trial site, regardless of treatment discontinuation or rescue intervention) and the on-treatment period. All results from statistical analyses were accompanied by a two-sided 95% confidence interval and corresponding P values (with significance defined as P<0.05). Supportive secondary endpoint analyses were not controlled for multiple comparisons and should not be used to infer definitive treatment effects." | 1306 (HT: 177) | 1306 (HT: 177) | 1306 (HT: 177) | 1306 (HT: 177) | 100 | 68 weeks |
| | Placebo | | 655 (HT: 97) | 655 (HT: 97) | 655 (HT: 97) | 655 (HT: 97) | 100 | |
| Tirzepatide vs placebo | | | | | | | | |
| SURMOUNT-1 2024 | Tirzepatide | "Power analysis based on data from a previous study suggested that a sample size of 2400 participants would provide an effective power of > 90% to demonstrate the superiority of tirzepatide (10mg, 15mg, or both) to placebo, relative to the coprimary end points, each at a two-sided significance level of 0.025. The sample size calculation as- | 1896<br>(HT: SBP 153;<br>S-MED 500; DBP 168;<br>D-MED 500) | 1896[a] | | 1689[a] | | |
| | Placebo | | 643 | 643[a] | | 495[a] | | |

**Table 2. Overview of study populations** *(Continued)*

| | | | |
|---|---|---|---|
| | | sumed at least an 11% point difference in the mean percentage weight reduction from baseline at 72 weeks for tirzepatide (10mg, 15 mg, or both) as compared with placebo, a common standard deviation of 10% and a dropout rate of 25%." | (HT: SBP 42; S-MED 500;<br>DBP: 39;<br><br>D-MED 135) |

DBP: diastolic blood pressure; HR: hazard ratio; HT: hypertensive subgroup; ITT: intention-to-treat; N: number of participants analysed; nr: not reported; OD: orlistat and diastolic blood pressure ≥ 90 mm Hg; OS: orlistat and systolic blood pressure ≥ 140 mm Hg; Phen/Top (HD): phentermine/topiramate high dose (15 mg/92 mg); Phen/Top (LD): phentermine/topiramate low dose (7.5 mg/46 mg); PD: placebo and diastolic blood pressure ≥ 90 mm Hg; PS: placebo and systolic blood pressure ≥ 140 mm Hg; SBP: systolic blood pressure; D-MED: systolic BP of participants on hypertensive medication; S-MED: diastolic BP of participants on hypertensive medication; vs: versus

[a]Only data for the entire population reported

**Table 3. Baseline characteristics (part 1)**

| Study | Intervention(s) and comparator(s) | Age (mean years (SD)) | Sex (female %) | BMI (mean kg/m² (SD)) | Body weight (mean kg (SD)) | Sitting systolic blood pressure (mean mm Hg (SD)) | Sitting diastolic blood pressure (mean mm Hg (SD)) | Comorbid conditions (%) |
|---|---|---|---|---|---|---|---|---|
| **Orlistat vs placebo** | | | | | | | | |
| Bakris 2002 | Orlistat | 53.2 (0.5) | 63 | 35.8 (3.9) | 101.2 (1.0) | 154.2 (13.4) | 98.4 (3.7) | diabetes (8) |
| | Placebo | 52.5 (0.5) | 59 | 35.4 (4.0) | 101.5 (1.0) | 150.8 (12.7) | 98.3 (3.5) | diabetes (8) |
| Cocco 2005 | Orlistat | 54.9 (5.1) | 51 | 36.5 (1.9) | 107.0 (5.7) | 145.8 (9.8) | 87.8 (7.3) | metabolic syndrome (100)<br>coronary heart disease (77)<br>myocardial infarction (47) |
| | Placebo | 54.5 (4.5) | 51 | 36.1 (1.8) | 106.0 (5.9) | 142.1 (6.2) | 85.3 (5.6) | |
| Guy-Grand 2004 | Orlistat | 49.1 (0.6) | 69 | 34.3 (0.2) | 93.9 (0.8) | 150.0 (0.8) | 96.9 (0.3) | nr |
| | Placebo | 49.5 (0.5) | 65 | 33.9 (0.2) | 93.5 (0.8) | 152.2 (0.9) | 97.0 (0.3) | nr |
| XENDOS 2001-2006 | Orlistat (DBP ≥ 90 mm Hg) | 46 (7) | 3862 | nr | 116 (17) | 146 (13) | 95 (5) | nr |

**Table 3. Baseline characteristics (part 1)** *(Continued)*

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | Orlistat (SBP ≥ 140 mm Hg) | 47(7) | 42 | nr | 116 (17) | 149 (10) | 91 (9) | nr |
| | Placebo (DBP ≥ 90 mm Hg) | 46 (7) | 4456 | nr | 114 (18) | 142 (126) | 95 (5) | nr |
| | Placebo (SBP ≥ 140 mm Hg) | 47(7) | 42 | nr | 115 (18) | 149 (8) | 91 (8) | nr |
| **Phentermine/topiramate vs placebo** | | | | | | | | |
| CONQUER 2013 | Phen/Top (LD) | 53.0 (9.8) | 65.9% | 36.7 (4.6) | 104.4 (18.4) | 134.2 (13.0) | 83.7 (9.1) | nr |
| | Phen/Top (HD) | | | | | | | |
| | Placebo | | | | | | | |
| **Naltrexone/bupropion vs placebo** | | | | | | | | |
| Nissen 2016 | Nal/Bup | 61.8 (7.27) | 54.5 | 37.1 (5.27) | 106 (19.09) | 126.1 (12.55) | 74.5 (9.01) | History of cardiovascular disease (30.3)<br>History of Type 2 diabetes (86.4)<br>History of dyslipidaemia (92.4)<br>History of low LDL (27.8)<br>Current smoker (8.4) |
| | Placebo | 61.6 (7.38) | 54.4 | 37.3 (5.42) | 106.6 (19.17) | 125.7 (12.62) | 74.4 (9.14) | History of cardiovascular disease (31.3)<br>History of type 2 diabetes (86.9)<br>History of dyslipidemia (92.1)<br>History of low LDL (28.2)<br>Current smoker (8.5) |
| Semaglutide vs placebo | | | | | | | | |

**Table 3. Baseline characteristics (part 1)** *(Continued)*

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| STEP 1 2023[a] | Semaglutide | 46 (13) | 73.1 | 37.8 (6.7) | 105.4 (22.1) | 126 (14) | 80 (10) | Prediabetes (45.4)<br>History of the following conditions:<br>Dyslipidemia (38.2)<br>Hypertension (36.1)<br>Knee osteoarthritis (13.2)<br>Obstructive sleep apnea (12.2)<br>Astma or chronic obstructive pulmonary disease (11.3)<br>Nonalcoholic fatty liver disease (7.7)<br>Polycystic ovarian syndrome (6.5; 62/955)<br>Coronary artery disease (2.5) |
| | Placebo | 47 (12) | 76.0 | 38.0 (6.5) | 105.2 (21.5) | 127 (14) | 80 (10) | Prediabetes (40.2)<br>History of the following conditions:<br>Dyslipidemia (34.5)<br>Hypertension (35.7)<br>Knee osteoarthritis (15.6)<br>Obstructive sleep apnea (10.8)<br>Astma or chronic obstructive pulmonary disease (12.2)<br>Nonalcoholic fatty liver disease (9.5)<br>Polycystic ovarian syndrome (6.8; 34/498)<br>Coronary artery disease (2.6) |
| Tirzepatide vs placebo | | | | | | | | |
| SURMOUNT-1 2024[a] | Tirzepatide | 45.0 (12.4) | 1278 (67.4) | 37.9 (6.8) | 104.8 (22.4) | 123.5 (12.7) | 79.5 (8.2) | Prediabetes (40.2) |
| | Placebo | 44.4 (12.5) | 436 (67.8) | 38.2 (6.9) | 104.8 (21.4) | 122.9 (12.8) | 79.6 (8.0) | Prediabetes (42.0) |

**Table 3. Baseline characteristics (part 1)** *(Continued)*

BMI: body mass index; LDL: low-density lipoprotein; nr: not reported; Phen/Top (HD): phentermine/topiramate high dose (15 mg/92 mg); Phen/Top (LD): phentermine/topiramate low dose (7.5 mg/46 mg); SBD: systolic blood pressure; SD: standard deviation; vs: versus

[a]Only data for the entire population reported

**Table 4. Adverse events (part 1)**

| Study | Intervention(s) and comparator(s) | Randomised/safety (N) | Death (n (%)) | All adverse events (n (%)) | Leading to withdrawal (n (%)) | Serious adverse events (n (%)) |
|---|---|---|---|---|---|---|
| **Orlistat vs placebo** | | | | | | |
| Bakris 2002 | Orlistat | 278/268[a] | 0 (0) | 239 (89) | 18 (6.7) | 31 (11.7) |
| | Placebo | 276/274[a] | 0 (0) | 195 (71) | 20 (7.3) | 24 (8.6) |
| Cocco 2005 | Orlistat | 45/45 | 0 (0) | nr | nr | 0 (0) |
| | Placebo | 45/45 | 0 (0) | nr | nr | 0 (0) |
| Guy-Grand 2004 | Orlistat | 304/304 | nr | nr | nr | nr |
| | Placebo | 310/310 | nr | nr | nr | nr |
| XENDOS 2001-2006 | Orlistat (DBP ≥ 90 mm Hg) | 408/407 | 2 (0.5) | 403 (99) | 37 (9) | 73 (18) |
| | Orlistat (SBP ≥ 140mm Hg) | 516/513 | 1 (0.2) | 508 (99) | 46 (9) | 92 (18) |
| | Placebo (DBP ≥ 90 mm Hg) | 441/437 | 0 (0) | 420 (96) | 17 (4) | 52 (12) |
| | Placebo (SBP ≥ 140 mm Hg) | 509/508 | 0 (0) | 493 (97) | 20 (4) | 60 (12) |
| **Phentermine/topiramate vs placebo** | | | | | | |
| CONQUER 2013 | Phen/Top (LD) | 261/261 | 0 (0) | 223 (85.4) | 31 (11.9) | 9 (3.4) |
| | Phen/Top (HD) | 520/520 | 0 (0) | 462 (88.8) | 103 (19.8) | 19 (3.7) |
| | Placebo | 524/524 | 0 (0) | 405 (77.3) | 51 (9.7) | 22 (4.2) |
| **Naltrexone/bupropion vs placebo** | | | | | | |
| Nissen 2016 | Nal/Bup | 4164/4164 | 63 (1.5)/P = 0.916 | 1796 (43.1)/ P < 0.001 | 1273 (30.6)/ P < 0.001 | 891 (21.4)/P = 0.297 |
| | Placebo | 4119/4119 | 63 (1.5) | 1053 (25.6) | 379 (9.2) | 843 (20.5) |
| **Semaglutide vs placebo** | | | | | | |
| STEP 1 2023 | Semaglutide | 177/177 | nr | nr | nr | nr |
| | Placebo | 97/97 | nr | nr | nr | nr |
| **Tirzepatide vs placebo** | | | | | | |
| SURMOUNT-1 2024 | Tirzepatide (SBP ≥ 140 mm Hg) | 153/153 | nr | nr | nr | nr |
| | Tirzepatide (DBP ≥ 90 mm Hg) | 168/168 | nr | nr | nr | nr |

**Table 4. Adverse events (part 1)** *(Continued)*

| | | | | | | |
|---|---|---|---|---|---|---|
| | Tirzepatide (antihypertensive medication) | 500/500 | nr | nr | nr | nr |
| | Placebo (SBP ≥ 140 mm Hg) | 42/42 | nr | nr | nr | nr |
| | Placebo (DBP ≥ 90 mm Hg) | 39/39 | nr | nr | nr | nr |
| | Placebo (antihypertensive medication) | 135/135 | nr | nr | nr | nr |

[a]Calculated from percentage rates
DBP: diastolic blood pressure; N: number of participants; n: number of events; nr: not reported; P: probability; Phen/Top (HD): phentermine/topiramate high dose (15 mg/92 mg); Phen/Top (LD): phentermine/topiramate low dose (7.5 mg/46 mg); SBD: systolic blood pressure; vs: versus